



ASSESSMENT COVERSHEET

Attach this coversheet as the cover of your submission. All sections must be completed.

Section A: Submission Details

Programme : BACHELOR OF COMPUTER ENGINEERING TECHNOLOGY (COMPUTER SYSTEMS) WITH HONOURS

Course Code & Name : IPB 49906 FINAL YEAR PROJECT 2

Course Lecturer(s) : Ts. DIYANA BT AB KADIR (FYP2 BCE COORDINATOR)

Submission Title : PROGRESS REPORT

Deadline : Day 5 Month APR Year 2026 Time 11:50PM

Penalties :

- 5% will be deducted per day to a maximum of four (4) working days, after which the submission will **not** be accepted.
- Plagiarised work is an Academic Offence in University Rules & Regulations and will be penalised accordingly.

Section B: Academic Integrity

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☒	This submission is my own, unless indicated with proper referencing.
☒	This submission has not been previously submitted or published.
☒	This submission follows the requirements stated in the course.

Section C: Submission Receipt

(must be filled in manually)

Office Receipt of Submission

Date & Time of Submission (stamp)	Student Name(s)	Student ID(s)
4/8/2026 9:59 PM	MD EJAJ MAHMUD	5222222123

Student Receipt of Submission

This is your submission receipt, the only accepted evidence that you have submitted your work. After this is stamped by the appointed staff & filled in, cut along the dotted lines above & retain this for your record.

Date & Time of Submission (stamp)	Course Code	Submission Title	Student ID(s) & Signature(s)
4/8/2026 9:59 PM	IPB 49906	Progress Report 2	5222222123



**UNIVERSITI KUALA LUMPUR
RUBRIC FOR PROJECT PROGRESS**

COURSE CODE	IPB 49906	TOTAL MARKS (50 MARKS)
COURSE NAME	FINAL YEAR PROJECT 2	
COURSE SEMESTER/ YEAR	MARCH 2026	
STUDENT NAME	MD EAJ MAHMUD	
STUDENT ID	52222222123	
ASSESSMENT DATE	WEEK 1 – WEEK 11	Smart Recycling: Reward-Based Reverse Vending Machine Using Proximity Sensor Detection for Plastic Waste Management
ASSESSMENT DUE	29 MAY 2026 4PM	
NAME OF SUPERVISOR	Dr. Grad. Engr. Hannah Binti Sofian	

NO.	CRITERIA	Description	WEIGHTAGE	SCORE 1-10	ACTUAL SCORE
1	Consultation with Supervisor	Meet Supervisor with significant progress	1		
2	Timelines	Record of research progress with respect to the Timelines	1		
3	Research progress	Show progress in research with related information and resources	1		
4	Proposed Idea	Ability to apply higher order thinking	1		
5	Self-driven & Independency	Ability to perform independently	1		
TOTAL SCORE (50 MARKS)					



PROGRESS REPORT NO. 2

Student's Name: MD EJAJ MAHMUD

FYP Code : IPB49906

Semester	07	Week	02
Meeting No.	01	Date	08/04/2026

Progress (fill in by Student)

Week 3 was the mid-semester break, so after the holiday I continued the project work in Week 4 and Week 5 by focusing on simulation while waiting for the remaining ordered items to arrive. Since some of the hardware is still pending delivery, I could not start the full prototype assembly yet. Instead of waiting without doing anything, I used this time to simulate the project in Proteus based on the planned reverse vending machine design. This helped me continue the project progress even though the physical setup is not fully ready yet.

The simulation includes the main controller and the important components that will be used in the actual project, such as the Arduino Mega 2560, LCD, ultrasonic sensor, IR sensor, proximity sensors, buzzer, LEDs, and servo motor. Through this simulation, I was able to review the circuit connection, pin arrangement, and the overall sequence of the system more clearly.

At the current stage in Week 5, the project is still in the preparation and simulation phase because a few ordered items have not been received yet. Once the remaining components arrive, I will continue with the actual wiring, component testing, and prototype assembly.

Appendix 1 shows the current simulation image prepared for Progress Report 2.

Appendix 1: Project Simulation

Current Proteus simulation prepared while waiting for the remaining hardware items.

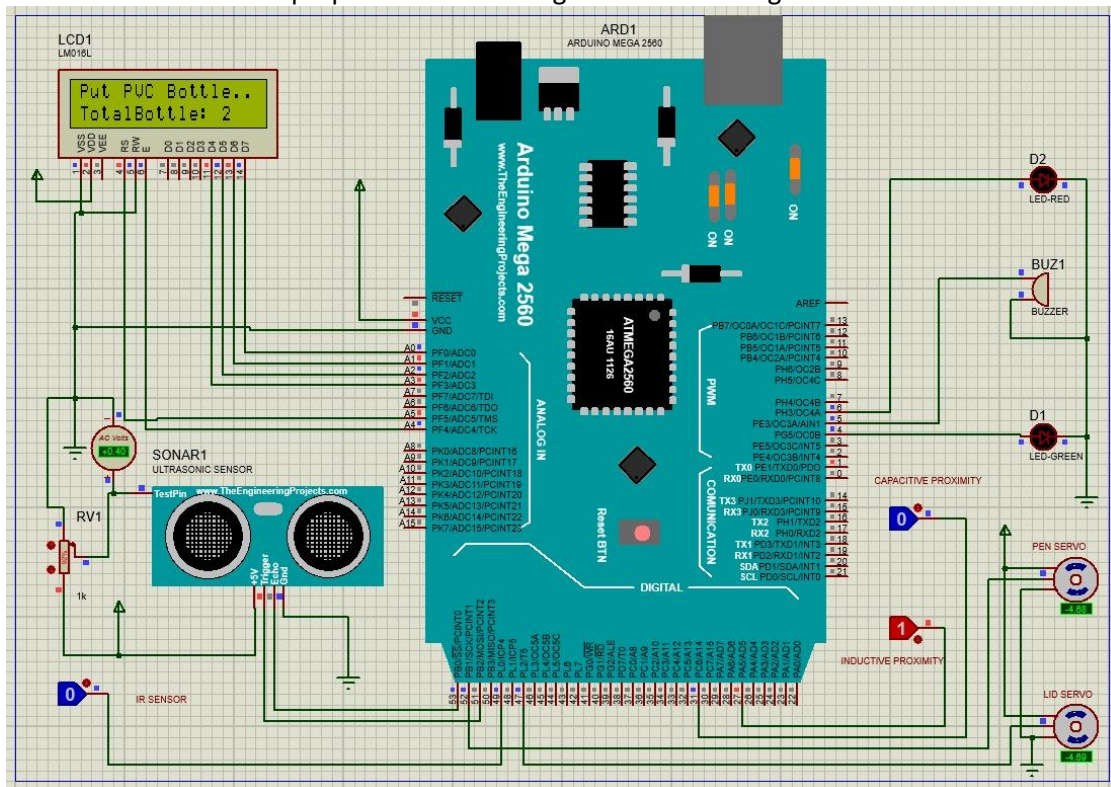
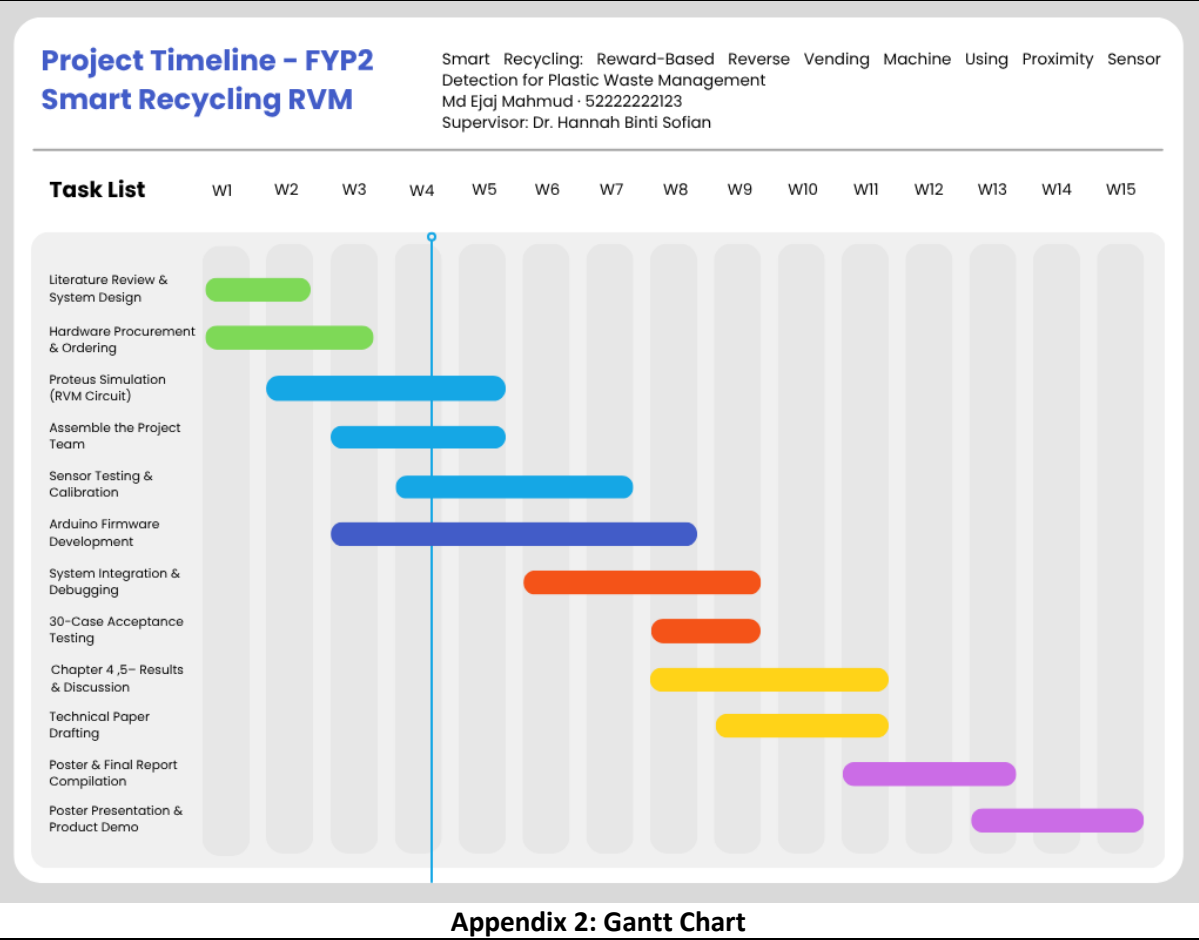
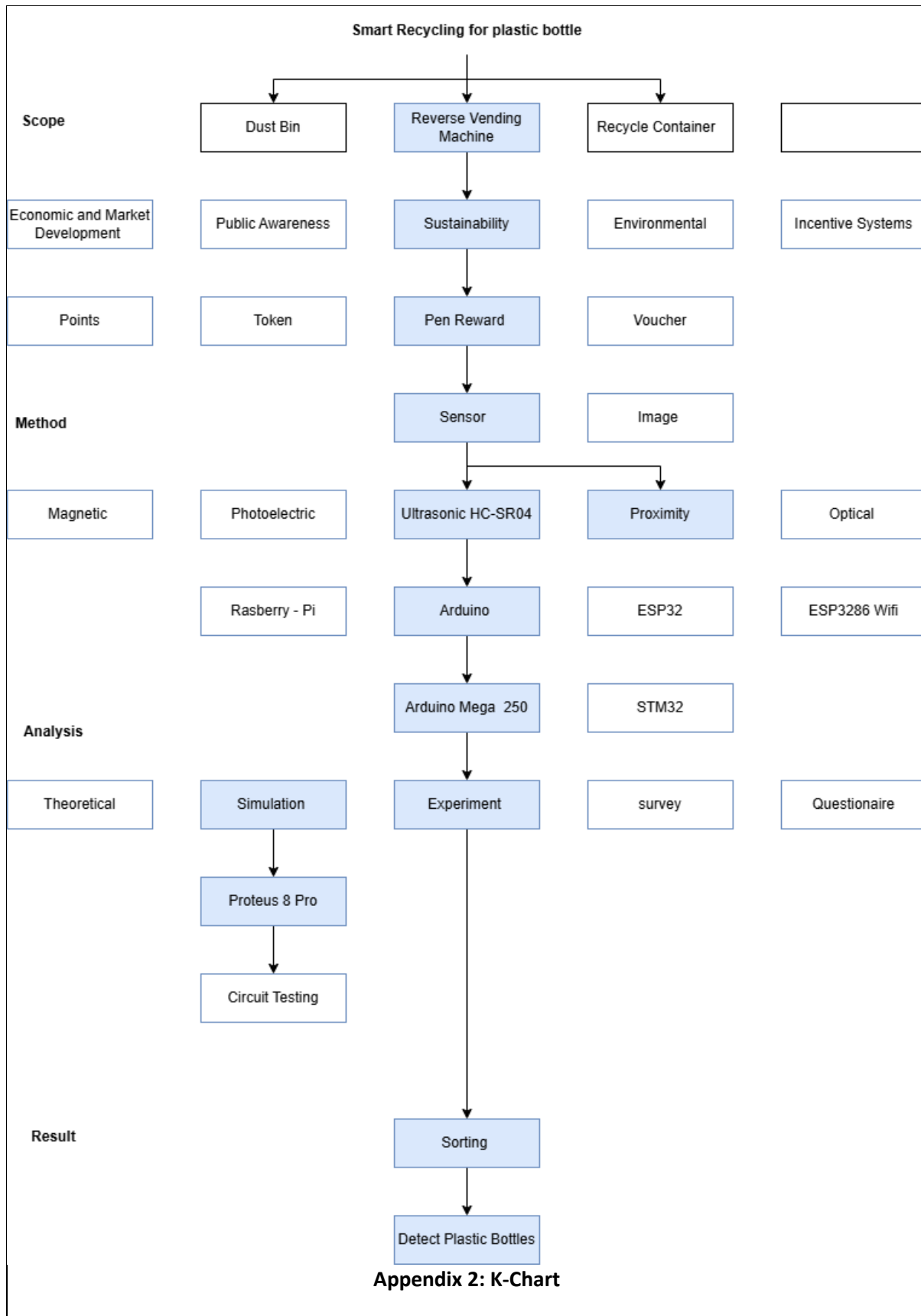


Figure A1. Reverse vending machine simulation used for Progress Report 2.







Comments (by Supervisor)	Signature and Name
ok. pls explain in PR 2	

COMPONENT DATASHEETS

Appendix — Progress Report 2

Project : Smart Recycling: Reward-Based Reverse Vending Machine
Using Proximity Sensor Detection for Plastic Waste Management

Student : Md Ejaj Mahmud

ID : 52222222123

Course : IPB 49906 — Final Year Project 2

SV : Dr. Hannah Binti Sofian

INCLUDED DATASHEETS:

1

Arduino Mega 2560

Microcontroller Board — ATmega2560, 16 MHz, 54 digital I/O pins

2

HC-SR04 Ultrasonic Sensor

Bin Level Detection — Range: 2 cm–400 cm | 5V | 4-pin

3

SG90 Micro Servo Motor

Lid & Pen Dispenser Actuator — Torque: 1.6 kg/cm | 4.8–6V

4

TCRT5000 IR Sensor

Bottle Presence Detection — IR reflective | 950 nm emitter

5

LJ12A3-4-Z/BX Proximity Sensor

Metal/Plastic Detection — NPN 3-wire | 6–36V | 4 mm sensing

6

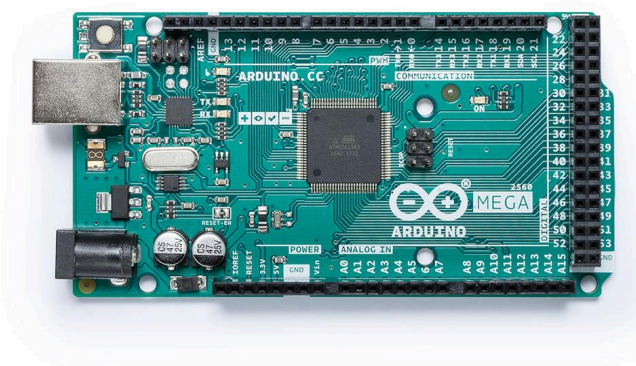
HD44780 LCD 16×2 Controller

Status Display — Hitachi driver | 16-char × 2-line | parallel



Arduino Mega 2560

Microcontroller — ATmega2560, 16 MHz, 54 I/O



Description

Arduino® Mega 2560 Rev3 is an exemplary development board dedicated for building extensive applications as compared to other maker boards by Arduino. The board accommodates the ATmega2560 microcontroller, which operates at a frequency of 16 MHz. The board contains 54 digital input/output pins, 16 analog inputs, 4 UARTs (hardware serial ports), a USB connection, a power jack, an ICSP header, and a reset button.

Target Areas

3D Printing, Robotics, Maker



Features

▪ ATmega2560 Processor

- Up to 16 MIPS Throughput at 16MHz
- 256k bytes (of which 8k is used for the bootloader)
- 4k bytes EEPROM
- 8k bytes Internal SRAM
- 32 × 8 General Purpose Working Registers
- Real Time Counter with Separate Oscillator
- Four 8-bit PWM Channels
- Four Programmable Serial USART
- Controller/Peripheral SPI Serial Interface

▪ ATmega16U2

- Up to 16 MIPS Throughput at 16 MHz
- 16k bytes ISP Flash Memory
- 512 bytes EEPROM
- 512 bytes SRAM
- USART with SPI master only mode and hardware flow control (RTS/CTS)
- Master/Slave SPI Serial Interface

▪ Sleep Modes

- Idle
- ADC Noise Reduction
- Power-save
- Power-down
- Standby
- Extended Standby

▪ Power

- USB Connection
- External AC/DC Adapter

▪ I/O

- 54 Digital
- 16 Analog
- 15 PWM Output



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**14 电路板简介**

14.1 应用示例

14.2 配件

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15 额定值

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17 电路板操作

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17.2 入门指南 - Arduino Cloud Editor

17.3 示例程序

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18.6 数字引脚 D22 - D53 LHS

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19.2 电路板安装孔

20 符合性声明 CE DoC (欧盟)**21 声明符合欧盟 RoHS 和 REACH 211 01/19/2021****22 冲突矿产声明****23 FCC 警告****24 公司信息****25 参考资料****26 修订记录**

1 The Board

Mega 2560 Rev3 is a successor board of Arduino Mega, it is dedicated to applications and projects that require large number of input output pins and the use cases which need high processing power. The Mega 2560 Rev3 comes with a much larger set of IOs when we compare it with the traditional Arduino® UNO board considering the form factor of both the boards.

1.1 Application Examples

- **Robotics:** Featuring the high processing capacity, the Mega 2560 Rev3 can handle the extensive robotic applications. It is compatible with the motor controller shield that enables it to control multiple motors at an instance, thus making it perfect of robotic applications. The large number of I/O pins can accommodate many robotic sensors as well.
- **3D Printing:** Algorithms play a significant role in implementation of 3D printers. Mega 2560 Rev3 has the power to process these complex algorithms required for 3D printing. Additionally, the slight changes to the code is easily possible with the Arduino IDE and thus 3D printing programs can be customized according to user requirements.
- **Wi-Fi:** Integrating wireless functionality enhances the utility of the applications. Mega 2560 Rev3 is compatible with Wi-Fi® shields hence allowing the wireless features for the applications in 3D printing and Robotics.

1.2 Accessories

1.3 Related Products

- Arduino® UNO R3
- Arduino® Nano
- Arduino® Due without headers

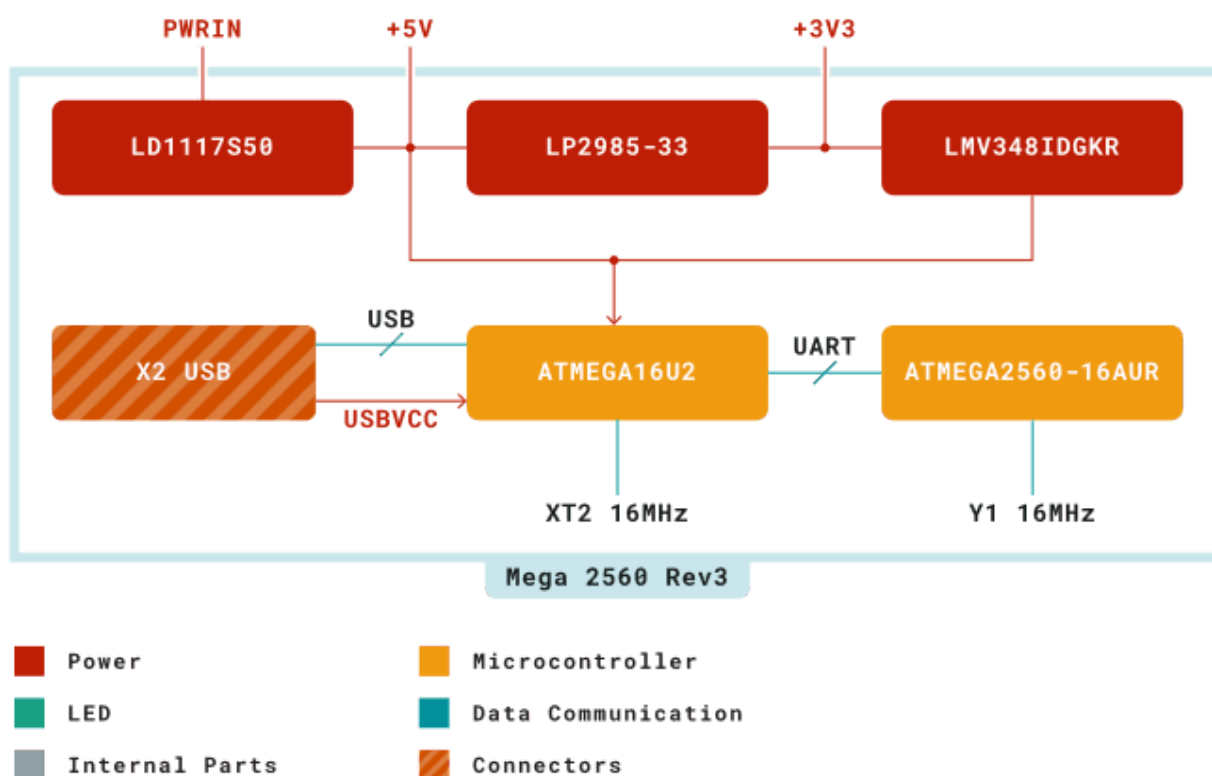
2 Ratings

2.1 Recommended Operating Conditions

Symbol	Description	Min	Typ	Max	Unit
V_{IN}	Input voltage from VIN pad / DC Jack	7	7.0	12	V
V_{USB}	Input voltage from USB connector	4.8	5.0	5.5	V
T_{OP}	Operating Temperature	-40	25	85	°C

3 Functional Overview

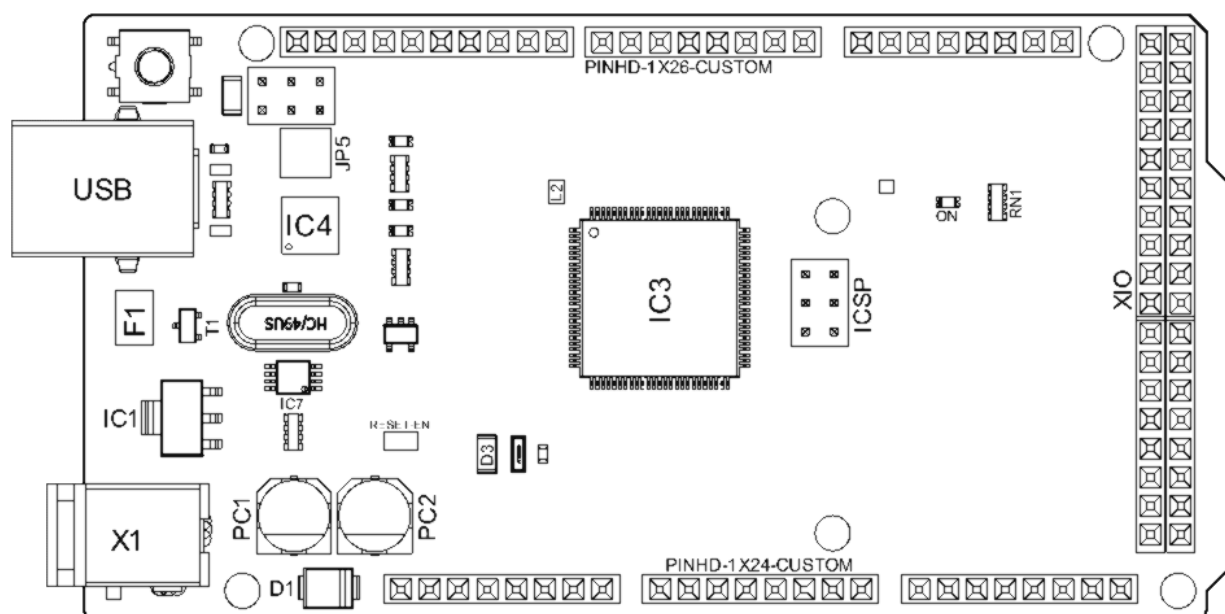
3.1 Block Diagram



Arduino Mega 2560 Rev3 Block Diagram

3.2 Board Topology

Front View



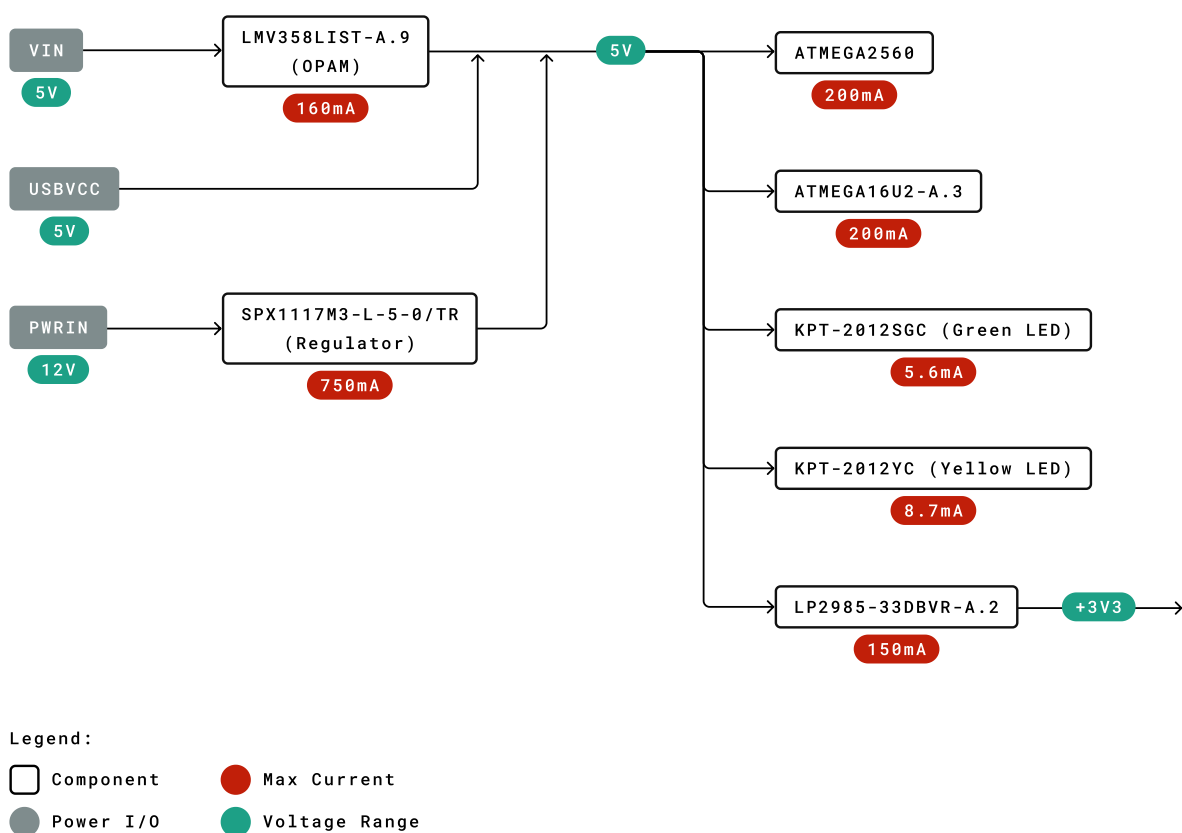
Arduino Mega 2560 Rev3 Top View

Ref.	Description	Ref.	Description
USB	USB B Connector	F1	Chip Capacitor
IC1	5V Linear Regulator	X1	Power Jack Connector
JP5	Plated Holes	IC4	ATmega16U2 chip
PC1	Electrolytic Aluminum Capacitor	PC2	Electrolytic Aluminum Capacitor
D1	General Purpose Rectifier	D3	General Purpose Diode
L2	Fixed Inductor	IC3	ATmega2560 chip
ICSP	Connector Header	ON	Green LED
RN1	Resistor Array	XIO	Connector

3.3 Processor

Primary processor of Mega 2560 Rev3 board is ATmega2560 chip which operates at a frequency of 16 MHz. It accommodates a large number of input and output lines which gives the provision of interfacing many external devices. At the same time the operations and processing is not slowed due to its significantly larger RAM than the other processors. The board also features a USB serial processor ATmega16U2 which acts an interface between the USB input signals and the main processor. This increases the flexibility of interfacing and connecting peripherals to the Mega 2560 Rev3 board.

3.4 Power Tree



Power Tree



4 Board Operation

4.1 Getting Started - IDE

If you want to program your Mega 2560 Rev3 while offline you need to install the Arduino Desktop IDE [1] To connect the Mega 2560 Rev3 to your computer, you'll need a Type-B USB cable. This also provides power to the board, as indicated by the LED.

4.2 Getting Started - Arduino Cloud Editor

All Arduino boards, including this one, work out-of-the-box on the Arduino Cloud Editor [2], by just installing a simple plugin.

The Arduino Cloud Editor is hosted online, therefore it will always be up-to-date with the latest features and support for all boards. Follow [3] to start coding on the browser and upload your sketches onto your board.

4.3 Sample Sketches

Sample sketches for the Mega 2560 Rev3 can be found either in the "Examples" menu in the Arduino IDE or under the "Documentation" menu on the Arduino website [4].

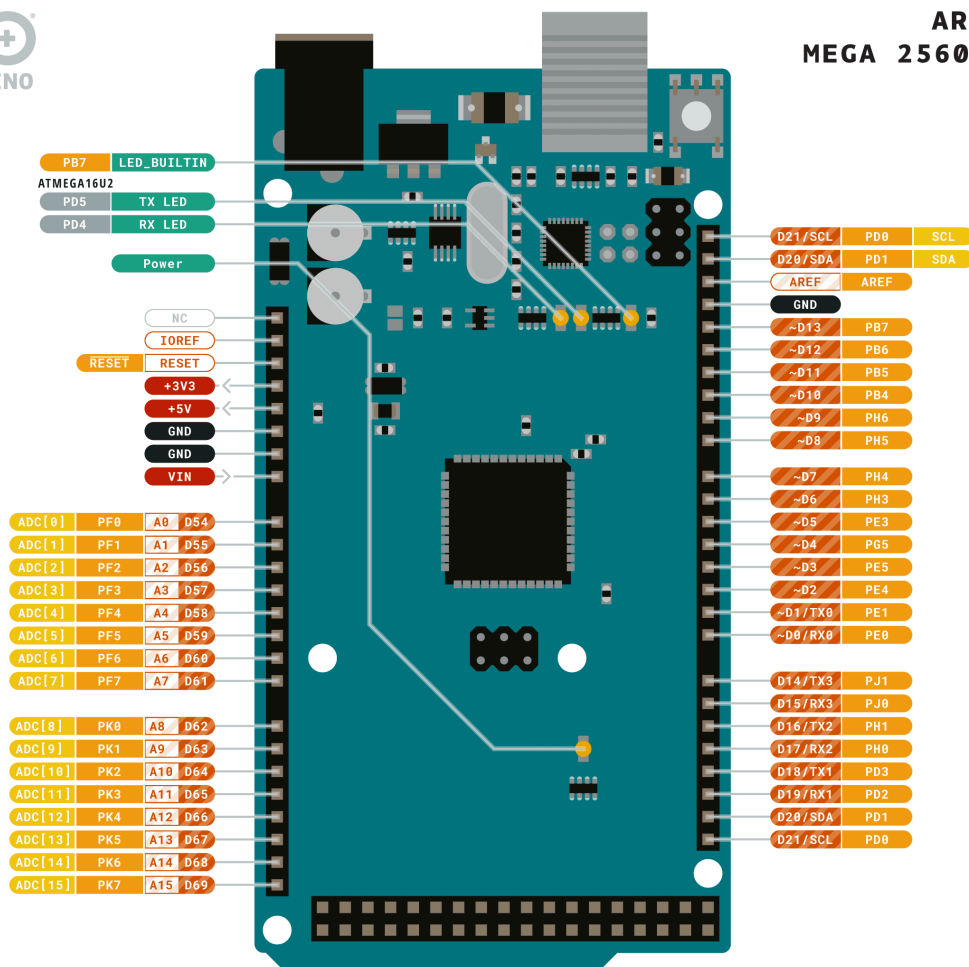
4.4 Online Resources

Now that you have gone through the basics of what you can do with the board you can explore the endless possibilities it provides by checking exciting projects on Arduino Project Hub [5], the Arduino Library Reference [6] and the online store [7] where you will be able to complement your board with sensors, actuators and more.

5 Connector Pinouts



ARDUINO MEGA 2560 REV3



Ground	Internal Pin	Digital Pin	Microcontroller's Port
Power	SWD Pin	Analog Pin	
LED	Other Pin	Default	

ARDUINO.CC



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Arduino Mega 2560 Rev3 Pinout

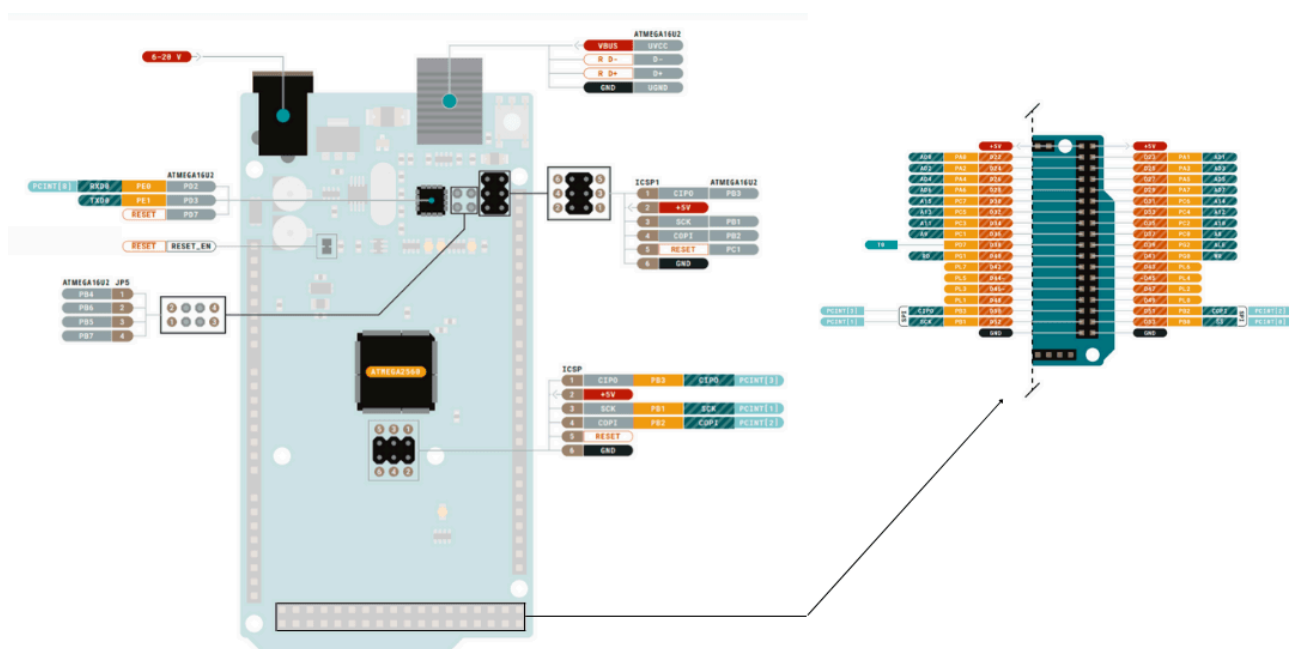
5.1 Analog

Pin	Function	Type	Description
1	NC	NC	Not Connected
2	IOREF	IOREF	Reference for digital logic V - connected to 5V
3	Reset	Reset	Reset
4	+3V3	Power	+3V3 Power Rail
5	+5V	Power	+5V Power Rail
6	GND	Power	Ground
7	GND	Power	Ground
8	VIN	Power	Voltage Input
9	A0	Analog	Analog input 0 /GPIO
10	A1	Analog	Analog input 1 /GPIO
11	A2	Analog	Analog input 2 /GPIO
12	A3	Analog	Analog input 3 /GPIO
13	A4	Analog	Analog input 4 /GPIO
14	A5	Analog	Analog input 5 /GPIO
15	A6	Analog	Analog input 6 /GPIO
16	A7	Analog	Analog input 7 /GPIO
17	A8	Analog	Analog input 8 /GPIO
18	A9	Analog	Analog input 9 /GPIO
19	A10	Analog	Analog input 10 /GPIO
20	A11	Analog	Analog input 11 /GPIO
21	A12	Analog	Analog input 12 /GPIO
22	A13	Analog	Analog input 13 /GPIO
23	A14	Analog	Analog input 14 /GPIO
24	A15	Analog	Analog input 15 /GPIO

5.2 Digital

Pin	Function	Type	Description
1	D21/SCL	Digital Input/I2C	Digital input 21/I2C Dataline
2	D20/SDA	Digital Input/I2C	Digital input 20/I2C Dataline
3	AREF	Digital	Analog Reference Voltage
4	GND	Power	Ground
5	D13	Digital/GPIO	Digital input 13/GPIO
6	D12	Digital/GPIO	Digital input 12/GPIO
7	D11	Digital/GPIO	Digital input 11/GPIO
8	D10	Digital/GPIO	Digital input 10/GPIO
9	D9	Digital/GPIO	Digital input 9/GPIO
10	D8	Digital/GPIO	Digital input 8/GPIO
11	D7	Digital/GPIO	Digital input 7/GPIO
12	D6	Digital/GPIO	Digital input 6/GPIO
13	D5	Digital/GPIO	Digital input 5/GPIO
14	D4	Digital/GPIO	Digital input 4/GPIO

Pin	Function	Type	Description
15	D3	Digital/GPIO	Digital input 3/GPIO
16	D2	Digital/GPIO	Digital input 2/GPIO
17	D1/TX0	Digital/GPIO	Digital input 1 /GPIO
18	D0/Tx1	Digital/GPIO	Digital input 0 /GPIO
19	D14	Digital/GPIO	Digital input 14 /GPIO
20	D15	Digital/GPIO	Digital input 15 /GPIO
21	D16	Digital/GPIO	Digital input 16 /GPIO
22	D17	Digital/GPIO	Digital input 17 /GPIO
23	D18	Digital/GPIO	Digital input 18 /GPIO
24	D19	Digital/GPIO	Digital input 19 /GPIO
25	D20	Digital/GPIO	Digital input 20 /GPIO
26	D21	Digital/GPIO	Digital input 21 /GPIO



Arduino Mega 2560 Rev3 Pinout

5.3 ATMEGA16U2 JP5

Pin	Function	Type	Description
1	PB4	Internal	Serial Wire Debug
2	PB6	Internal	Serial Wire Debug
3	PB5	Internal	Serial Wire Debug
4	PB7	Internal	Serial Wire Debug

5.4 ATMEGA16U2 ICSP1

Pin	Function	Type	Description
1	CIPO	Internal	Controller In Peripheral Out
2	+5V	Internal	Power Supply of 5V
3	SCK	Internal	Serial Clock
4	COPI	Internal	Controller Out Peripheral In
5	RESET	Internal	Reset
6	GND	Internal	Ground

5.5 Digital Pins D22 - D53 LHS

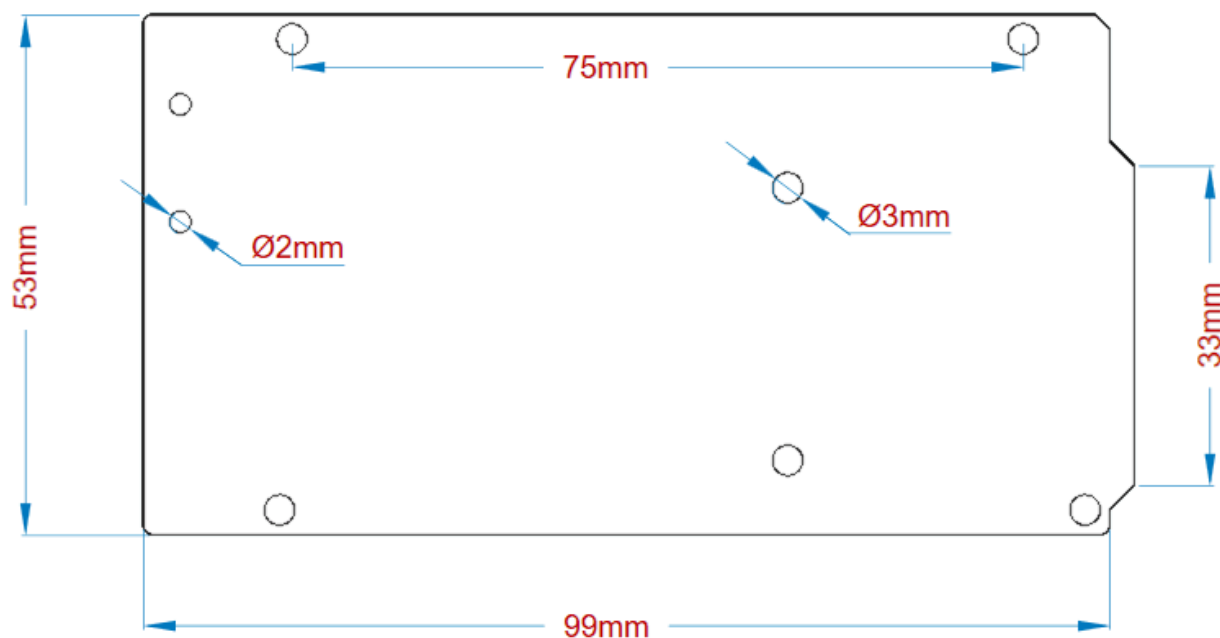
Pin	Function	Type	Description
1	+5V	Power	Power Supply of 5V
2	D22	Digital	Digital input 22/GPIO
3	D24	Digital	Digital input 24/GPIO
4	D26	Digital	Digital input 26/GPIO
5	D28	Digital	Digital input 28/GPIO
6	D30	Digital	Digital input 30/GPIO
7	D32	Digital	Digital input 32/GPIO
8	D34	Digital	Digital input 34/GPIO
9	D36	Digital	Digital input 36/GPIO
10	D38	Digital	Digital input 38/GPIO
11	D40	Digital	Digital input 40/GPIO
12	D42	Digital	Digital input 42/GPIO
13	D44	Digital	Digital input 44/GPIO
14	D46	Digital	Digital input 46/GPIO
15	D48	Digital	Digital input 48/GPIO
16	D50	Digital	Digital input 50/GPIO
17	D52	Digital	Digital input 52/GPIO
18	GND	Power	Ground

5.6 Digital Pins D22 - D53 RHS

Pin	Function	Type	Description
1	+5V	Power	Power Supply of 5V
2	D23	Digital	Digital input 23/GPIO
3	D25	Digital	Digital input 25/GPIO
4	D27	Digital	Digital input 27/GPIO
5	D29	Digital	Digital input 29/GPIO
6	D31	Digital	Digital input 31/GPIO
7	D33	Digital	Digital input 33/GPIO
8	D35	Digital	Digital input 35/GPIO
9	D37	Digital	Digital input 37/GPIO
10	D39	Digital	Digital input 39/GPIO
11	D41	Digital	Digital input 41/GPIO
12	D43	Digital	Digital input 43/GPIO
13	D45	Digital	Digital input 45/GPIO
14	D47	Digital	Digital input 47/GPIO
15	D49	Digital	Digital input 49/GPIO
16	D51	Digital	Digital input 51/GPIO
17	D53	Digital	Digital input 53/GPIO
18	GND	Power	Ground

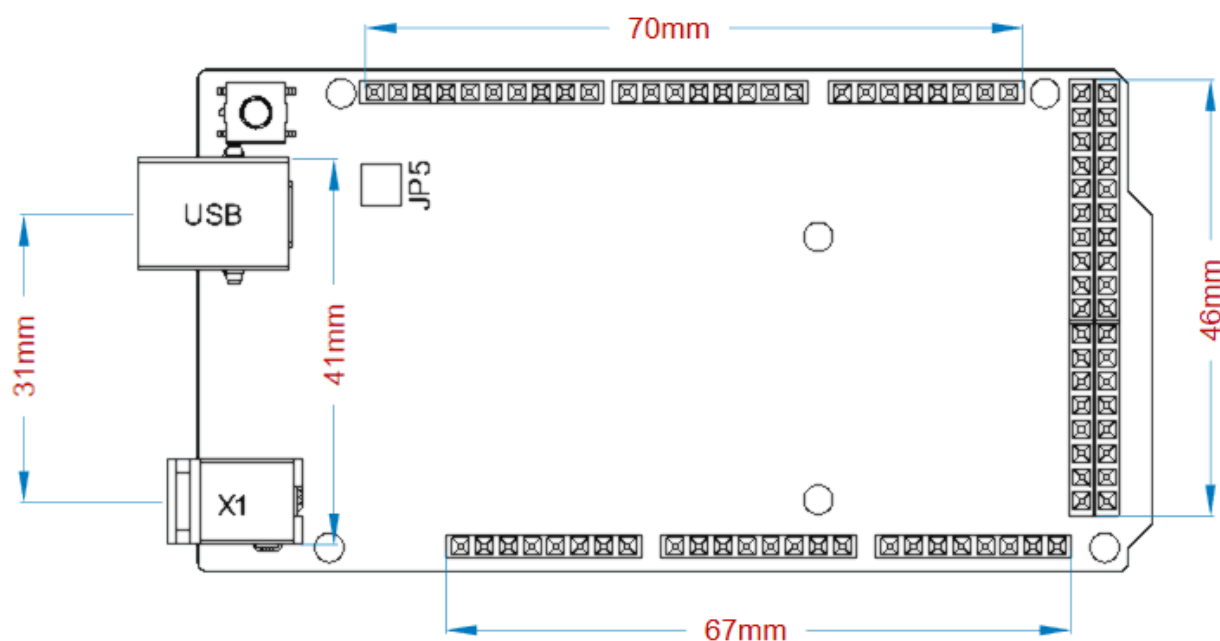
6 Mechanical Information

6.1 Board Outline



Arduino Mega 2560 Rev3 Outline

6.2 Board Mount Holes

*Arduino Mega 2560 Rev3 Mount Holes*

Certifications

7 Declaration of Conformity CE DoC (EU)

We declare under our sole responsibility that the products above are in conformity with the essential requirements of the following EU Directives and therefore qualify for free movement within markets comprising the European Union (EU) and European Economic Area (EEA).



8 Declaration of Conformity to EU RoHS & REACH 211 01/19/2021

Arduino boards are in compliance with RoHS 2 Directive 2011/65/EU of the European Parliament and RoHS 3 Directive 2015/863/EU of the Council of 4 June 2015 on the restriction of the use of certain hazardous substances in electrical and electronic equipment.

Substance	Maximum Limit (ppm)
Lead (Pb)	1000
Cadmium (Cd)	100
Mercury (Hg)	1000
Hexavalent Chromium (Cr6+)	1000
Poly Brominated Biphenyls (PBB)	1000
Poly Brominated Diphenyl ethers (PBDE)	1000
Bis(2-Ethylhexyl} phthalate (DEHP)	1000
Benzyl butyl phthalate (BBP)	1000
Dibutyl phthalate (DBP)	1000
Diisobutyl phthalate (DIBP)	1000

Exemptions : No exemptions are claimed.

Arduino Boards are fully compliant with the related requirements of European Union Regulation (EC) 1907 /2006 concerning the Registration, Evaluation, Authorization and Restriction of Chemicals (REACH). We declare none of the SVHCs (<https://echa.europa.eu/web/guest/candidate-list-table>), the Candidate List of Substances of Very High Concern for authorization currently released by ECHA, is present in all products (and also package) in quantities totaling in a concentration equal or above 0.1%. To the best of our knowledge, we also declare that our products do not contain any of the substances listed on the "Authorization List" (Annex XIV of the REACH regulations) and Substances of Very High Concern (SVHC) in any significant amounts as specified by the Annex XVII of Candidate list published by ECHA (European Chemical Agency) 1907 /2006/EC.

9 Conflict Minerals Declaration

As a global supplier of electronic and electrical components, Arduino is aware of our obligations with regards to laws and regulations regarding Conflict Minerals, specifically the Dodd-Frank Wall Street Reform and Consumer Protection Act, Section 1502. Arduino does not directly source or process conflict minerals such as Tin, Tantalum, Tungsten, or Gold. Conflict minerals are contained in our products in the form of solder, or as a component in metal alloys. As part of our reasonable due diligence Arduino has contacted component suppliers within our supply chain to verify their continued compliance with the regulations. Based on the information received thus far we declare that our products contain Conflict Minerals sourced from conflict-free areas.

10 FCC Caution

Any Changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

This device complies with part 15 of the FCC Rules. Operation is subject to the following two conditions:

- (1) This device may not cause harmful interference
- (2) this device must accept any interference received, including interference that may cause undesired operation.

FCC RF Radiation Exposure Statement:

1. This Transmitter must not be co-located or operating in conjunction with any other antenna or transmitter.
2. This equipment complies with RF radiation exposure limits set forth for an uncontrolled environment.
3. This equipment should be installed and operated with minimum distance 20cm between the radiator & your body.

English: User manuals for licence-exempt radio apparatus shall contain the following or equivalent notice in a conspicuous location in the user manual or alternatively on the device or both. This device complies with Industry Canada licence-exempt RSS standard(s). Operation is subject to the following two conditions:

- (1) this device may not cause interference
- (2) this device must accept any interference, including interference that may cause undesired operation of the device.

French: Le présent appareil est conforme aux CNR d'Industrie Canada applicables aux appareils radio exempts de licence. L'exploitation est autorisée aux deux conditions suivantes :

- (1) l'appareil ne doit pas produire de brouillage
- (2) l'utilisateur de l'appareil doit accepter tout brouillage radioélectrique subi, même si le brouillage est susceptible d'en compromettre le fonctionnement.

IC SAR Warning:

English This equipment should be installed and operated with minimum distance 20 cm between the radiator and your body.



French: Lors de l'installation et de l'exploitation de ce dispositif, la distance entre le radiateur et le corps est d'au moins 20 cm.

Important: The operating temperature of the EUT can't exceed 85°C and shouldn't be lower than -40°C.

Hereby, Arduino S.r.l. declares that this product is in compliance with essential requirements and other relevant provisions of Directive 201453/EU. This product is allowed to be used in all EU member states.

11 Company Information

Company name	Arduino S.r.l.
Company Address	Arduino SRL, Via Andrea Appiani 25, 20900 Monza MB, Italy

12 Reference Documentation

Ref	Link
Arduino IDE (Desktop)	https://www.arduino.cc/en/Main/Software
Arduino Cloud Editor	https://create.arduino.cc/editor
Arduino Cloud Editor - Getting Started	https://docs.arduino.cc/arduino-cloud/guides/editor/
Arduino Website	https://www.arduino.cc/
Arduino Project Hub	https://create.arduino.cc/projecthub?by=part&part_id=11332&sort=trending
Library Reference	https://www.arduino.cc/reference/en/libraries/
Online Store	https://store.arduino.cc/

13 Revision History

Date	Revision	Changes
25/04/2024	3	Updated link to new Cloud Editor
09/10/2023	2	Updated recommended operating conditions
29/09/2020	1	First Release

中文 (ZH)

描述

与 Arduino 的其他开发板相比，Arduino® Mega 2560 Rev3 是一款用于构建广泛应用的示范性开发板。该电路板内置 ATmega2560 微控制器，工作频率为 16 MHz。电路板包含 54 个数字输入/输出引脚、16 个模拟输入、4 个 UART（硬件串行端口）、一个 USB 连接、一个电源插孔、一个 ICSP 接头和一个复位按钮。

目标领域：

3D 打印、机器人、创客

特点

- **ATmega2560 处理器**
 - 在 16MHz 的工作频率下吞吐量高达 16 MIPS
 - 256k 字节（其中 8k 字节用于引导加载程序）
 - 4k 字节 EEPROM
 - 8k 字节内部 SRAM
 - 32 × 8 通用工作寄存器
 - 带有独立振荡器的实时计数器
 - 四个 8 位 PWM 通道
 - 4 个可编程串行 USART
 - 控制器/外设 SPI 串行接口
- **ATmega16U2**
 - 在 16MHz 的工作频率下吞吐量高达 16 MIPS
 - 16k 字节 ISP 闪存
 - 512 字节 EEPROM
 - 512 字节 SRAM
 - 仅支持 SPI 主设备模式和硬件流控制 (RTS/CTS) 的 USART
 - 主/从 SPI 串行接口
- **休眠模式**
 - 空闲模式
 - ADC 降噪模式
 - 节能模式
 - 掉电模式
 - 待机模式
 - 扩展待机模式
- **电源**
 - USB 连接

- 外部交流/直流适配器
- 输入/输出
 - 54 个数字输入/输出引脚
 - 16 个模拟输入
 - 15 个 PWM 输出

目录

14 电路板简介

Mega 2560 Rev3 是 Arduino Mega 的一款升级版电路板，专门用于需要大量输入输出引脚和需要高处理能力的应用和项目。与传统的 Arduino® UNO 电路板相比，尽管两款电路板外形尺寸相似，但 Mega 2560 Rev3 的输入输出引脚数量要多得多。

14.1 应用示例

- **机器人：** Mega 2560 Rev3 具有高处理能力，可以处理大量机器人应用。它与电机控制器扩展板兼容，能够同时控制多个电机，因此非常适合机器人应用。其大量的 I/O 引脚也能容纳许多机器人传感器。
- **3D 打印：** 算法在 3D 打印机的实施过程中发挥着重要作用。Mega 2560 Rev3 能够处理 3D 打印所需的复杂算法。此外，使用 Arduino IDE 可以轻松地对代码进行细微修改，因此可以根据用户要求定制 3D 打印程序。
- **Wi-Fi：** 该电路板集成无线功能，增强了应用的实用性。Mega 2560 Rev3 与 Wi-Fi® 扩展板兼容，因此可以为 3D 打印和机器人应用提供无线功能。

14.2 配件

14.3 相关产品

- Arduino® UNO R3
- Arduino® Nano
- Arduino® Due（不带接头）

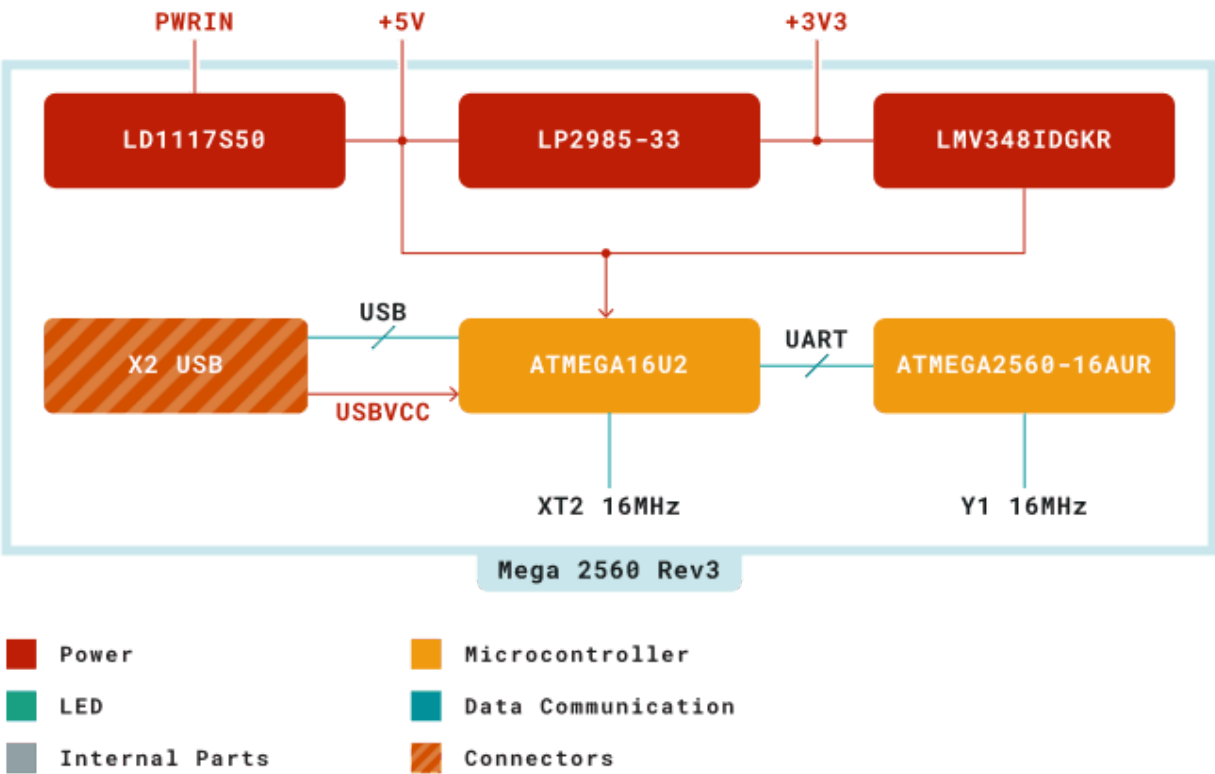
15 额定值

15.1 建议运行条件

符号	描述	最小值	典型值	最大值	单位
V _{IN}	来自 VIN 焊盘/DC 插孔的输入电压	7	7.0	12	V
V _{USB}	来自 USB 连接器的输入电压	4.8	5.0	5.5	V
T _{OP}	工作温度	-40	25	85	°C

16 功能概述

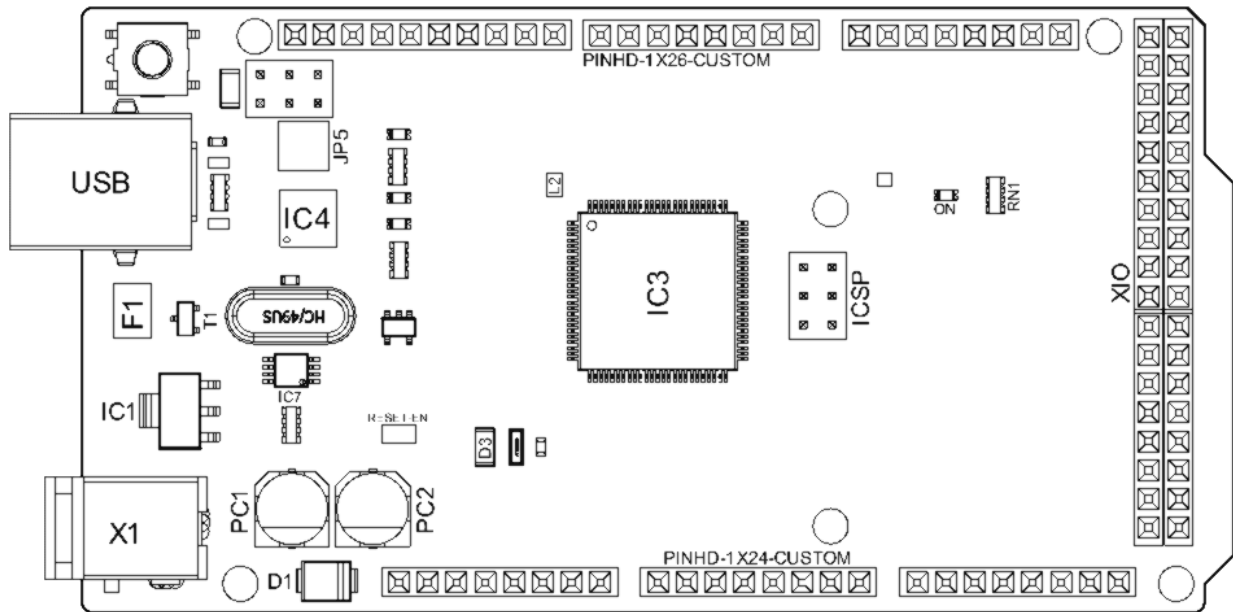
16.1 方框图



Arduino Mega 2560 Rev3 方框图

16.2 电路板拓扑结构

前视图



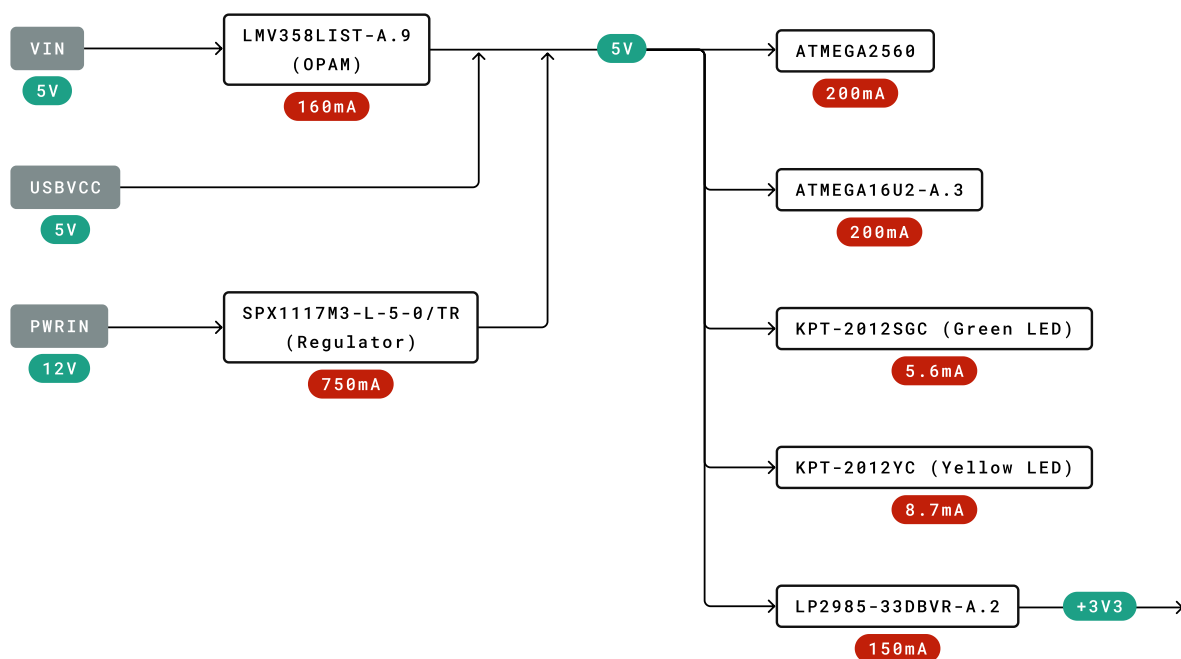
Arduino Mega 2560 Rev3 俯视图

编号	描述	编号	描述
USB	USB B 连接器	F1	片式电容器
IC1	5V 线性稳压器	X1	电源插孔连接器
JP5	电镀通孔	IC4	ATmega16U2 芯片
PC1	电解铝电容	PC2	电解铝电容
D1	通用整流器	D3	通用二极管
L2	固定电感器	IC3	ATmega2560 芯片
ICSP	连接器接头	ON	绿色 LED
RN1	电阻器阵列	XIO	连接器

16.3 处理器

Mega 2560 Rev3 电路板的主处理器是 ATmega2560 芯片，工作频率为 16 MHz。它提供了大量的输入和输出线路，可以连接许多外部设备。同时，由于它的 RAM 比其他处理器大得多，因此操作和处理速度不会减慢。该电路板上还配备一个 USB 串行处理器 ATmega16U2，它是 USB 输入信号与主处理器之间的接口。这增加了将外设连接到 Mega 2560 Rev3 电路板的灵活性。

16.4 电源树



Legend :

- Component
- Max Current
- Power I/O
- Voltage Range

Power Tree

17 电路板操作

17.1 入门指南 - IDE

如需在离线状态下对 Mega 2560 Rev3 进行编程，则需要安装 Arduino Desktop IDE **[1]** 若要将 Mega 2560 Rev3 连接到计算机，则需要使用 Type-B USB 电缆。如 LED 指示灯所示，该电缆还可以为电路板提供电源。

17.2 入门指南 - Arduino Cloud Editor

包括本电路板在内的所有 Arduino 电路板，都可以在 Arduino Cloud Editor **[2]** 上开箱即用，只需安装一个简单的插件即可。

Arduino Cloud Editor 是在线托管的，因此它将始终提供最新功能并支持所有电路板。接下来**[3]**开始在浏览器上编码并将程序上传到您的电路板上。

17.3 示例程序

Mega 2560 Rev3 的示例程序可以在 Arduino IDE 的“示例”菜单或 Arduino 网站 **[4]** 的“文档”部分找到。

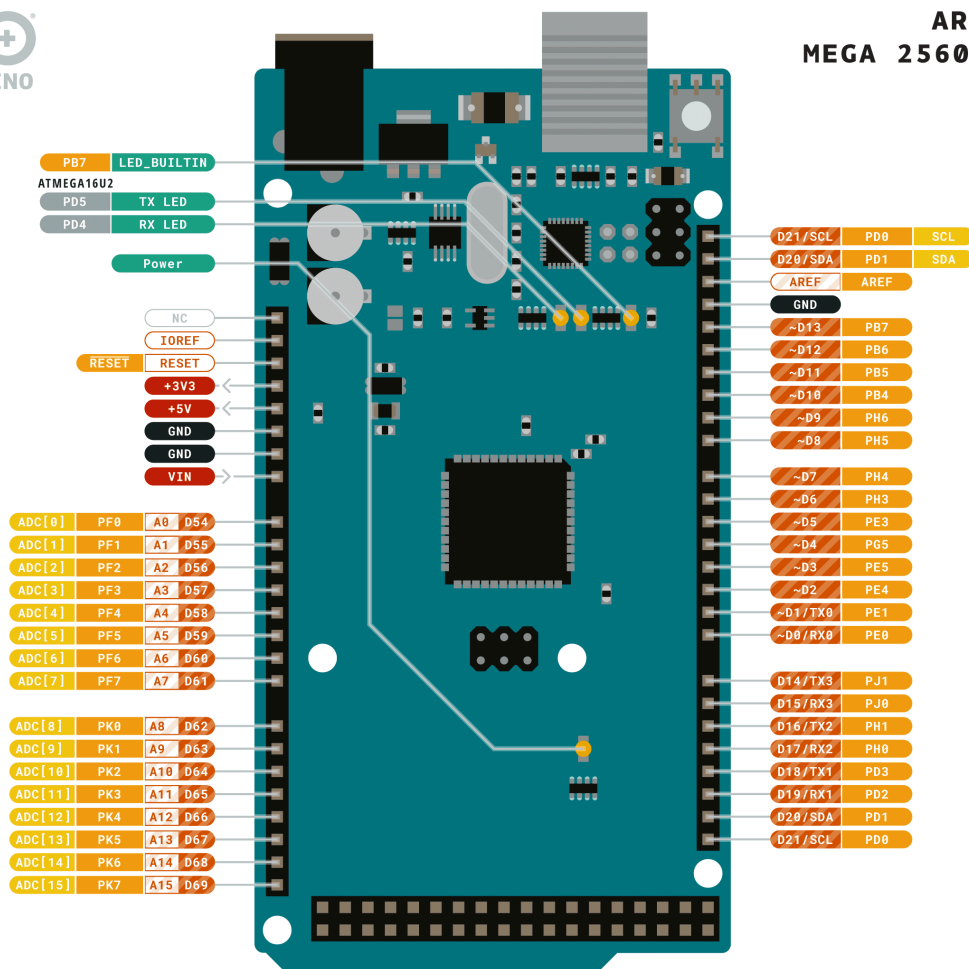
17.4 在线资源

现在，您已经了解该电路板的基本功能，就可以通过查看 Arduino Project Hub **[5]**、Arduino Library Reference **[6]** 和在线商店 **[7]** 上的精彩项目来探索它所提供的无限可能性；在这些项目中，您可以为电路板配备传感器、执行器等。

18 连接器引脚布局



ARDUINO MEGA 2560 REV3



Ground	Internal Pin	Digital Pin	Microcontroller's Port
Power	SWD Pin	Analog Pin	
LED	Other Pin	Default	

ARDUINO.CC



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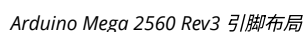
Arduino Mega 2560 Rev3 引脚布局

18.1 模拟

引脚	功能	类型	描述
1	NC	NC	未连接
2	IOREF	IOREF	数字逻辑参考电压 V - 连接至 5V
3	Reset	复位	复位
4	+3V3	电源	+3V3 电源轨
5	+5V	电源	+5V 电源轨
6	GND	电源	接地
7	GND	电源	接地
8	VIN	电源	电压输入
9	A0	模拟	模拟输入0 / GPIO
10	A1	模拟	模拟输入1 / GPIO
11	A2	模拟	模拟输入2 / GPIO
12	A3	模拟	模拟输入3 / GPIO
13	A4	模拟	模拟输入4 / GPIO
14	A5	模拟	模拟输入5 / GPIO
15	A6	模拟	模拟输入6 / GPIO
16	A7	模拟	模拟输入7 / GPIO
17	A8	模拟	模拟输入8 / GPIO
18	A9	模拟	模拟输入9 / GPIO
19	A10	模拟	模拟输入10 / GPIO
20	A11	模拟	模拟输入11 / GPIO
21	A12	模拟	模拟输入12 / GPIO
22	A13	模拟	模拟输入13 / GPIO
23	A14	模拟	模拟输入14 / GPIO
24	A15	模拟	模拟输入15 / GPIO

18.2 数字

引脚	功能	类型	描述
1	D21/SCL	数字输入/I2C	数字输入 21/I2C 数据线
2	D20/SDA	数字输入/I2C	数字输入 20/I2C 数据线
3	AREF	数字	模拟参考电压
4	GND	电源	接地
5	D13	数字/GPIO	数字输入 13/GPIO
6	D12	数字/GPIO	数字输入 12/GPIO
7	D11	数字/GPIO	数字输入 11/GPIO
8	D10	数字/GPIO	数字输入 10/GPIO
9	D9	数字/GPIO	数字输入 9/GPIO
10	D8	数字/GPIO	数字输入 8/GPIO
11	D7	数字/GPIO	数字输入 7/GPIO
12	D6	数字/GPIO	数字输入 6/GPIO
13	D5	数字/GPIO	数字输入 5/GPIO
14	D4	数字/GPIO	数字输入 4/GPIO
15	D3	数字/GPIO	数字输入 3/GPIO
16	D2	数字/GPIO	数字输入 2/GPIO
17	D1/TX0	数字/GPIO	数字输入 1/GPIO
18	D0/Tx1	数字/GPIO	数字输入 0/GPIO
19	D14	数字/GPIO	数字输入 14 /GPIO
20	D15	数字/GPIO	数字输入 15 /GPIO
21	D16	数字/GPIO	数字输入 16 /GPIO
22	D17	数字/GPIO	数字输入 17 /GPIO
23	D18	数字/GPIO	数字输入 18 /GPIO
24	D19	数字/GPIO	数字输入 19/GPIO
25	D20	数字/GPIO	数字输入 20/GPIO
26	D21	数字/GPIO	数字输入 21/GPIO



引脚	功能	类型	描述
1	PB4	内部	串行线调试
2	PB6	内部	串行线调试
3	PB5	内部	串行线调试
4	PB7	内部	串行线调试

引脚	功能	类型	描述
1	CIPO	内部	控制器输入外设输出
2	+5V	内部	5 V的电源
3	SCK	内部	串行时钟
4	COPI	内部	控制器输出外设输入
5	RESET	内部	复位
6	GND	内部	接地

18.5 数字引脚 D22 - D53 LHS

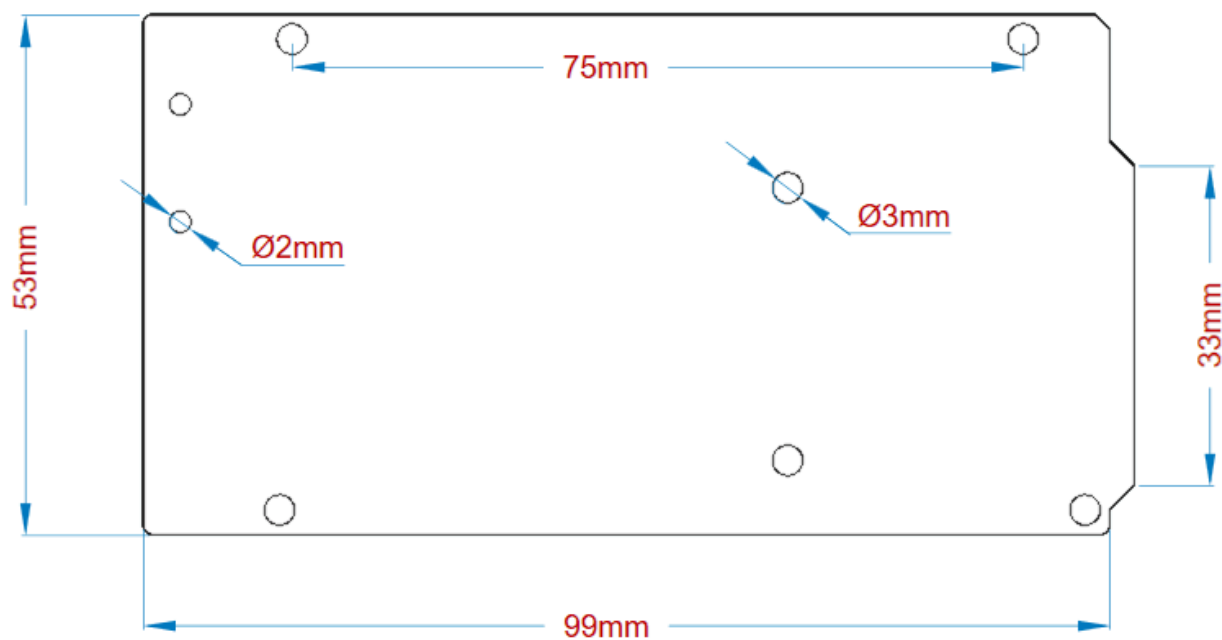
引脚	功能	类型	描述
1	+5V	电源	5 V的电源
2	D22	数字	数字输入 22/GPIO
3	D24	数字	数字输入 24/GPIO
4	D26	数字	数字输入 26/GPIO
5	D28	数字	数字输入 28/GPIO
6	D30	数字	数字输入 30/GPIO
7	D32	数字	数字输入 32/GPIO
8	D34	数字	数字输入 34/GPIO
9	D36	数字	数字输入 36/GPIO
10	D38	数字	数字输入 38/GPIO
11	D40	数字	数字输入 40/GPIO
12	D42	数字	数字输入 42/GPIO
13	D44	数字	数字输入 44/GPIO
14	D46	数字	数字输入 46/GPIO
15	D48	数字	数字输入 48/GPIO
16	D50	数字	数字输入 50/GPIO
17	D52	数字	数字输入 52/GPIO
18	GND	电源	接地

18.6 数字引脚 D22 - D53 LHS

引脚	功能	类型	描述
1	+5V	电源	5 V的电源
2	D23	数字	数字输入 23/GPIO
3	D25	数字	数字输入 25/GPIO
4	D27	数字	数字输入 27/GPIO
5	D29	数字	数字输入 29/GPIO
6	D31	数字	数字输入 31/GPIO
7	D33	数字	数字输入 33/GPIO
8	D35	数字	数字输入 35/GPIO
9	D37	数字	数字输入 37/GPIO
10	D39	数字	数字输入 39/GPIO
11	D41	数字	数字输入 41/GPIO
12	D43	数字	数字输入 43/GPIO
13	D45	数字	数字输入 45/GPIO
14	D47	数字	数字输入 47/GPIO
15	D49	数字	数字输入 49/GPIO
16	D51	数字	数字输入 51/GPIO
17	D53	数字	数字输入 53/GPIO
18	GND	电源	接地

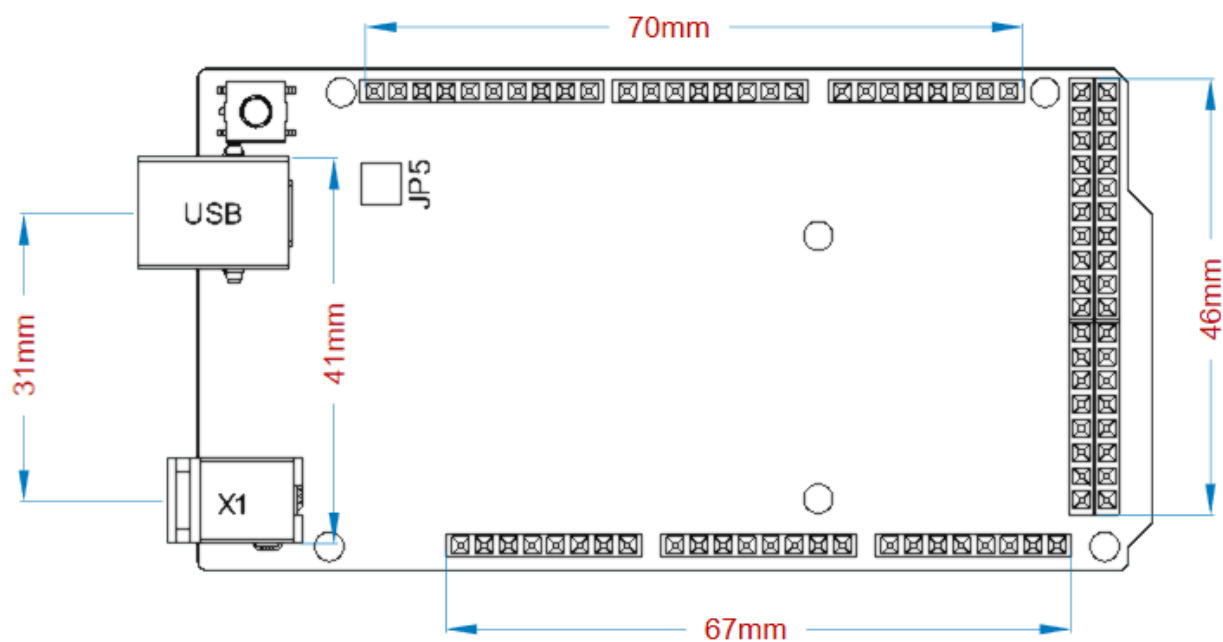
19 机械层信息

19.1 电路板外形图



Arduino Mega 2560 Rev3 外形图

19.2 电路板安装孔



Arduino Mega 2560 Rev3 安装孔

认证

20 符合性声明 CE DoC (欧盟)

我们在此郑重声明，上述产品符合以下欧盟指令的基本要求，因此有资格在包括欧盟（EU）和欧洲经济区（EEA）在内的市场内自由流通。

21 声明符合欧盟 RoHS 和 REACH 211 01/19/2021

Arduino 电路板符合欧洲议会关于限制在电子电气设备中使用某些有害物质的 RoHS 2 指令 2011/65/EU 和欧盟理事会于 2015 年 6 月 4 日颁布的关于限制在电子电气设备中使用某些有害物质的 RoHS 3 指令 2015/863/EU。

物质	最大限值 (ppm)
铅 (Pb)	1000
镉 (Cd)	100
汞 (Hg)	1000
六价铬 (Cr6+)	1000
多溴联苯 (PBB)	1000
多溴联苯醚 (PBDE)	1000
邻苯二甲酸二(2-乙基己)酯 (DEHP)	1000
邻苯二甲酸丁苄酯 (BBP)	1000
邻苯二甲酸二丁酯 (DBP)	1000
邻苯二甲酸二异丁酯 (DIBP)	1000

豁免：未申请任何豁免。

Arduino 电路板完全符合欧盟法规 (EC) 1907/2006 中关于化学品注册、评估、许可和限制 (REACH) 的相关要求。我们声明，所有产品（包括包装）中的 SVHC (<https://echa.europa.eu/web/guest/candidate-list-table>)，（欧洲化学品管理局目前发布的《高度关注物质候选授权清单》）含量总浓度均未超过 0.1%。据我们所知，我们还声明，我们的产品不含 ECHA（欧洲化学品管理局）1907/2006/EC 公布的候选清单附件 XVII 中规定的“授权清单”（REACH 法规附件 XIV）和高度关注物质 (SVHC) 所列的任何物质。

22 冲突矿产声明

作为电子和电气元件的全球供应商，Arduino 意识到我们有义务遵守有关冲突矿产的法律法规，特别是《多德-弗兰克华尔街改革与消费者保护法案》第 1502 条。Arduino 不直接采购或加工锡、钽、钨或金等冲突矿物。冲突矿物以焊料的形式或作为金属合金的组成部分存在于我们的产品中。作为我们合理尽职调查的一部分，Arduino 已联系供应链中的元件供应商，以核实他们是否始终遵守法规的相关规定。根据迄今收到的信息，我们声明我们的产品中含有来自非冲突地区的冲突矿物。

23 FCC 警告

任何未经合规性负责方明确批准的更改或修改都可能导致用户无权操作设备。

本设备符合 FCC 规则第 15 部分的规定。操作须满足以下两个条件：

- (1) 此设备不会造成有害干扰
- (2) 此设备必须接受接收到的任何干扰，包括可能导致不良操作的干扰。

FCC 射频辐射暴露声明：

1. 此发射器不得与任何其他天线或发射器放置在同一位置或同时运行。
2. 此设备符合为非受控环境规定的射频辐射暴露限值。
3. 安装和操作本设备时，辐射源与您的身体之间至少应保持 20 厘米的距离。

English: User manuals for license-exempt radio apparatus shall contain the following or equivalent notice in a conspicuous location in the user manual or alternatively on the device or both. This device complies with Industry Canada license-exempt RSS standard(s). Operation is subject to the following two conditions:

- (1) this device may not cause interference
- (2) this device must accept any interference, including interference that may cause undesired operation of the device.

French: Le présent appareil est conforme aux CNR d'Industrie Canada applicables aux appareils radio exempts de licence. L'exploitation est autorisée aux deux conditions suivantes :

- (1) l'appareil nedit pas produire de brouillage
- (2) l'utilisateur de l'appareil doit accepter tout brouillage radioélectrique subi, même si le brouillage est susceptible d'en compromettre le fonctionnement.

IC SAR警告：

English This equipment should be installed and operated with a minimum distance of 20 cm between the radiator and your body.

French: Lors de l' installation et de l' exploitation de ce dispositif, la distance entre le radiateur et le corps est d'au moins 20 cm.

重要提示： EUT 的工作温度不能超过 85°C，也不能低于 -40°C。

Arduino S.r.l. 特此声明，本产品符合 201453/EU 指令的基本要求和和其他相关规定。本产品允许在所有欧盟成员国使用。



24 公司信息

公司名称	Arduino S.r.l.
公司地址	Via Andrea Appiani 25, 20900 MONZA MB, Italy

25 参考资料

参考资料	链接
Arduino IDE (Desktop)	https://www.arduino.cc/en/Main/Software
Arduino Cloud Editor	https://create.arduino.cc/editor
Arduino Cloud Editor - 入门指南	https://docs.arduino.cc/arduino-cloud/guides/editor/
Arduino 网站	https://www.arduino.cc/
Arduino Project Hub	https://create.arduino.cc/projecthub?by=part&part_id=11332&sort=trending
库参考	https://www.arduino.cc/reference/en/libraries/
在线商店	https://store.arduino.cc/

26 修订记录

日期	版次	变更
25/04/2024	3	新的 Cloud Editor 的更新链接
09/10/2023	2	更新的建议运行条件
29/09/2020	1	首次发布



HC-SR04 Ultrasonic Sensor

Range: 2 cm–400 cm | 5V | 4-pin



Tech Support: services@elecfreaks.com

Ultrasonic Ranging Module HC - SR04

Product features:

Ultrasonic ranging module HC - SR04 provides 2cm - 400cm non-contact measurement function, the ranging accuracy can reach to 3mm. The modules includes ultrasonic transmitters, receiver and control circuit. The basic principle of work:

- (1) Using IO trigger for at least 10us high level signal,
- (2) The Module automatically sends eight 40 kHz and detect whether there is a pulse signal back.
- (3) IF the signal back, through high level , time of high output IO duration is the time from sending ultrasonic to returning.

Test distance = (high level time×velocity of sound (340M/S) / 2,

Wire connecting direct as following:

- 5V Supply
- Trigger Pulse Input
- Echo Pulse Output
- 0V Ground

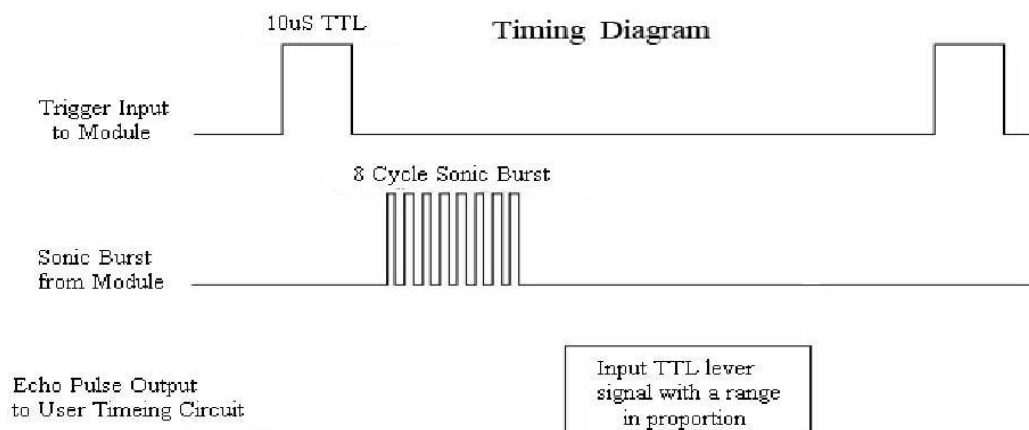
Electric Parameter

Working Voltage	DC 5 V
Working Current	15mA
Working Frequency	40Hz
Max Range	4m
Min Range	2cm
MeasuringAngle	15 degree
Trigger Input Signal	10uS TTL pulse
Echo Output Signal	Input TTL lever signal and the range in proportion
Dimension	45*20*15mm



Timing diagram

The Timing diagram is shown below. You only need to supply a short 10uS pulse to the trigger input to start the ranging, and then the module will send out an 8 cycle burst of ultrasound at 40 kHz and raise its echo. The Echo is a distance object that is pulse width and the range in proportion. You can calculate the range through the time interval between sending trigger signal and receiving echo signal. Formula: $\mu\text{S} / 58 = \text{centimeters}$ or $\mu\text{S} / 148 = \text{inch}$; or: the range = high level time * velocity (340M/S) / 2; we suggest to use over 60ms measurement cycle, in order to prevent trigger signal to the echo signal.



Attention:

- The module is not suggested to connect directly to electric, if connected electric, the GND terminal should be connected the module first, otherwise, it will affect the normal work of the module.
- When tested objects, the range of area is not less than 0.5 square meters and the plane requests as smooth as possible, otherwise ,it will affect the results of measuring.

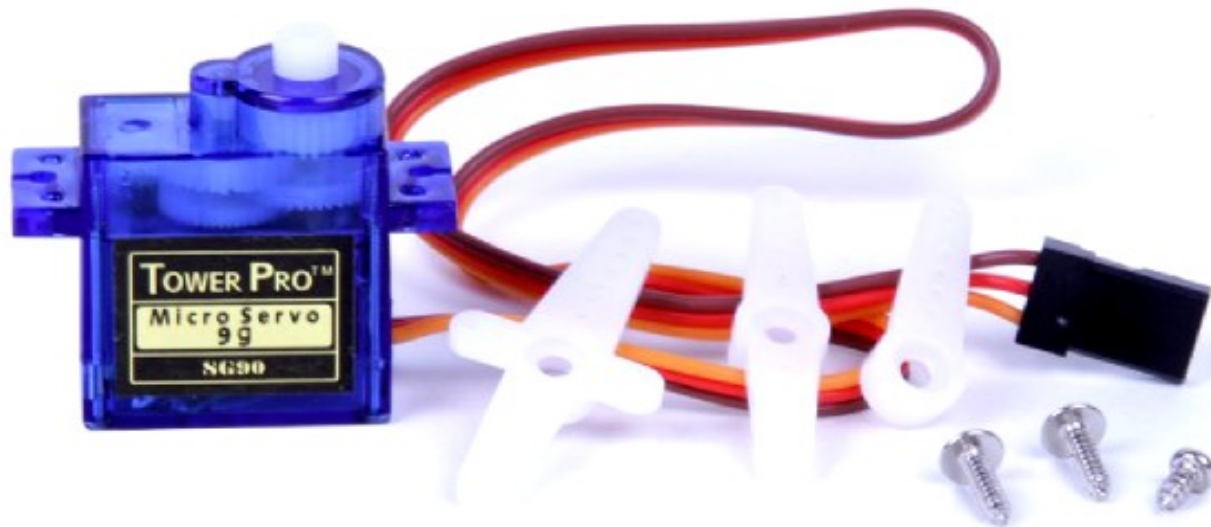
www.ElecFreaks.com



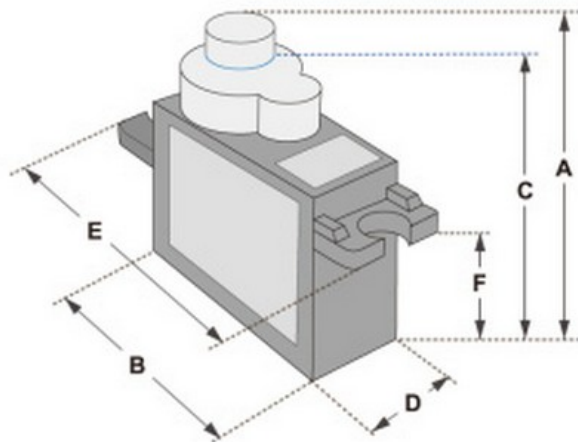


SG90 Micro Servo Motor

Torque: 1.6 kg/cm | 4.8–6V | 180°



Tiny and lightweight with high output power. Servo can rotate approximately 180 degrees (90 in each direction), and works just like the standard kinds but smaller. You can use any servo code, hardware or library to control these servos. Good for beginners who want to make stuff move without building a motor controller with feedback & gear box, especially since it will fit in small places. It comes with a 3 horns (arms) and hardware.

**Dimensions & Specifications**

A (mm) : 32

B (mm) : 23

C (mm) : 28.5

D (mm) : 12

E (mm) : 32

F (mm) : 19.5

Speed (sec) : 0.1

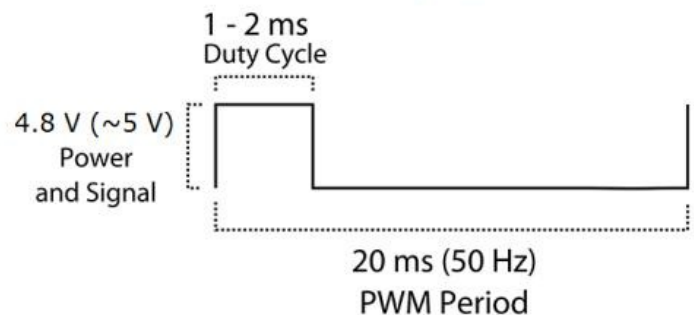
Torque (kg-cm) : 2.5

Weight (g) : 14.7

Voltage : 4.8 - 6

Position "0" (1.5 ms pulse) is middle, "90" (~2ms pulse) is middle, is all the way to the right, "-90" (~1ms pulse) is all the way to the left.

PWM=Orange (⏏)
Vcc=Red (+)
Ground=Brown (-)

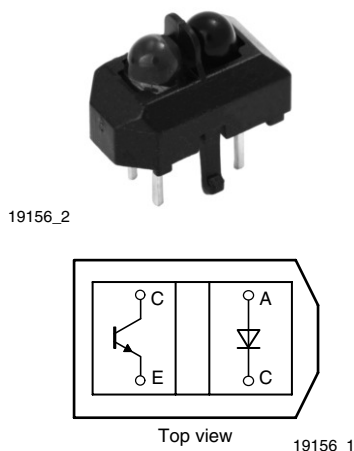




TCRT5000 IR Sensor

IR reflective sensor | 950 nm emitter

Reflective Optical Sensor with Transistor Output



FEATURES

- Package type: leaded
- Detector type: phototransistor
- Dimensions (L x W x H in mm): 10.2 x 5.8 x 7
- Peak operating distance: 2.5 mm
- Operating range within > 20 % relative collector current: 0.2 mm to 15 mm
- Typical output current under test: $I_C = 1$ mA
- Daylight blocking filter
- Emitter wavelength: 950 nm
- Lead (Pb)-free soldering released
- Compliant to RoHS directive 2002/95/EC and in accordance to WEEE 2002/96/EC



RoHS
COMPLIANT

DESCRIPTION

The TCRT5000 and TCRT5000L are reflective sensors which include an infrared emitter and phototransistor in a leaded package which blocks visible light. The package includes two mounting clips. TCRT5000L is the long lead version.

APPLICATIONS

- Position sensor for shaft encoder
- Detection of reflective material such as paper, IBM cards, magnetic tapes etc.
- Limit switch for mechanical motions in VCR
- General purpose - wherever the space is limited

PRODUCT SUMMARY

PART NUMBER	DISTANCE FOR MAXIMUM CTR _{rel} ⁽¹⁾ (mm)	DISTANCE RANGE FOR RELATIVE $I_{out} > 20\%$ (mm)	TYPICAL OUTPUT CURRENT UNDER TEST ⁽²⁾ (mA)	DAYLIGHT BLOCKING FILTER INTEGRATED
TCRT5000	2.5	0.2 to 15	1	Yes
TCRT5000L	2.5	0.2 to 15	1	Yes

Notes

⁽¹⁾ CTR: current transference ratio, I_{out}/I_{in}

⁽²⁾ Conditions like in table basic characteristics/sensors

ORDERING INFORMATION

ORDERING CODE	PACKAGING	VOLUME ⁽¹⁾	REMARKS
TCRT5000	Tube	MOQ: 4500 pcs, 50 pcs/tube	3.5 mm lead length
TCRT5000L	Tube	MOQ: 2400 pcs, 48 pcs/tube	15 mm lead length

Note

⁽¹⁾ MOQ: minimum order quantity

ABSOLUTE MAXIMUM RATINGS ⁽¹⁾

PARAMETER	TEST CONDITION	SYMBOL	VALUE	UNIT
INPUT (EMITTER)				
Reverse voltage		V_R	5	V
Forward current		I_F	60	mA
Forward surge current	$t_p \leq 10 \mu s$	I_{FSM}	3	A
Power dissipation	$T_{amb} \leq 25^\circ C$	P_V	100	mW
Junction temperature		T_j	100	$^\circ C$

ABSOLUTE MAXIMUM RATINGS ⁽¹⁾				
PARAMETER	TEST CONDITION	SYMBOL	VALUE	UNIT
OUTPUT (DETECTOR)				
Collector emitter voltage		V_{CEO}	70	V
Emitter collector voltage		V_{ECO}	5	V
Collector current		I_C	100	mA
Power dissipation	$T_{amb} \leq 55^\circ\text{C}$	P_V	100	mW
Junction temperature		T_j	100	$^\circ\text{C}$
SENSOR				
Total power dissipation	$T_{amb} \leq 25^\circ\text{C}$	P_{tot}	200	mW
Ambient temperature range		T_{amb}	- 25 to + 85	$^\circ\text{C}$
Storage temperature range		T_{stg}	- 25 to + 100	$^\circ\text{C}$
Soldering temperature	2 mm from case, $t \leq 10$ s	T_{sd}	260	$^\circ\text{C}$

Note

⁽¹⁾ $T_{amb} = 25^\circ\text{C}$, unless otherwise specified

ABSOLUTE MAXIMUM RATINGS

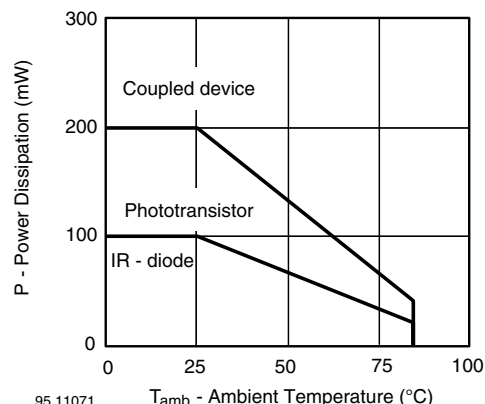


Fig. 1 - Power Dissipation Limit vs. Ambient Temperature

BASIC CHARACTERISTICS ⁽¹⁾						
PARAMETER	TEST CONDITION	SYMBOL	MIN.	TYP.	MAX.	UNIT
INPUT (EMITTER)						
Forward voltage	$I_F = 60$ mA	V_F		1.25	1.5	V
Junction capacitance	$V_R = 0$ V, $f = 1$ MHz	C_j		17		pF
Radiant intensity	$I_F = 60$ mA, $t_p = 20$ ms	I_e			21	mW/sr
Peak wavelength	$I_F = 100$ mA	λ_P	940			nm
Virtual source diameter	Method: 63 % encircled energy	d		2.1		mm
OUTPUT (DETECTOR)						
Collector emitter voltage	$I_C = 1$ mA	V_{CEO}	70			V
Emitter collector voltage	$I_e = 100$ μ A	V_{ECO}	7			V
Collector dark current	$V_{CE} = 20$ V, $I_F = 0$ A, $E = 0$ lx	I_{CEO}		10	200	nA
SENSOR						
Collector current	$V_{CE} = 5$ V, $I_F = 10$ mA, $D = 12$ mm	I_C ^{(2) (3)}	0.5	1	2.1	mA
Collector emitter saturation voltage	$I_F = 10$ mA, $I_C = 0.1$ mA, $D = 12$ mm	V_{CEsat} ^{(2) (3)}			0.4	V

Note

⁽¹⁾ $T_{amb} = 25^\circ\text{C}$, unless otherwise specified

⁽²⁾ See figure 3

⁽³⁾ Test surface: mirror (Mfr. Spindler a. Hoyer, Part No. 340005)

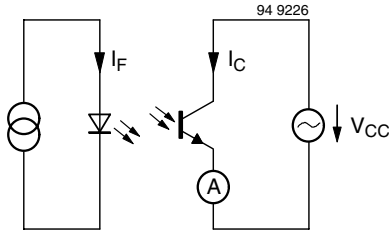


Fig. 2 - Test Circuit

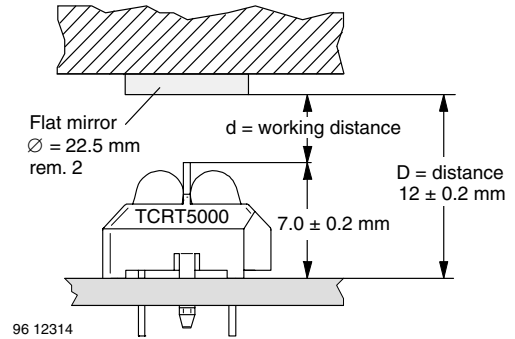


Fig. 3 - Test Circuit

BASIC CHARACTERISTICS

$T_{amb} = 25^\circ\text{C}$, unless otherwise specified

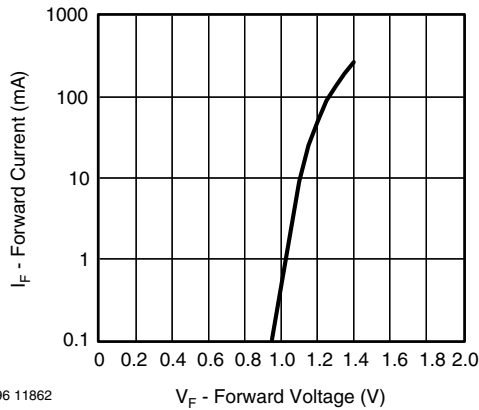


Fig. 4 - Forward Current vs. Forward Voltage

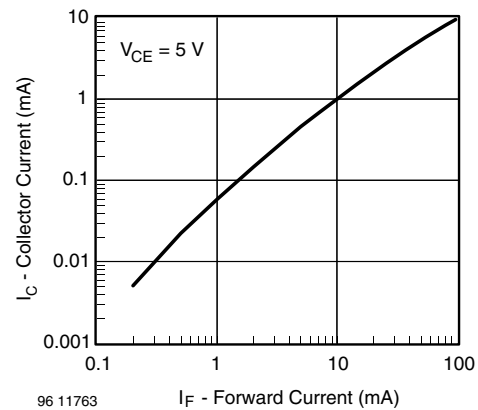


Fig. 6 - Collector Current vs. Forward Current

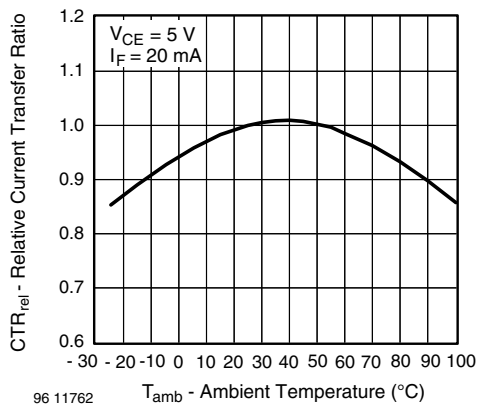


Fig. 5 - Relative Current Transfer Ratio vs. Ambient Temperature

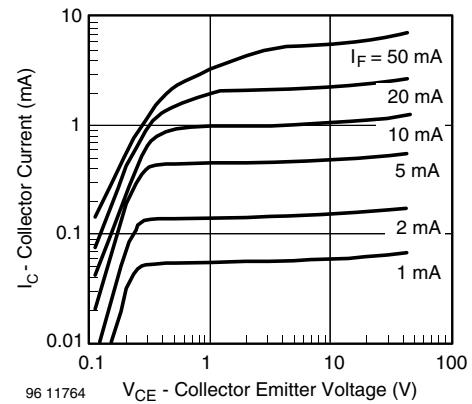


Fig. 7 - Collector Emitter Saturation Voltage vs. Collector Current

TCRT5000, TCRT5000L

Vishay Semiconductors

Reflective Optical Sensor with
Transistor Output

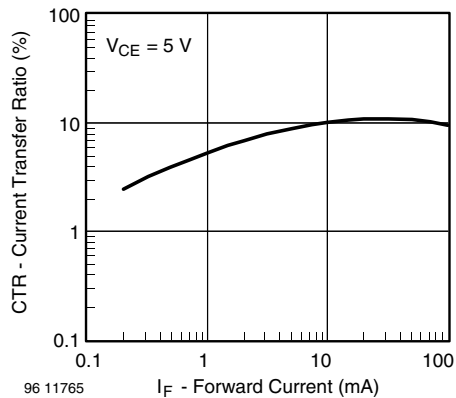


Fig. 8 - Current Transfer Ratio vs. Forward Current

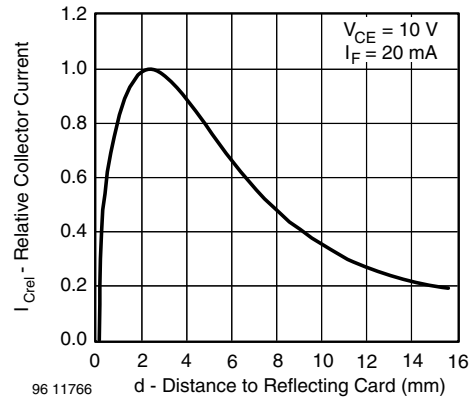
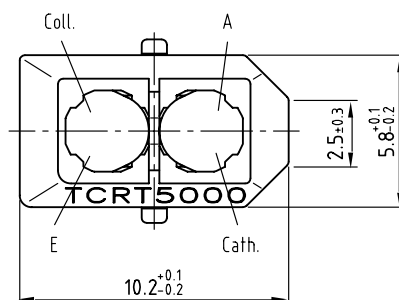
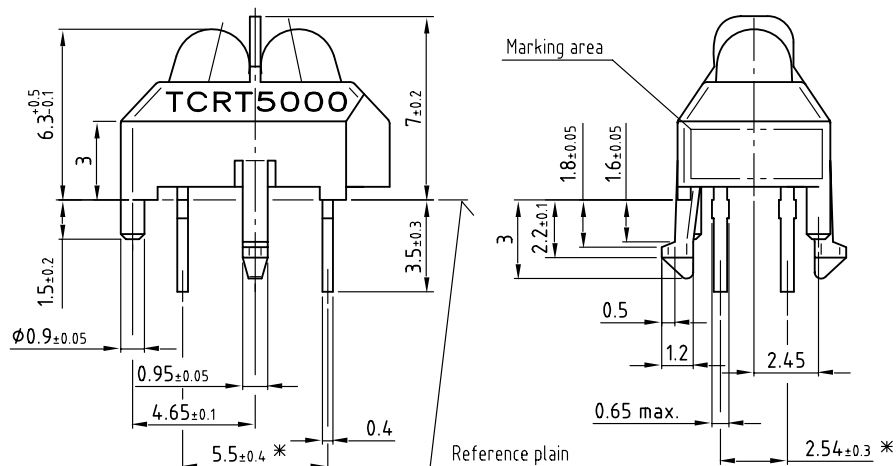


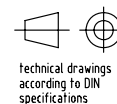
Fig. 9 - Relative Collector Current vs. Distance

PACKAGE DIMENSIONS in millimeters, TCRT5000

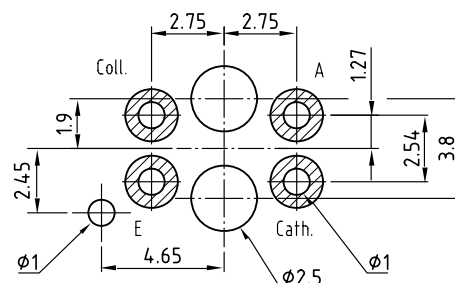


* Tolerances related to reference plain

weight: ca. 0.23g



Footprint Top View

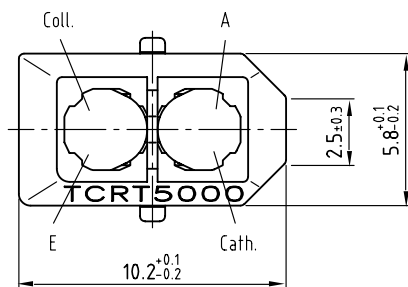
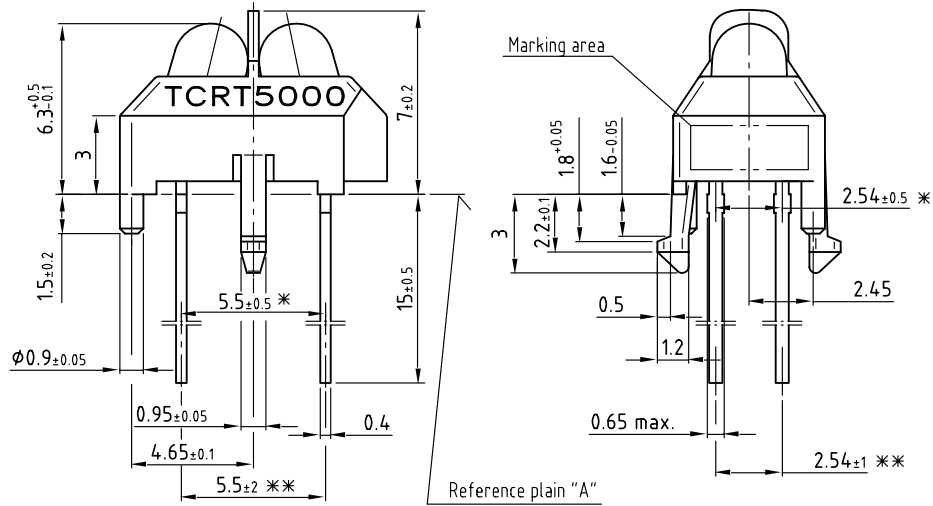


Drawing-No.: 6.550-5096.01-4

Issue: 4; 11.04.02

96 12073

PACKAGE DIMENSIONS in millimeters, TCRT5000L



weight: ca. 0.23g

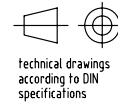
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Issue: 4; 11.04.02

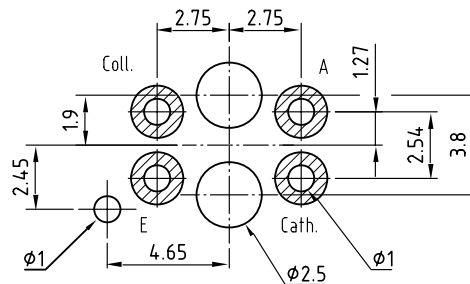
95 11267

* Tolerances related to reference plain "A"

** Tolerances related on lead end



Footprint Top View



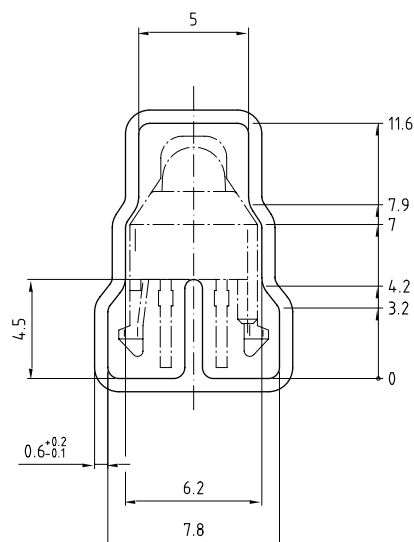
TCRT5000, TCRT5000L

Vishay Semiconductors

Reflective Optical Sensor with
Transistor Output



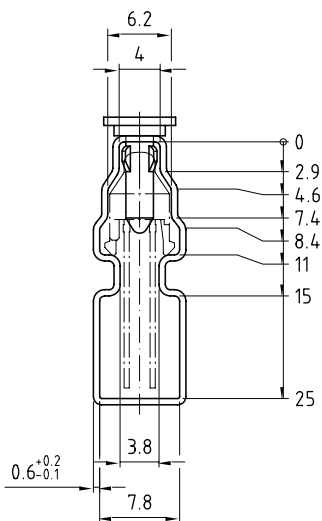
TUBE DIMENSIONS in millimeters, TCRT5000



With rubber stopper
Tolerance: $\pm 0.5\text{mm}$
Length: $575 \pm 1\text{mm}$

Drawing-No.: 9.700-5139.01-4
Issue: 1; 10.05.00
20298

TUBE DIMENSIONS in millimeters, TCRT5000L



With stopper pins
Tolerance: $\pm 0.5\text{mm}$
Length: $575 \pm 1\text{mm}$

Drawing-No.: 9.700-5178.01-4
Issue: 1; 25.02.00
20299



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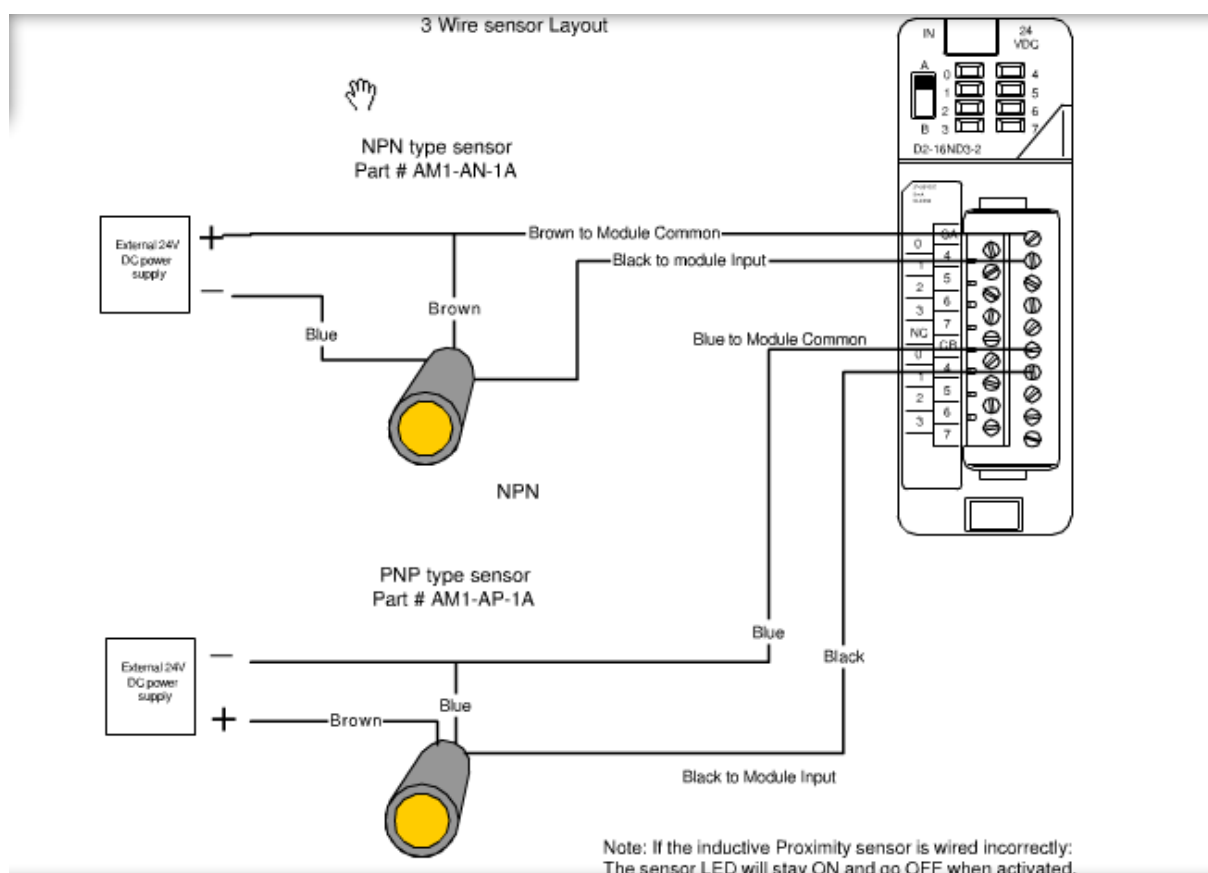


LJ12A3-4-Z/BX Proximity Sensor

NPN 3-wire | 6–36V | 4 mm sensing range

Inductive Proximity Sensor,LJ12A3-4-Z/BX ,NPN,3-wire NO,diameter 12mm,Proximity Switch

Inductive Proximity sensor/switch			
Model:	LJ12A3-4-Z/BX	Output:	NPN,3 wire,NO
Size:	12mm	Work Voltage:	6VDC-36VDC
Detection Distance:	4mm	Output Current:	300mA
		Crust Material:	Metal



Connect the wires in the following order by finger twisting the bare ends together and covering the bare connection with the small wire nut. The black common should be connected to the power feed common of the control power system. The brown wire will be tied to the power feed voltage, 12VDC to 36 VDC, depending on the type of switch. The blue wire will be connected to the device that is being electrically controlled, usually a small electric relay.



HD44780 LCD 16×2 Controller

Hitachi | 16-char × 2-line | parallel

HD44780U (LCD-II)

(Dot Matrix Liquid Crystal Display Controller/Driver)

HITACHI

ADE-207-272(Z)

'99.9

Rev. 0.0

Description

The HD44780U dot-matrix liquid crystal display controller and driver LSI displays alphanumerics, Japanese kana characters, and symbols. It can be configured to drive a dot-matrix liquid crystal display under the control of a 4- or 8-bit microprocessor. Since all the functions such as display RAM, character generator, and liquid crystal driver, required for driving a dot-matrix liquid crystal display are internally provided on one chip, a minimal system can be interfaced with this controller/driver.

A single HD44780U can display up to one 8-character line or two 8-character lines.

The HD44780U has pin function compatibility with the HD44780S which allows the user to easily replace an LCD-II with an HD44780U. The HD44780U character generator ROM is extended to generate 208 5×8 dot character fonts and 32 5×10 dot character fonts for a total of 240 different character fonts.

The low power supply (2.7V to 5.5V) of the HD44780U is suitable for any portable battery-driven product requiring low power dissipation.

Features

- 5×8 and 5×10 dot matrix possible
- Low power operation support:
 - 2.7 to 5.5V
- Wide range of liquid crystal display driver power
 - 3.0 to 11V
- Liquid crystal drive waveform
 - A (One line frequency AC waveform)
- Correspond to high speed MPU bus interface
 - 2 MHz (when $V_{CC} = 5V$)
- 4-bit or 8-bit MPU interface enabled
- 80×8 -bit display RAM (80 characters max.)
- 9,920-bit character generator ROM for a total of 240 character fonts
 - 208 character fonts (5×8 dot)
 - 32 character fonts (5×10 dot)

HD44780U

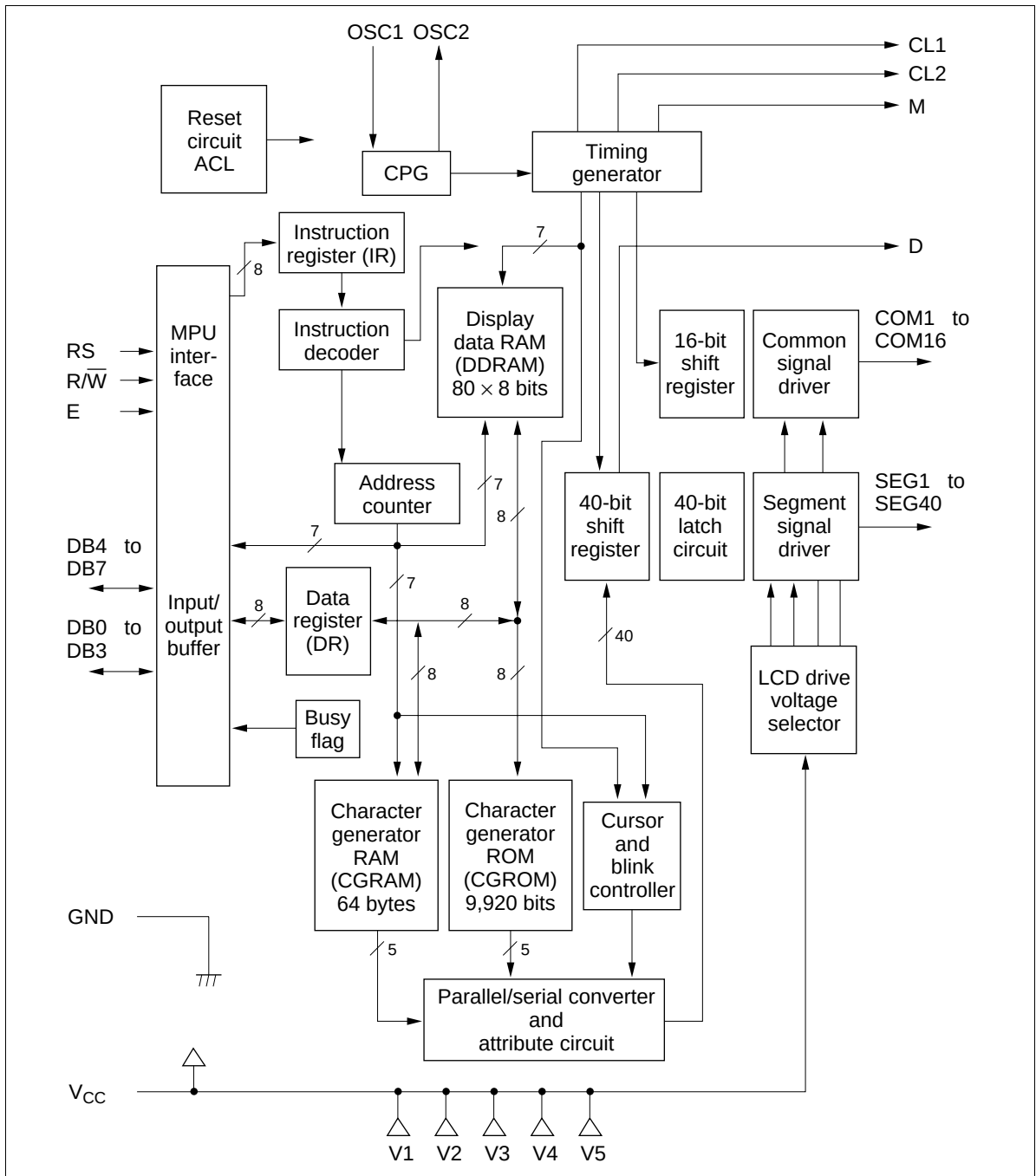
- 64 × 8-bit character generator RAM
 - 8 character fonts (5 × 8 dot)
 - 4 character fonts (5 × 10 dot)
- 16-common × 40-segment liquid crystal display driver
- Programmable duty cycles
 - 1/8 for one line of 5 × 8 dots with cursor
 - 1/11 for one line of 5 × 10 dots with cursor
 - 1/16 for two lines of 5 × 8 dots with cursor
- Wide range of instruction functions:
 - Display clear, cursor home, display on/off, cursor on/off, display character blink, cursor shift, display shift
- Pin function compatibility with HD44780S
- Automatic reset circuit that initializes the controller/driver after power on
- Internal oscillator with external resistors
- Low power consumption

Ordering Information

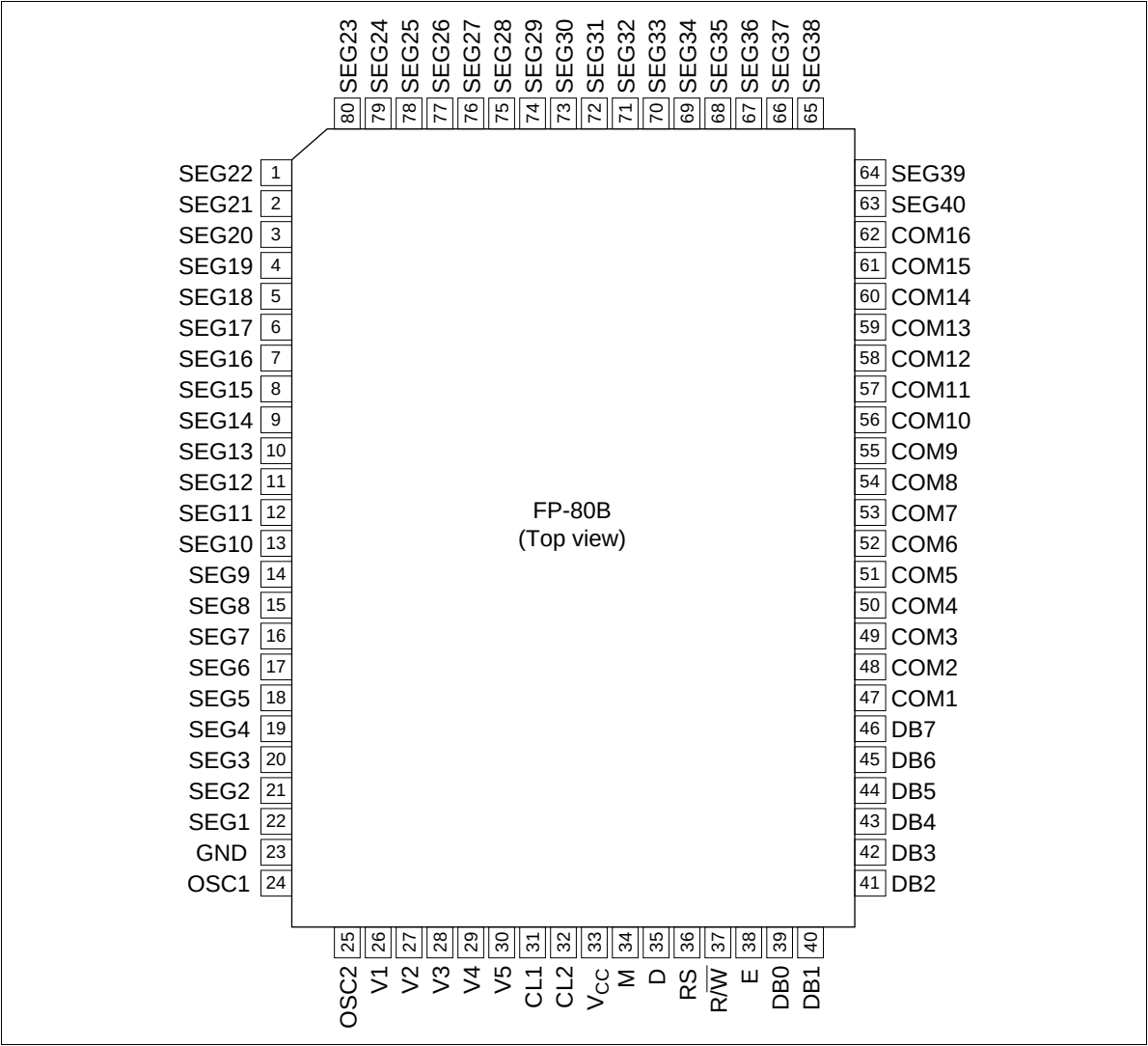
Type No.	Package	CGROM
HD44780UA00FS HCD44780UA00 HD44780UA00TF	FP-80B Chip TFP-80F	Japanese standard font
HD44780UA02FS HCD44780UA02 HD44780UA02TF	FP-80B Chip TFP-80F	European standard font
HD44780UBxxFS HCD44780UBxx HD44780UBxxTF	FP-80B Chip TFP-80F	Custom font

Note: xx: ROM code No.

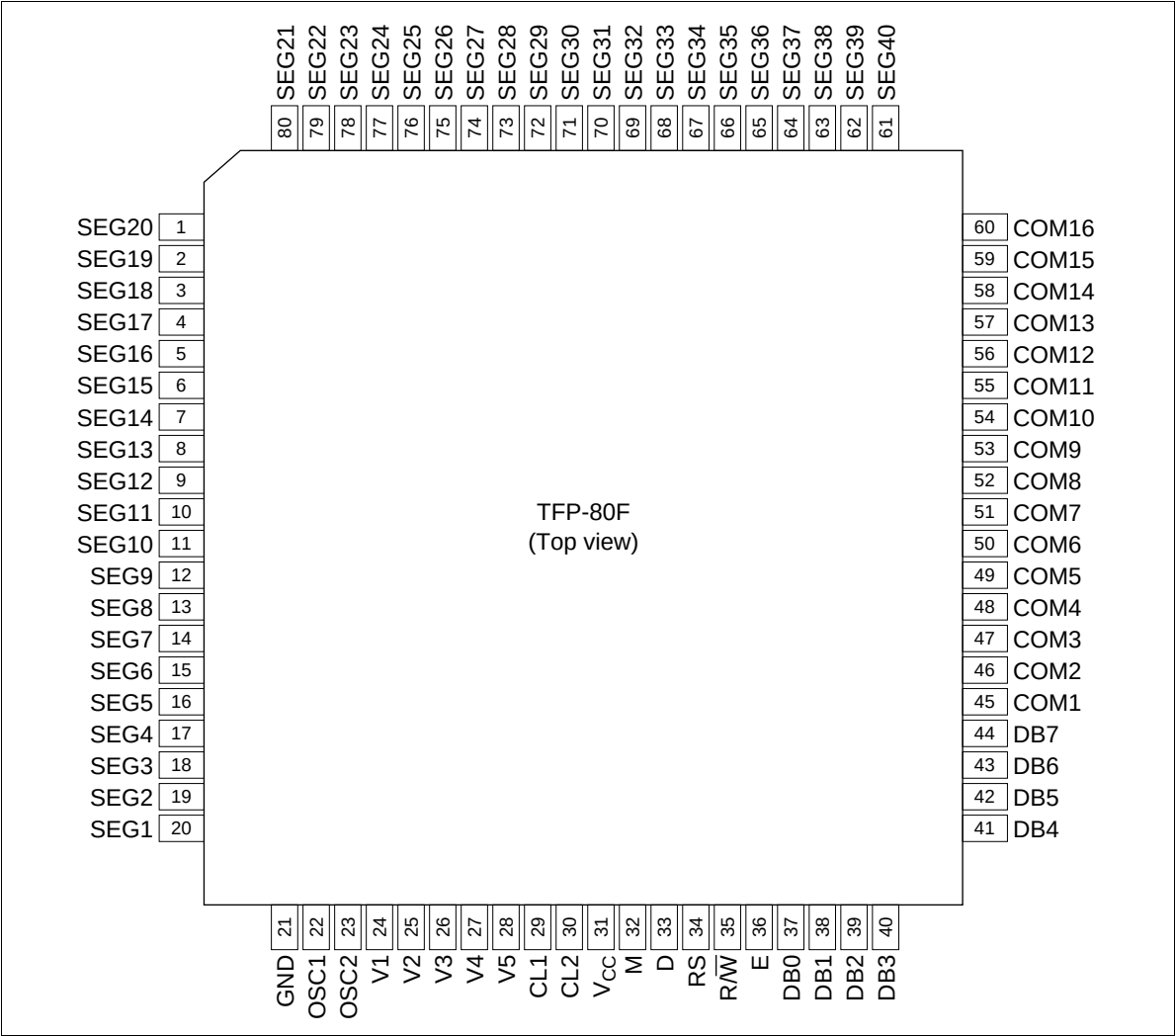
HD44780U Block Diagram



HD44780U Pin Arrangement (FP-80B)

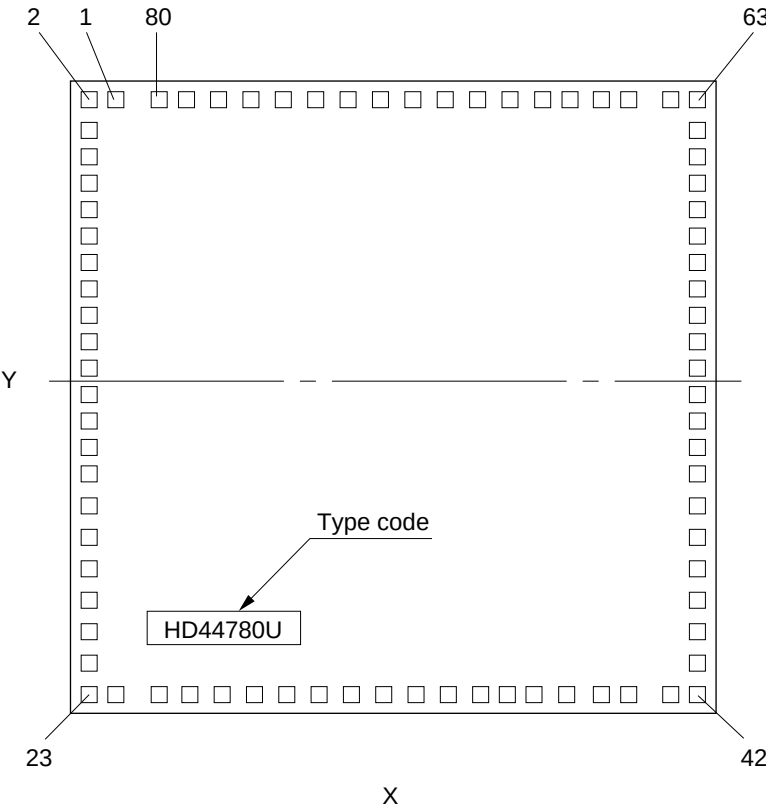


HD44780U Pin Arrangement (TFP-80F)



HD44780U Pad Arrangement

Chip size: 4.90 × 4.90 mm²
Coordinate: Pad center (μm)
Origin: Chip center
Pad size: 114 × 114 μm²



HCD44780U Pad Location Coordinates

Pad No.	Function	Coordinate		Pad No.	Function	Coordinate	
		X (um)	Y (um)			X (um)	Y (um)
1	SEG22	-2100	2313	41	DB2	2070	-2290
2	SEG21	-2280	2313	42	DB3	2260	-2290
3	SEG20	-2313	2089	43	DB4	2290	-2099
4	SEG19	-2313	1833	44	DB5	2290	-1883
5	SEG18	-2313	1617	45	DB6	2290	-1667
6	SEG17	-2313	1401	46	DB7	2290	-1452
7	SEG16	-2313	1186	47	COM1	2313	-1186
8	SEG15	-2313	970	48	COM2	2313	-970
9	SEG14	-2313	755	49	COM3	2313	-755
10	SEG13	-2313	539	50	COM4	2313	-539
11	SEG12	-2313	323	51	COM5	2313	-323
12	SEG11	-2313	108	52	COM6	2313	-108
13	SEG10	-2313	-108	53	COM7	2313	108
14	SEG9	-2313	-323	54	COM8	2313	323
15	SEG8	-2313	-539	55	COM9	2313	539
16	SEG7	-2313	-755	56	COM10	2313	755
17	SEG6	-2313	-970	57	COM11	2313	970
18	SEG5	-2313	-1186	58	COM12	2313	1186
19	SEG4	-2313	-1401	59	COM13	2313	1401
20	SEG3	-2313	-1617	60	COM14	2313	1617
21	SEG2	-2313	-1833	61	COM15	2313	1833
22	SEG1	-2313	-2073	62	COM16	2313	2095
23	GND	-2280	-2290	63	SEG40	2296	2313
24	OSC1	-2080	-2290	64	SEG39	2100	2313
25	OSC2	-1749	-2290	65	SEG38	1617	2313
26	V1	-1550	-2290	66	SEG37	1401	2313
27	V2	-1268	-2290	67	SEG36	1186	2313
28	V3	-941	-2290	68	SEG35	970	2313
29	V4	-623	-2290	69	SEG34	755	2313
30	V5	-304	-2290	70	SEG33	539	2313
31	CL1	-48	-2290	71	SEG32	323	2313
32	CL2	142	-2290	72	SEG31	108	2313
33	V _{CC}	309	-2290	73	SEG30	-108	2313
34	M	475	-2290	74	SEG29	-323	2313
35	D	665	-2290	75	SEG28	-539	2313
36	RS	832	-2290	76	SEG27	-755	2313
37	R/W	1022	-2290	77	SEG26	-970	2313
38	E	1204	-2290	78	SEG25	-1186	2313
39	DB0	1454	-2290	79	SEG24	-1401	2313
40	DB1	1684	-2290	80	SEG23	-1617	2313

Pin Functions

Signal	No. of Lines	I/O	Device Interfaced with	Function
RS	1	I	MPU	Selects registers. 0: Instruction register (for write) Busy flag: address counter (for read) 1: Data register (for write and read)
R/W	1	I	MPU	Selects read or write. 0: Write 1: Read
E	1	I	MPU	Starts data read/write.
DB4 to DB7	4	I/O	MPU	Four high order bidirectional tristate data bus pins. Used for data transfer and receive between the MPU and the HD44780U. DB7 can be used as a busy flag.
DB0 to DB3	4	I/O	MPU	Four low order bidirectional tristate data bus pins. Used for data transfer and receive between the MPU and the HD44780U. These pins are not used during 4-bit operation.
CL1	1	O	Extension driver	Clock to latch serial data D sent to the extension driver
CL2	1	O	Extension driver	Clock to shift serial data D
M	1	O	Extension driver	Switch signal for converting the liquid crystal drive waveform to AC
D	1	O	Extension driver	Character pattern data corresponding to each segment signal
COM1 to COM16	16	O	LCD	Common signals that are not used are changed to non-selection waveforms. COM9 to COM16 are non-selection waveforms at 1/8 duty factor and COM12 to COM16 are non-selection waveforms at 1/11 duty factor.
SEG1 to SEG40	40	O	LCD	Segment signals
V1 to V5	5	—	Power supply	Power supply for LCD drive $V_{cc} - V5 = 11\text{ V (max)}$
V_{cc} , GND	2	—	Power supply	V_{cc} : 2.7V to 5.5V, GND: 0V
OSC1, OSC2	2	—	Oscillation resistor clock	When crystal oscillation is performed, a resistor must be connected externally. When the pin input is an external clock, it must be input to OSC1.

Function Description

Registers

The HD44780U has two 8-bit registers, an instruction register (IR) and a data register (DR).

The IR stores instruction codes, such as display clear and cursor shift, and address information for display data RAM (DDRAM) and character generator RAM (CGRAM). The IR can only be written from the MPU.

The DR temporarily stores data to be written into DDRAM or CGRAM and temporarily stores data to be read from DDRAM or CGRAM. Data written into the DR from the MPU is automatically written into DDRAM or CGRAM by an internal operation. The DR is also used for data storage when reading data from DDRAM or CGRAM. When address information is written into the IR, data is read and then stored into the DR from DDRAM or CGRAM by an internal operation. Data transfer between the MPU is then completed when the MPU reads the DR. After the read, data in DDRAM or CGRAM at the next address is sent to the DR for the next read from the MPU. By the register selector (RS) signal, these two registers can be selected (Table 1).

Busy Flag (BF)

When the busy flag is 1, the HD44780U is in the internal operation mode, and the next instruction will not be accepted. When $RS = 0$ and $R/\overline{W} = 1$ (Table 1), the busy flag is output to DB7. The next instruction must be written after ensuring that the busy flag is 0.

Address Counter (AC)

The address counter (AC) assigns addresses to both DDRAM and CGRAM. When an address of an instruction is written into the IR, the address information is sent from the IR to the AC. Selection of either DDRAM or CGRAM is also determined concurrently by the instruction.

After writing into (reading from) DDRAM or CGRAM, the AC is automatically incremented by 1 (decremented by 1). The AC contents are then output to DB0 to DB6 when $RS = 0$ and $R/\overline{W} = 1$ (Table 1).

Table 1 Register Selection

RS	$R/\overline{W}$	Operation
0	0	IR write as an internal operation (display clear, etc.)
0	1	Read busy flag (DB7) and address counter (DB0 to DB6)
1	0	DR write as an internal operation (DR to DDRAM or CGRAM)
1	1	DR read as an internal operation (DDRAM or CGRAM to DR)

Display Data RAM (DDRAM)

Display data RAM (DDRAM) stores display data represented in 8-bit character codes. Its extended capacity is 80×8 bits, or 80 characters. The area in display data RAM (DDRAM) that is not used for display can be used as general data RAM. See Figure 1 for the relationships between DDRAM addresses and positions on the liquid crystal display.

The DDRAM address (A_{DD}) is set in the address counter (AC) as hexadecimal.

- 1-line display ($N = 0$) (Figure 2)
 - When there are fewer than 80 display characters, the display begins at the head position. For example, if using only the HD44780, 8 characters are displayed. See Figure 3.
 - When the display shift operation is performed, the DDRAM address shifts. See Figure 3.

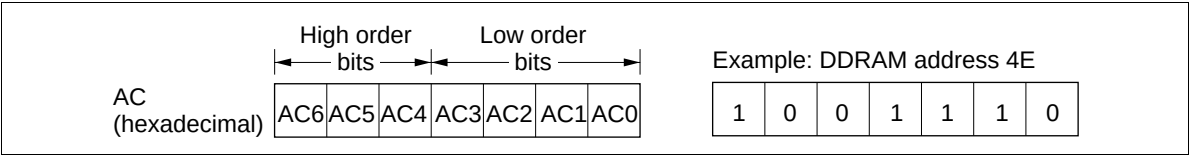


Figure 1 DDRAM Address

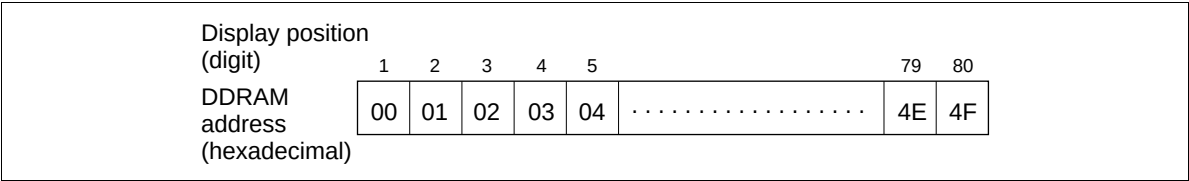


Figure 2 1-Line Display

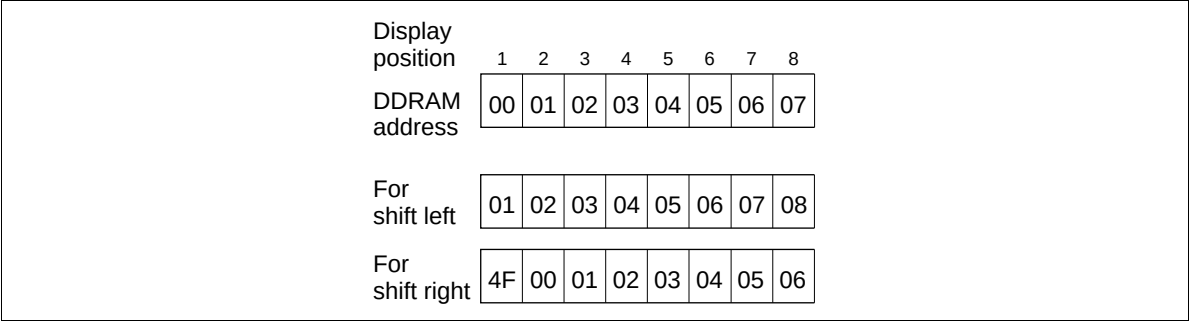


Figure 3 1-Line by 8-Character Display Example

- 2-line display (N = 1) (Figure 4)
 - Case 1: When the number of display characters is less than 40×2 lines, the two lines are displayed from the head. Note that the first line end address and the second line start address are not consecutive. For example, when just the HD44780 is used, 8 characters $\times$ 2 lines are displayed. See Figure 5.

When display shift operation is performed, the DDRAM address shifts. See Figure 5.

Display position	1	2	3	4	5		39	40
DDR4 address (hexadecimal)	00	01	02	03	04		26	27
	40	41	42	43	44		66	67

Figure 4 2-Line Display

Display position	1	2	3	4	5	6	7	8
DDRAM address	00	01	02	03	04	05	06	07
	40	41	42	43	44	45	46	47
For shift left	01	02	03	04	05	06	07	08
	41	42	43	44	45	46	47	48
For shift right	27	00	01	02	03	04	05	06
	67	40	41	42	43	44	45	46

Figure 5 2-Line by 8-Character Display Example

- Case 2: For a 16-character × 2-line display, the HD44780 can be extended using one 40-output extension driver. See Figure 6.
- When display shift operation is performed, the DDRAM address shifts. See Figure 6.

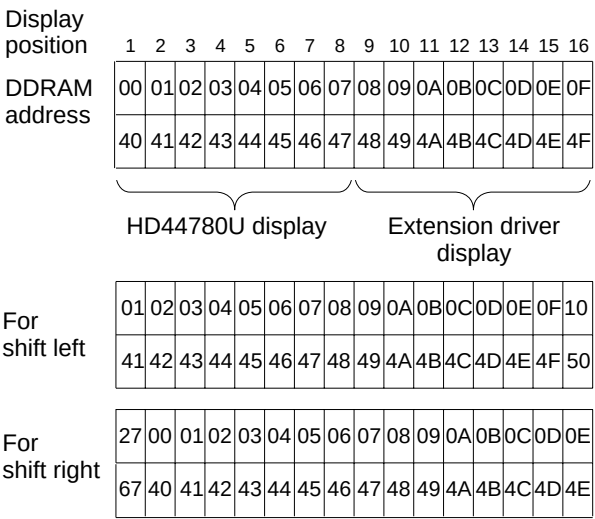


Figure 6 2-Line by 16-Character Display Example

Character Generator ROM (CGROM)

The character generator ROM generates 5×8 dot or 5×10 dot character patterns from 8-bit character codes (Table 4). It can generate 208 5×8 dot character patterns and 32 5×10 dot character patterns. User-defined character patterns are also available by mask-programmed ROM.

Character Generator RAM (CGRAM)

In the character generator RAM, the user can rewrite character patterns by program. For 5×8 dots, eight character patterns can be written, and for 5×10 dots, four character patterns can be written.

Write into DDRAM the character codes at the addresses shown as the left column of Table 4 to show the character patterns stored in CGRAM.

See Table 5 for the relationship between CGRAM addresses and data and display patterns.

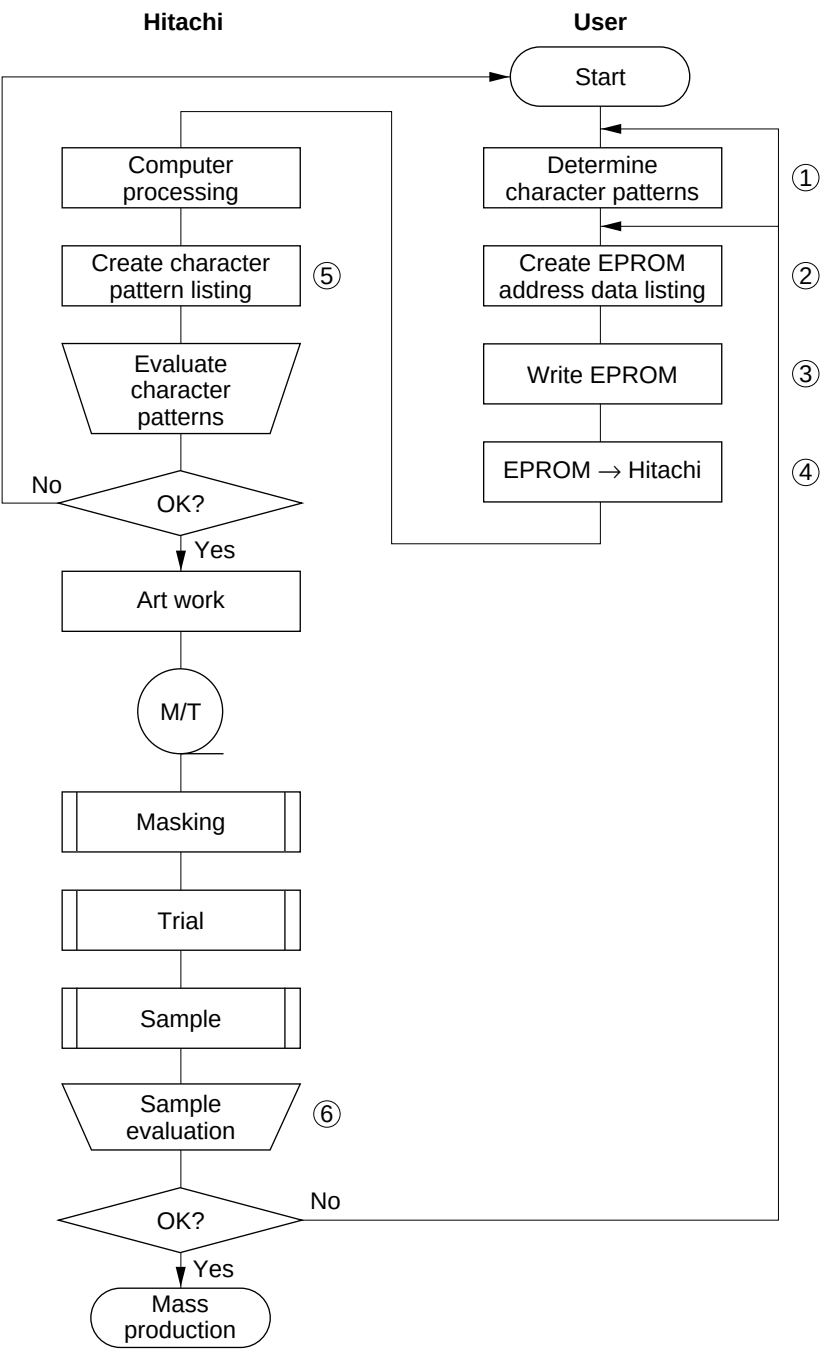
Areas that are not used for display can be used as general data RAM.

Modifying Character Patterns

- Character pattern development procedure

The following operations correspond to the numbers listed in Figure 7:

1. Determine the correspondence between character codes and character patterns.
2. Create a listing indicating the correspondence between EPROM addresses and data.
3. Program the character patterns into the EPROM.
4. Send the EPROM to Hitachi.
5. Computer processing on the EPROM is performed at Hitachi to create a character pattern listing, which is sent to the user.
6. If there are no problems within the character pattern listing, a trial LSI is created at Hitachi and samples are sent to the user for evaluation. When it is confirmed by the user that the character patterns are correctly written, mass production of the LSI proceeds at Hitachi.



Note: For a description of the numbers used in this figure, refer to the preceding page.

Figure 7 Character Pattern Development Procedure

• Programming character patterns

This section explains the correspondence between addresses and data used to program character patterns in EPROM. The HD44780U character generator ROM can generate 208 5 × 8 dot character patterns and 32 5 × 10 dot character patterns for a total of 240 different character patterns.

— Character patterns

EPROM address data and character pattern data correspond with each other to form a 5 × 8 or 5 × 10 dot character pattern (Tables 2 and 3).

Table 2 Example of Correspondence between EPROM Address Data and Character Pattern (5 × 8 Dots)

EPROM Address												Data				
												LSB				
A11	A10	A9	A8	A7	A6	A5	A4	A3	A2	A1	A0	O4	O3	O2	O1	O0
0	1	1	0	0	0	1	0	0	0	0	0	1	0	0	0	0
								0	0	0	1	1	0	0	0	
								0	0	1	0	1	0	1	1	0
								0	0	1	1	1	1	0	0	1
								0	1	0	0	1	0	0	0	1
								0	1	0	1	1	0	0	0	1
								0	1	1	0	1	1	1	1	0
								0	1	1	1	1	0	0	0	0
								1	0	0	0	0	0	0		
								1	0	0	1	0	0	0		
								1	0	1	0	0	0	0		
								1	0	1	1	0	0	0		
								1	1	0	0	0	0	0		
								1	1	0	1	0	0	0		
								1	1	1	0	0	0	0		
								1	1	1	1	0	0	0		

← Cursor position

- Notes:
- 1. EPROM addresses A11 to A4 correspond to a character code.
 - 2. EPROM addresses A3 to A0 specify a line position of the character pattern.
 - 3. EPROM data O4 to O0 correspond to character pattern data.
 - 4. EPROM data O5 to O7 must be specified as 0.
 - 5. A lit display position (black) corresponds to a 1.
 - 6. Line 9 and the following lines must be blanked with 0s for a 5 × 8 dot character fonts.

- Handling unused character patterns
1. EPROM data outside the character pattern area: Always input 0s.
 2. EPROM data in CGRAM area: Always input 0s. (Input 0s to EPROM addresses 00H to FFH.)
 3. EPROM data used when the user does not use any HD44780U character pattern: According to the user application, handled in one of the two ways listed as follows.
 - a. When unused character patterns are not programmed: If an unused character code is written into DDRAM, all its dots are lit. By not programming a character pattern, all of its bits become lit. (This is due to the EPROM being filled with 1s after it is erased.)
 - b. When unused character patterns are programmed as 0s: Nothing is displayed even if unused character codes are written into DDRAM. (This is equivalent to a space.)

Table 3 Example of Correspondence between EPROM Address Data and Character Pattern
(5 × 10 Dots)

EPROM Address											Data				
A11A10A9 A8 A7 A6 A5 A4 A3 A2 A1 A0											LSB O4 O3 O2 O1 O0				
0 1 0 1 0 0 1 0	0 0 0 0										0	0	0	0	0
	0 0 0 1										0	0	0	0	0
	0 0 1 0										0	1	1	0	1
	0 0 1 1										1	0	0	1	1
	0 1 0 0										1	0	0	0	1
	0 1 0 1										1	0	0	0	1
	0 1 1 0										0	1	1	1	1
	0 1 1 1										0	0	0	0	1
	1 0 0 0										0	0	0	0	1
	1 0 0 1										0	0	0	0	1
	1 0 1 0										0	0	0	0	0
	1 0 1 1										0	0	0	0	0
	1 1 0 0										0	0	0	0	0
	1 1 0 1										0	0	0	0	0
	1 1 1 0										0	0	0	0	0
	1 1 1 1										0	0	0	0	0

Cursor position

- Notes:
1. EPROM addresses A11 to A3 correspond to a character code.
 2. EPROM addresses A3 to A0 specify a line position of the character pattern.
 3. EPROM data O4 to O0 correspond to character pattern data.
 4. EPROM data O5 to O7 must be specified as 0.
 5. A lit display position (black) corresponds to a 1.
 6. Line 11 and the following lines must be blanked with 0s for a 5 × 10 dot character fonts.

Table 4 Correspondence between Character Codes and Character Patterns (ROM Code: A00)

Lower 4 Bits \ Upper 4 Bits	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111
xxxx0000	CG RAM (1)			0	@	P	`	P				-	9	E	o	p
xxxx0001	(2)		!	1	A	Q	a	9			.	7	+	4	ä	q
xxxx0010	(3)		"	2	B	R	b	r			「	イ	ツ	×	ß	θ
xxxx0011	(4)		#	3	C	S	c	s			」	ウ	チ	モ	ε	ω
xxxx0100	(5)		\$	4	D	T	d	t			、	エ	ト	ホ	μ	Ω
xxxx0101	(6)		%	5	E	U	e	u			・	オ	ナ	1	ς	Ü
xxxx0110	(7)		&	6	F	V	f	v			ヲ	カ	ニ	ヨ	ρ	Σ
xxxx0111	(8)		'	7	G	W	g	w			フ	チ	ヌ	ラ	g	π
xxxx1000	(1)		(	8	H	X	h	x			イ	ウ	ネ	リ	フ	×
xxxx1001	(2)		)	9	I	Y	i	y			ウ	ク	ル	ル	´	y
xxxx1010	(3)		*	:	J	Z	j	z			エ	コ	ハ	レ	j	+
xxxx1011	(4)		+	;	K	C	k	c			オ	サ	ヒ	ロ	*	π
xxxx1100	(5)		,	<	L	*	l	l			ホ	シ	フ	ワ	φ	π
xxxx1101	(6)		-	=	M	J	m	}			ユ	ズ	ハ	ン	±	÷
xxxx1110	(7)		.	>	N	^	n	+			ヨ	セ	ホ	°	ñ	
xxxx1111	(8)		/	?	O	_	o	+			ッ	ソ	マ	°	ö	

Note: The user can specify any pattern for character-generator RAM.

Table 4 Correspondence between Character Codes and Character Patterns (ROM Code: A02)

Lower 4 Bits \ Upper 4 Bits		0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111
xxxx0000		CG RAM (1)			0	Q	P	`	P	E	o	W	°	À	Ä	À	À
xxxx0001	(2)			!	1	A	Q	a	q	A	J	i	±	À	Ä	À	Ä
xxxx0010	(3)		“	"	2	B	R	b	r	W	Γ	¢	²	À	Ö	À	Ö
xxxx0011	(4)		”	#	3	C	S	c	s	3	π	£	³	À	Ó	À	Ó
xxxx0100	(5)			\$	4	D	T	d	t	M	Σ	×	₴	À	Ö	À	Ö
xxxx0101	(6)			%	5	E	U	e	u	Ÿ	σ	¥	₤	À	Ö	À	Ö
xxxx0110	(7)			&	6	F	V	f	v	J	Ź	ı	₧	À	Ö	À	Ö
xxxx0111	(8)			'	7	G	W	g	w	Π	τ	§	•	Ç	×	Ç	÷
xxxx1000	(1)		↑	(	8	H	X	h	x	Y	#	ƒ	ω	È	Š	È	Š
xxxx1001	(2)		↓	)	9	I	Y	i	y	U	Θ	ƒ	¹	É	Ù	É	Ù
xxxx1010	(3)		÷	*	:	J	Z	j	z	4	Q	≡	Ω	Ê	Ú	Ê	Ú
xxxx1011	(4)		←	+	;	K	[	k	[	w	δ	⊗	⊗	Ë	Û	Ë	Û
xxxx1100	(5)		≤	,	<	L	\	l	ı	W	∞	W	℥	İ	Ü	İ	Ü
xxxx1101	(6)		≥	-	=	M	]	m	}	b	♣	9	℥	İ	Ý	İ	Ý
xxxx1110	(7)			.	>	N	^	n	~	W	ε	Q	4	İ	İ	İ	İ
xxxx1111	(8)			/	?	O	_	o	ó	3	0	'	¿	İ	İ	İ	İ

Table 5 Relationship between CGRAM Addresses, Character Codes (DDRAM) and Character Patterns (CGRAM Data)

For 5 × 8 dot character patterns

Character Codes (DDRAM data)								CGRAM Address								Character Patterns (CGRAM data)							
7	6	5	4	3	2	1	0	5	4	3	2	1	0	7	6	5	4	3	2	1	0		
High				Low				High				Low				High				Low			
0 0 0 0 * 0 0 0								0 0 0				0	0	0	<								

- Notes:
- Character code bits 0 to 2 correspond to CGRAM address bits 3 to 5 (3 bits: 8 types).
 - CGRAM address bits 0 to 2 designate the character pattern line position. The 8th line is the cursor position and its display is formed by a logical OR with the cursor.
Maintain the 8th line data, corresponding to the cursor display position, at 0 as the cursor display.
If the 8th line data is 1, 1 bits will light up the 8th line regardless of the cursor presence.
 - Character pattern row positions correspond to CGRAM data bits 0 to 4 (bit 4 being at the left).
 - As shown Table 5, CGRAM character patterns are selected when character code bits 4 to 7 are all 0. However, since character code bit 3 has no effect, the R display example above can be selected by either character code 00H or 08H.
 - 1 for CGRAM data corresponds to display selection and 0 to non-selection.
- * Indicates no effect.

Table 5 Relationship between CGRAM Addresses, Character Codes (DDRAM) and Character Patterns (CGRAM Data) (cont)

For 5 × 10 dot character patterns

Character Codes (DDRAM data)								CGRAM Address								Character Patterns (CGRAM data)							
7	6	5	4	3	2	1	0	5	4	3	2	1	0	7	6	5	4	3	2	1	0		
High				Low				High				Low				High				Low			
0 0 0 0 * 0 0 *								0 0				0	0	0	0	*	*	*	0	0	0	0	0
												0	0	0	1	↑	0 0 0 0 0				0	0	0
												0	0	1	0		0 0 0 0 0				0	0	0
												0	0	1	1		1	0	1	1	0		
												0	0	1	1		1	1	0	0	1		
												0	1	0	0		1	0	0	0	1		
												0	1	0	1		1	0	0	0	1		
												0	1	1	0		1	0	0	0	1		
												0	1	1	1		1	1	1	1	0		
												1	0	0	0		1	0	0	0	0		
												1	0	0	1		1	0	0	0	0		
0 0 0 0 * 1 1 *								1 1				1	0	1	0	↑	0 0 0 0 0				0	0	0
												1	0	1	1		0 0 0 0 0				0	0	0
												1	1	0	0		0 0 0 0 0				0	0	0
												1	1	0	1		0 0 0 0 0				0	0	0
												1	1	1	0		0 0 0 0 0				0	0	0
0 0 0 0 * 1 1 *								1 1				1	1	1	1	↑	0 0 0 0 0				0	0	0
												1	0	1	0		0 0 0 0 0				0	0	0
												1	0	1	1		0 0 0 0 0				0	0	0
												1	1	0	0		0 0 0 0 0				0	0	0
												1	1	1	0		0 0 0 0 0				0	0	0
0 0 0 0 * 1 1 *								1 1				1	1	1	1	↑	0 0 0 0 0				0	0	0
												1	1	1	1		0 0 0 0 0				0	0	0
												1	1	1	1		0 0 0 0 0				0	0	0
												1	1	1	1		0 0 0 0 0				0	0	0
												1	1	1	1		0 0 0 0 0				0	0	0

Character pattern

Cursor position

- Notes:
- Character code bits 1 and 2 correspond to CGRAM address bits 4 and 5 (2 bits: 4 types).
 - CGRAM address bits 0 to 3 designate the character pattern line position. The 11th line is the cursor position and its display is formed by a logical OR with the cursor.
Maintain the 11th line data corresponding to the cursor display position at 0 as the cursor display.
If the 11th line data is "1", "1" bits will light up the 11th line regardless of the cursor presence.
Since lines 12 to 16 are not used for display, they can be used for general data RAM.
 - Character pattern row positions are the same as 5 × 8 dot character pattern positions.
 - CGRAM character patterns are selected when character code bits 4 to 7 are all 0.
However, since character code bits 0 and 3 have no effect, the P display example above can be selected by character codes 00H, 01H, 08H, and 09H.
 - 1 for CGRAM data corresponds to display selection and 0 to non-selection.
- * Indicates no effect.

Timing Generation Circuit

The timing generation circuit generates timing signals for the operation of internal circuits such as DDRAM, CGROM and CGRAM. RAM read timing for display and internal operation timing by MPU access are generated separately to avoid interfering with each other. Therefore, when writing data to DDRAM, for example, there will be no undesirable interferences, such as flickering, in areas other than the display area.

Liquid Crystal Display Driver Circuit

The liquid crystal display driver circuit consists of 16 common signal drivers and 40 segment signal drivers. When the character font and number of lines are selected by a program, the required common signal drivers automatically output drive waveforms, while the other common signal drivers continue to output non-selection waveforms.

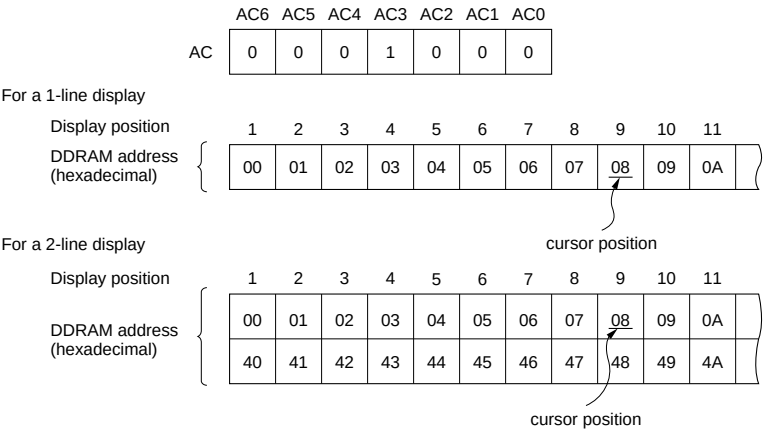
Sending serial data always starts at the display data character pattern corresponding to the last address of the display data RAM (DDRAM).

Since serial data is latched when the display data character pattern corresponding to the starting address enters the internal shift register, the HD44780U drives from the head display.

Cursor/Blink Control Circuit

The cursor/blink control circuit generates the cursor or character blinking. The cursor or the blinking will appear with the digit located at the display data RAM (DDRAM) address set in the address counter (AC).

For example (Figure 8), when the address counter is 08H, the cursor position is displayed at DDRAM address 08H.



Note: The cursor or blinking appears when the address counter (AC) selects the character generator RAM (CGRAM). However, the cursor and blinking become meaningless. The cursor or blinking is displayed in the meaningless position when the AC is a CGRAM address.

Figure 8 Cursor/Blink Display Example

Interfacing to the MPU

The HD44780U can send data in either two 4-bit operations or one 8-bit operation, thus allowing interfacing with 4- or 8-bit MPUs.

- For 4-bit interface data, only four bus lines (DB4 to DB7) are used for transfer. Bus lines DB0 to DB3 are disabled. The data transfer between the HD44780U and the MPU is completed after the 4-bit data has been transferred twice. As for the order of data transfer, the four high order bits (for 8-bit operation, DB4 to DB7) are transferred before the four low order bits (for 8-bit operation, DB0 to DB3). The busy flag must be checked (one instruction) after the 4-bit data has been transferred twice. Two more 4-bit operations then transfer the busy flag and address counter data.
- For 8-bit interface data, all eight bus lines (DB0 to DB7) are used.

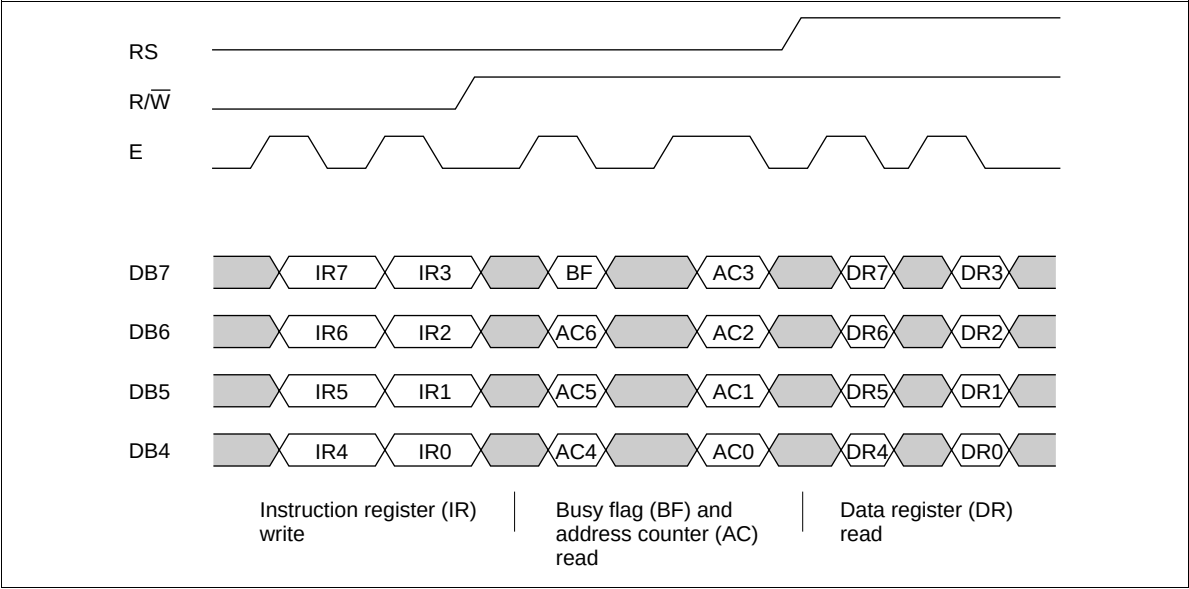


Figure 9 4-Bit Transfer Example

Reset Function

Initializing by Internal Reset Circuit

An internal reset circuit automatically initializes the HD44780U when the power is turned on. The following instructions are executed during the initialization. The busy flag (BF) is kept in the busy state until the initialization ends (BF = 1). The busy state lasts for 10 ms after V_{CC} rises to 4.5 V.

1. Display clear
2. Function set:
 - DL = 1; 8-bit interface data
 - N = 0; 1-line display
 - F = 0; 5×8 dot character font
3. Display on/off control:
 - D = 0; Display off
 - C = 0; Cursor off
 - B = 0; Blinking off
4. Entry mode set:
 - I/D = 1; Increment by 1
 - S = 0; No shift

Note: If the electrical characteristics conditions listed under the table Power Supply Conditions Using Internal Reset Circuit are not met, the internal reset circuit will not operate normally and will fail to initialize the HD44780U. For such a case, initialization must be performed by the MPU as explained in the section, Initializing by Instruction.

Instructions

Outline

Only the instruction register (IR) and the data register (DR) of the HD44780U can be controlled by the MPU. Before starting the internal operation of the HD44780U, control information is temporarily stored into these registers to allow interfacing with various MPUs, which operate at different speeds, or various peripheral control devices. The internal operation of the HD44780U is determined by signals sent from the MPU. These signals, which include register selection signal (RS), read/

write signal ($R/\overline{W}$), and the data bus (DB0 to DB7), make up the HD44780U instructions (Table 6). There are four categories of instructions that:

- Designate HD44780U functions, such as display format, data length, etc.
- Set internal RAM addresses
- Perform data transfer with internal RAM
- Perform miscellaneous functions

Normally, instructions that perform data transfer with internal RAM are used the most. However, auto-incrementation by 1 (or auto-decrementation by 1) of internal HD44780U RAM addresses after each data write can lighten the program load of the MPU. Since the display shift instruction (Table 11) can perform concurrently with display data write, the user can minimize system development time with maximum programming efficiency.

When an instruction is being executed for internal operation, no instruction other than the busy flag/address read instruction can be executed.

Because the busy flag is set to 1 while an instruction is being executed, check it to make sure it is 0 before sending another instruction from the MPU.

Note: Be sure the HD44780U is not in the busy state (BF = 0) before sending an instruction from the MPU to the HD44780U. If an instruction is sent without checking the busy flag, the time between the first instruction and next instruction will take much longer than the instruction time itself. Refer to Table 6 for the list of each instruction execution time.

Table 6 Instructions

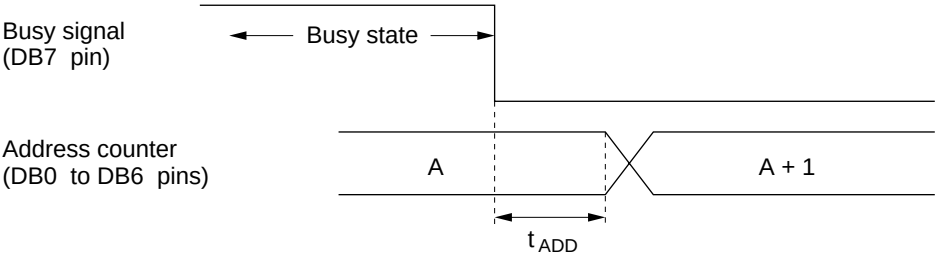
Instruction	Code										Description	Execution Time (max) (when f_{cp} or f_{osc} is 270 kHz)
	RS	R/W	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0		
Clear display	0	0	0	0	0	0	0	0	0	1	Clears entire display and sets DDRAM address 0 in address counter.	
Return home	0	0	0	0	0	0	0	0	1	—	Sets DDRAM address 0 in address counter. Also returns display from being shifted to original position. DDRAM contents remain unchanged.	1.52 ms
Entry mode set	0	0	0	0	0	0	0	1	I/D	S	Sets cursor move direction and specifies display shift. These operations are performed during data write and read.	37 μ s
Display on/off control	0	0	0	0	0	0	1	D	C	B	Sets entire display (D) on/off, cursor on/off (C), and blinking of cursor position character (B).	37 μ s
Cursor or display shift	0	0	0	0	0	1	S/C	R/L	—	—	Moves cursor and shifts display without changing DDRAM contents.	37 μ s
Function set	0	0	0	0	1	DL	N	F	—	—	Sets interface data length (DL), number of display lines (N), and character font (F).	37 μ s
Set CGRAM address	0	0	0	1	ACG	ACG	ACG	ACG	ACG	ACG	Sets CGRAM address. CGRAM data is sent and received after this setting.	37 μ s
Set DDRAM address	0	0	1	ADD	ADD	ADD	ADD	ADD	ADD	ADD	Sets DDRAM address. DDRAM data is sent and received after this setting.	37 μ s
Read busy flag & address	0	1	BF	AC	AC	AC	AC	AC	AC	AC	Reads busy flag (BF) indicating internal operation is being performed and reads address counter contents.	0 μ s

Table 6 Instructions (cont)

Instruction	Code										Description	Execution Time (max) (when f_{cp} or f_{OSC} is 270 kHz)
	RS	R/W	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0		
Write data to CG or DDRAM	1	0	Write data								Writes data into DDRAM or CGRAM.	37 μ s $t_{ADD} = 4 \mu$ s*
Read data from CG or DDRAM	1	1	Read data								Reads data from DDRAM or CGRAM.	37 μ s $t_{ADD} = 4 \mu$ s*
		I/D = 1:	Increment								DDRAM: Display data RAM	Execution time changes when frequency changes Example: When f_{cp} or f_{OSC} is 250 kHz, 37μ s $\times \frac{270}{250} = 40 \mu$ s
		I/D = 0:	Decrement								CGRAM: Character generator RAM	
		S = 1:	Accompanies display shift								ACG: CGRAM address	
		S/C = 1:	Display shift								ADD: DDRAM address	
		S/C = 0:	Cursor move								(corresponds to cursor address)	
		R/L = 1:	Shift to the right									
		R/L = 0:	Shift to the left									
		DL = 1:	8 bits, DL = 0: 4 bits								AC: Address counter used for both DD and CGRAM addresses	
		N = 1:	2 lines, N = 0: 1 line									
		F = 1:	5 $\times$ 10 dots, F = 0: 5 $\times$ 8 dots									
		BF = 1:	Internally operating									
		BF = 0:	Instructions acceptable									

Note: — indicates no effect.

* After execution of the CGRAM/DDRAM data write or read instruction, the RAM address counter is incremented or decremented by 1. The RAM address counter is updated after the busy flag turns off. In Figure 10, t_{ADD} is the time elapsed after the busy flag turns off until the address counter is updated.



Note: t_{ADD} depends on the operation frequency
 $t_{ADD} = 1.5/(f_{cp} \text{ or } f_{OSC})$ seconds

Figure 10 Address Counter Update

Instruction Description

Clear Display

Clear display writes space code 20H (character pattern for character code 20H must be a blank pattern) into all DDRAM addresses. It then sets DDRAM address 0 into the address counter, and returns the display to its original status if it was shifted. In other words, the display disappears and the cursor or blinking goes to the left edge of the display (in the first line if 2 lines are displayed). It also sets I/D to 1 (increment mode) in entry mode. S of entry mode does not change.

Return Home

Return home sets DDRAM address 0 into the address counter, and returns the display to its original status if it was shifted. The DDRAM contents do not change.

The cursor or blinking go to the left edge of the display (in the first line if 2 lines are displayed).

Entry Mode Set

I/D: Increments ($I/D = 1$) or decrements ($I/D = 0$) the DDRAM address by 1 when a character code is written into or read from DDRAM.

The cursor or blinking moves to the right when incremented by 1 and to the left when decremented by 1. The same applies to writing and reading of CGRAM.

S: Shifts the entire display either to the right ($I/D = 0$) or to the left ($I/D = 1$) when S is 1. The display does not shift if S is 0.

If S is 1, it will seem as if the cursor does not move but the display does. The display does not shift when reading from DDRAM. Also, writing into or reading out from CGRAM does not shift the display.

Display On/Off Control

D: The display is on when D is 1 and off when D is 0. When off, the display data remains in DDRAM, but can be displayed instantly by setting D to 1.

C: The cursor is displayed when C is 1 and not displayed when C is 0. Even if the cursor disappears, the function of I/D or other specifications will not change during display data write. The cursor is displayed using 5 dots in the 8th line for 5×8 dot character font selection and in the 11th line for the 5×10 dot character font selection (Figure 13).

B: The character indicated by the cursor blinks when B is 1 (Figure 13). The blinking is displayed as switching between all blank dots and displayed characters at a speed of 409.6-ms intervals when f_{cp} or f_{osc} is 250 kHz. The cursor and blinking can be set to display simultaneously. (The blinking frequency changes according to f_{osc} or the reciprocal of f_{cp} . For example, when f_{cp} is 270 kHz, $409.6 \times 250/270 = 379.2$ ms.)

Cursor or Display Shift

Cursor or display shift shifts the cursor position or display to the right or left without writing or reading display data (Table 7). This function is used to correct or search the display. In a 2-line display, the cursor moves to the second line when it passes the 40th digit of the first line. Note that the first and second line displays will shift at the same time.

When the displayed data is shifted repeatedly each line moves only horizontally. The second line display does not shift into the first line position.

The address counter (AC) contents will not change if the only action performed is a display shift.

Function Set

DL: Sets the interface data length. Data is sent or received in 8-bit lengths (DB7 to DB0) when DL is 1, and in 4-bit lengths (DB7 to DB4) when DL is 0. When 4-bit length is selected, data must be sent or received twice.

N: Sets the number of display lines.

F: Sets the character font.

Note: Perform the function at the head of the program before executing any instructions (except for the read busy flag and address instruction). From this point, the function set instruction cannot be executed unless the interface data length is changed.

Set CGRAM Address

Set CGRAM address sets the CGRAM address binary AAAAAA into the address counter.

Data is then written to or read from the MPU for CGRAM.

		RS	R/W	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0	
Clear display	Code	0	0	0	0	0	0	0	0	0	1	
Return home	Code	0	0	0	0	0	0	0	0	1	*	Note: * Don't care.
Entry mode set	Code	0	0	0	0	0	0	0	1	I/D	S	
Display on/off control	Code	0	0	0	0	0	0	1	D	C	B	
Cursor or display shift	Code	0	0	0	0	0	1	S/C	R/L	*	*	Note: * Don't care.
Function set	Code	0	0	0	0	1	DL	N	F	*	*	
Set CGRAM address	Code	0	0	0	1	A	A	A	A	A	A	

Higher order bit ← → Lower order bit

Set DDRAM Address

Set DDRAM address sets the DDRAM address binary AAAAAAA into the address counter.

Data is then written to or read from the MPU for DDRAM.

However, when N is 0 (1-line display), AAAAAAA can be 00H to 4FH. When N is 1 (2-line display), AAAAAAA can be 00H to 27H for the first line, and 40H to 67H for the second line.

Read Busy Flag and Address

Read busy flag and address reads the busy flag (BF) indicating that the system is now internally operating on a previously received instruction. If BF is 1, the internal operation is in progress. The next instruction will not be accepted until BF is reset to 0. Check the BF status before the next write operation. At the same time, the value of the address counter in binary AAAAAAA is read out. This address counter is used by both CG and DDRAM addresses, and its value is determined by the previous instruction. The address contents are the same as for instructions set CGRAM address and set DDRAM address.

Table 7 Shift Function

S/C	R/L	
0	0	Shifts the cursor position to the left. (AC is decremented by one.)
0	1	Shifts the cursor position to the right. (AC is incremented by one.)
1	0	Shifts the entire display to the left. The cursor follows the display shift.
1	1	Shifts the entire display to the right. The cursor follows the display shift.

Table 8 Function Set

N	F	No. of Display Lines	Character Font	Duty Factor	Remarks
0	0	1	5 × 8 dots	1/8	
0	1	1	5 × 10 dots	1/11	
1	*	2	5 × 8 dots	1/16	Cannot display two lines for 5 × 10 dot character font

Note: * Indicates don't care.

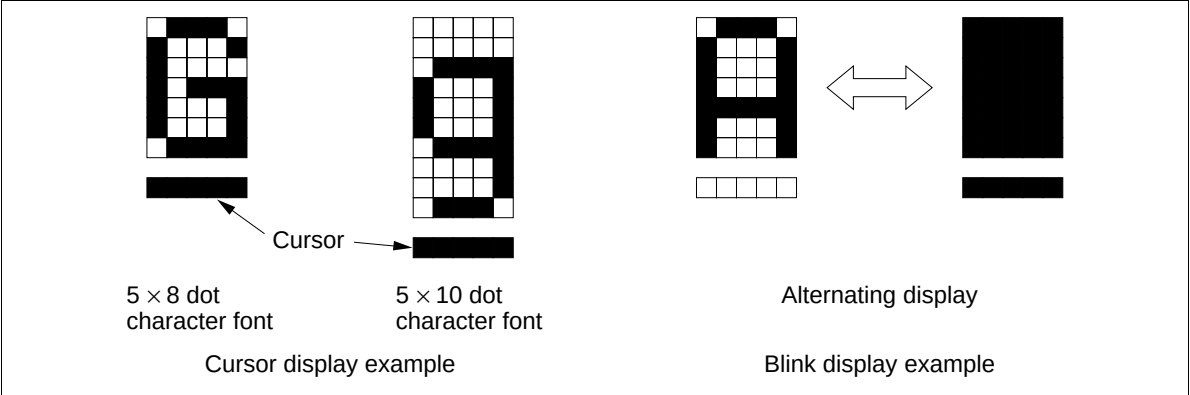


Figure 12 Cursor and Blinking

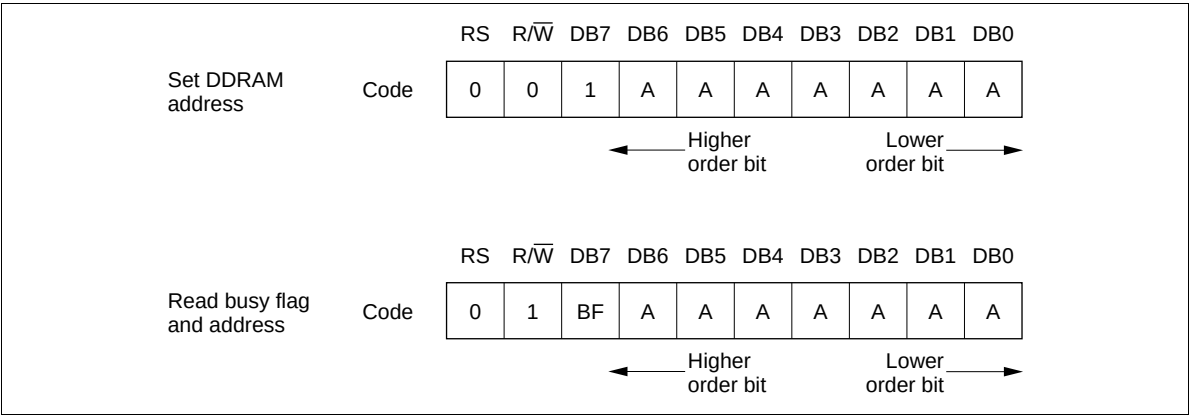


Figure 13 Instruction (2)

Write Data to CG or DDRAM

Write data to CG or DDRAM writes 8-bit binary data DDDDDDDD to CG or DDRAM.

To write into CG or DDRAM is determined by the previous specification of the CGRAM or DDRAM address setting. After a write, the address is automatically incremented or decremented by 1 according to the entry mode. The entry mode also determines the display shift.

Read Data from CG or DDRAM

Read data from CG or DDRAM reads 8-bit binary data DDDDDDDD from CG or DDRAM.

The previous designation determines whether CG or DDRAM is to be read. Before entering this read instruction, either CGRAM or DDRAM address set instruction must be executed. If not executed, the first read data will be invalid. When serially executing read instructions, the next address data is normally read from the second read. The address set instructions need not be executed just before this read instruction when shifting the cursor by the cursor shift instruction (when reading out DDRAM). The operation of the cursor shift instruction is the same as the set DDRAM address instruction.

After a read, the entry mode automatically increases or decreases the address by 1. However, display shift is not executed regardless of the entry mode.

Note: The address counter (AC) is automatically incremented or decremented by 1 after the write instructions to CGRAM or DDRAM are executed. The RAM data selected by the AC cannot be read out at this time even if read instructions are executed. Therefore, to correctly read data, execute either the address set instruction or cursor shift instruction (only with DDRAM), then just before reading the desired data, execute the read instruction from the second time the read instruction is sent.

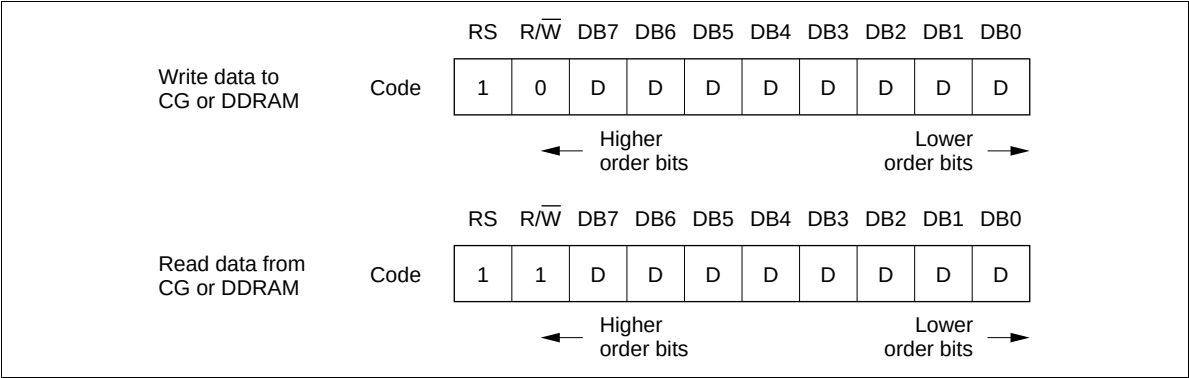


Figure 14 Instruction (3)

Interfacing the HD44780U

Interface to MPUs

- Interfacing to an 8-bit MPU
See Figure 16 for an example of using a I/O port (for a single-chip microcomputer) as an interface device.
In this example, P30 to P37 are connected to the data bus DB0 to DB7, and P75 to P77 are connected to E, R/W, and RS, respectively.

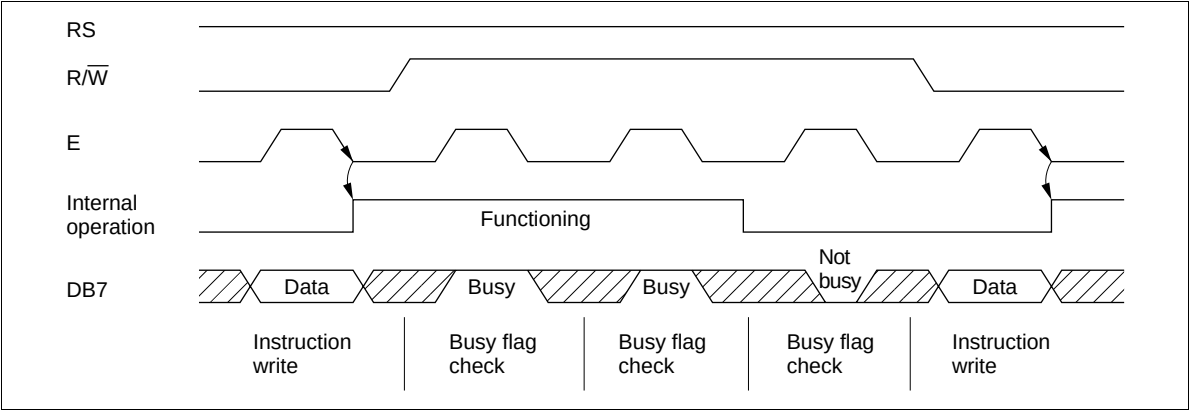


Figure 15 Example of Busy Flag Check Timing Sequence

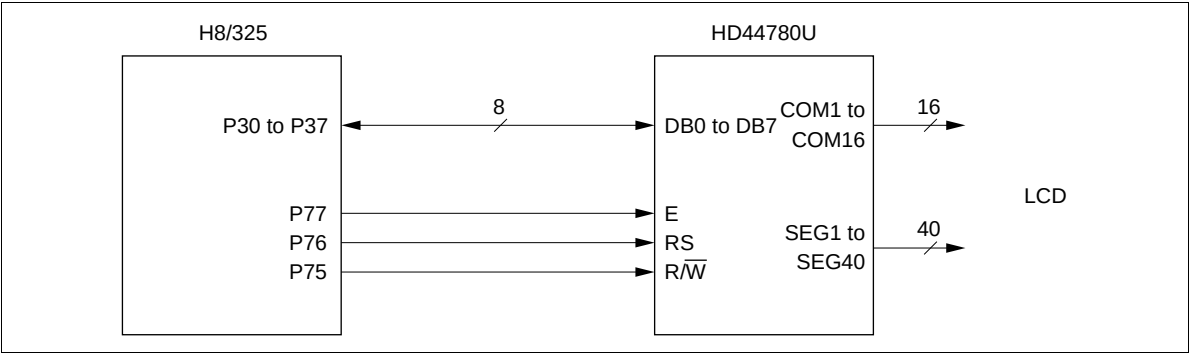


Figure 16 H8/325 Interface (Single-Chip Mode)

- Interfacing to a 4-bit MPU

The HD44780U can be connected to the I/O port of a 4-bit MPU. If the I/O port has enough bits, 8-bit data can be transferred. Otherwise, one data transfer must be made in two operations for 4-bit data. In this case, the timing sequence becomes somewhat complex. (See Figure 17.)

See Figure 18 for an interface example to the HMCS4019R.

Note that two cycles are needed for the busy flag check as well as for the data transfer. The 4-bit operation is selected by the program.

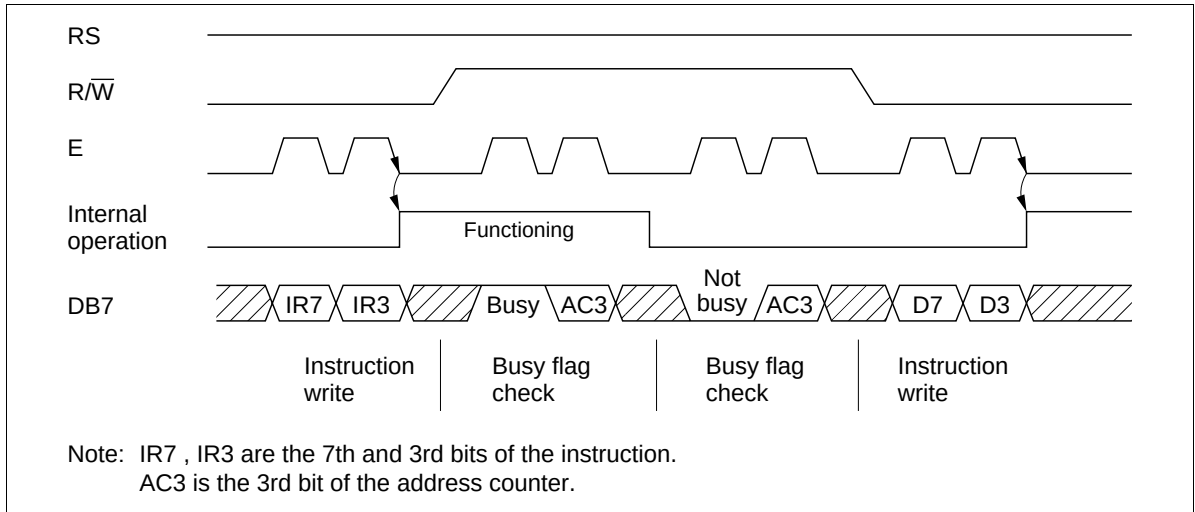


Figure 17 Example of 4-Bit Data Transfer Timing Sequence

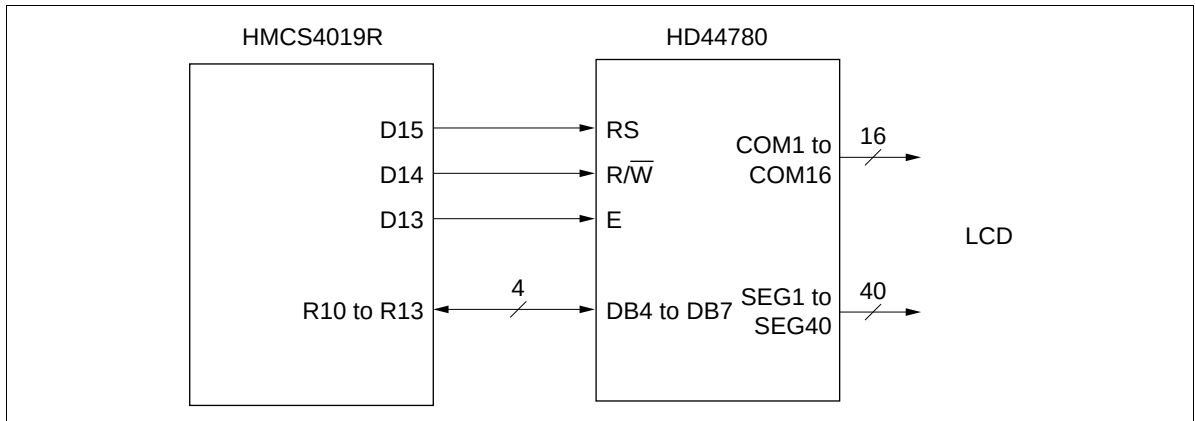


Figure 18 Example of Interface to HMCS4019R

Interface to Liquid Crystal Display

Character Font and Number of Lines: The HD44780U can perform two types of displays, 5×8 dot and 5×10 dot character fonts, each with a cursor.

Up to two lines are displayed for 5×8 dots and one line for 5×10 dots. Therefore, a total of three types of common signals are available (Table 9).

The number of lines and font types can be selected by the program. (See Table 6, Instructions.)

Connection to HD44780 and Liquid Crystal Display: See Figure 19 for the connection examples.

Table 9 Common Signals

Number of Lines	Character Font	Number of Common Signals	Duty Factor
1	5×8 dots + cursor	8	1/8
1	5×10 dots + cursor	11	1/11
2	5×8 dots + cursor	16	1/16

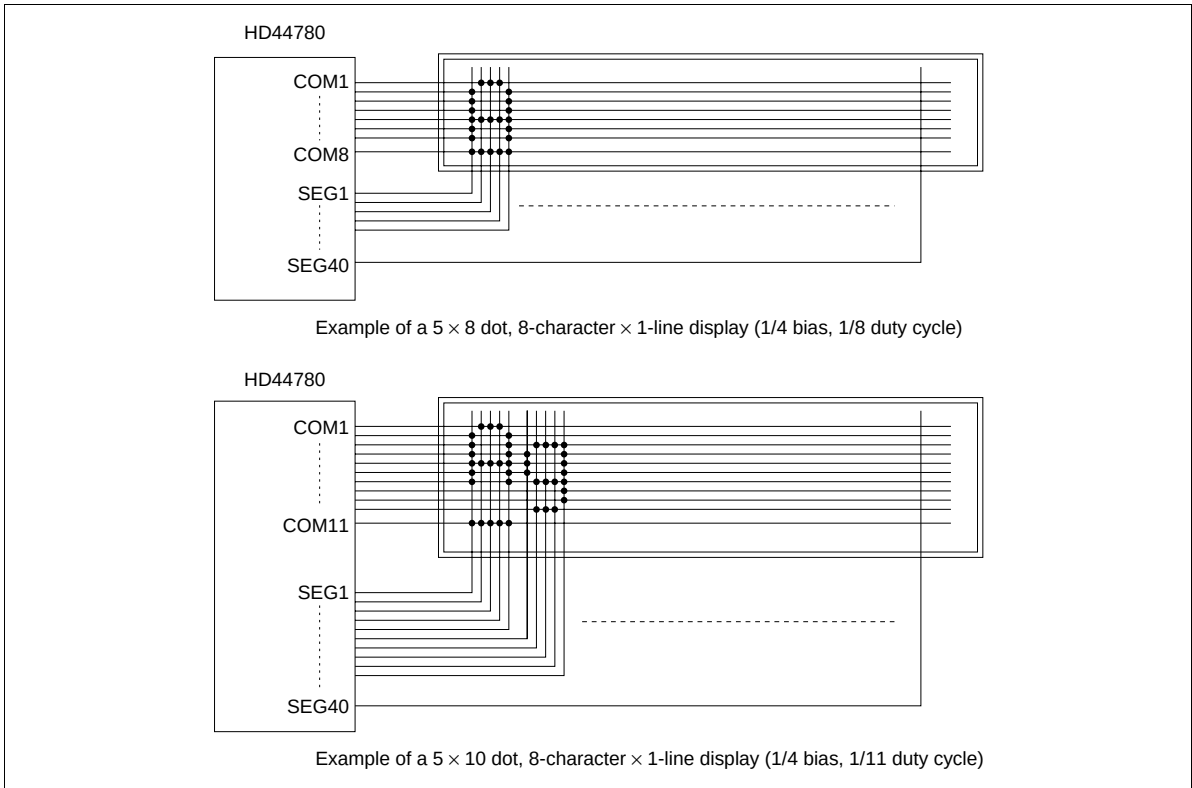
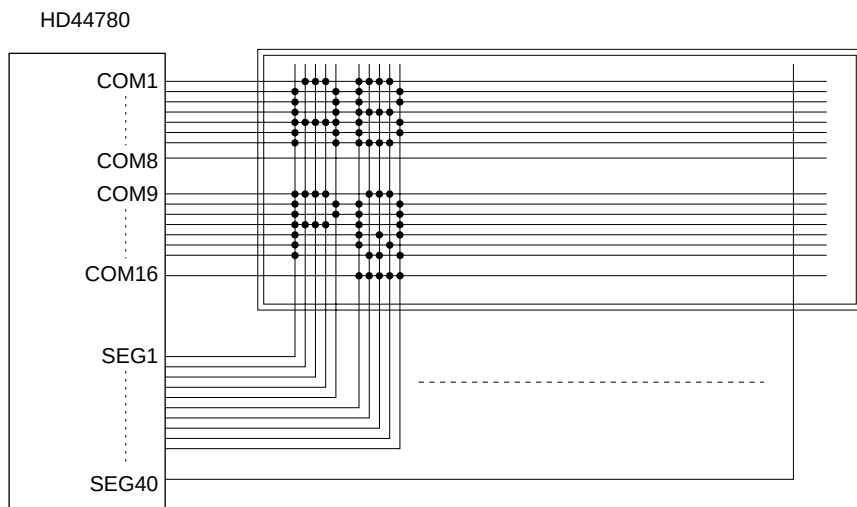


Figure 19 Liquid Crystal Display and HD44780 Connections

Since five segment signal lines can display one digit, one HD44780U can display up to 8 digits for a 1-line display and 16 digits for a 2-line display.

The examples in Figure 19 have unused common signal pins, which always output non-selection waveforms. When the liquid crystal display panel has unused extra scanning lines, connect the extra scanning lines to these common signal pins to avoid any undesirable effects due to crosstalk during the floating state.



Example of a 5 × 8 dot, 8-character × 2-line display (1/5 bias, 1/16 duty cycle)

Figure 19 Liquid Crystal Display and HD44780 Connections (cont)

Connection of Changed Matrix Layout: In the preceding examples, the number of lines correspond to the scanning lines. However, the following display examples (Figure 20) are made possible by altering the matrix layout of the liquid crystal display panel. In either case, the only change is the layout. The display characteristics and the number of liquid crystal display characters depend on the number of common signals or on duty factor. Note that the display data RAM (DDRAM) addresses for 4 characters $\times$ 2 lines and for 16 characters $\times$ 1 line are the same as in Figure 19.

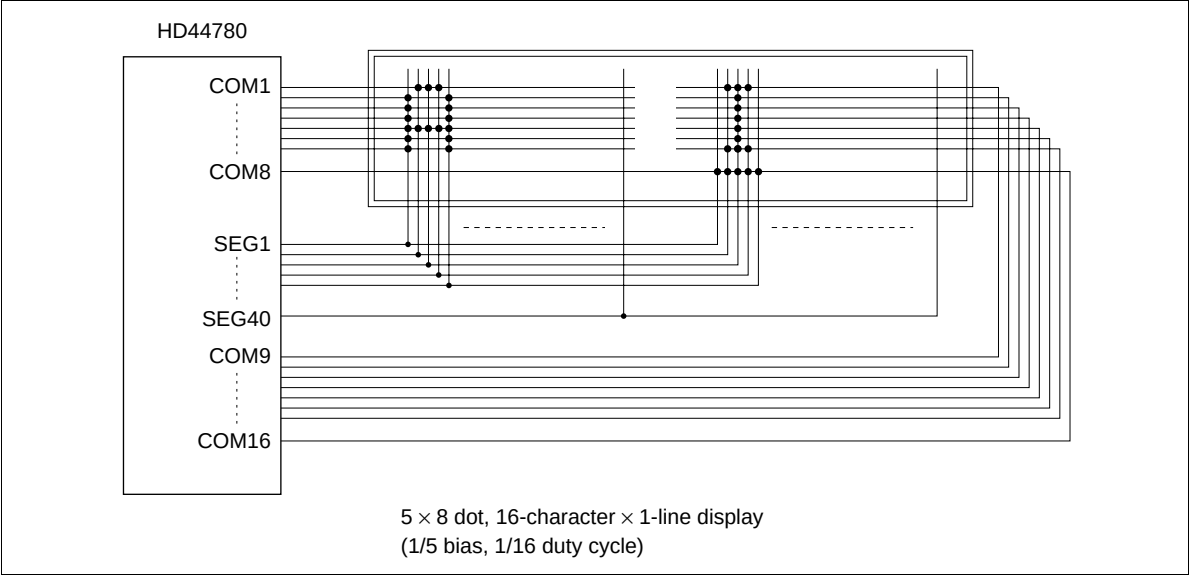


Figure 20 Changed Matrix Layout Displays

Power Supply for Liquid Crystal Display Drive

Various voltage levels must be applied to pins V1 to V5 of the HD44780U to obtain the liquid crystal display drive waveforms. The voltages must be changed according to the duty factor (Table 10).

VLCD is the peak value for the liquid crystal display drive waveforms, and resistance dividing provides voltages V1 to V5 (Figure 21).

Table 10 Duty Factor and Power Supply for Liquid Crystal Display Drive

Power Supply	Duty Factor	
	1/8, 1/11	1/16
	Bias	
	1/4	1/5
V1	$V_{cc}-1/4 \text{ VLCD}$	$V_{cc}-1/5 \text{ VLCD}$
V2	$V_{cc}-1/2 \text{ VLCD}$	$V_{cc}-2/5 \text{ VLCD}$
V3	$V_{cc}-1/2 \text{ VLCD}$	$V_{cc}-3/5 \text{ VLCD}$
V4	$V_{cc}-3/4 \text{ VLCD}$	$V_{cc}-4/5 \text{ VLCD}$
V5	$V_{cc}-\text{VLCD}$	$V_{cc}-\text{VLCD}$

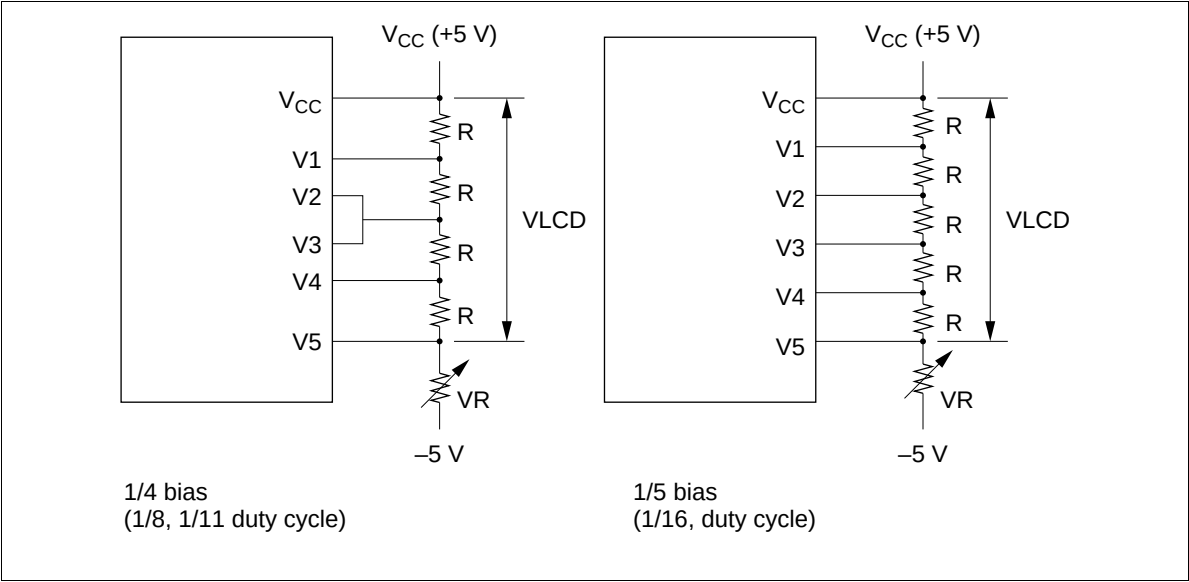


Figure 21 Drive Voltage Supply Example

Relationship between Oscillation Frequency and Liquid Crystal Display Frame Frequency

The liquid crystal display frame frequencies of Figure 22 apply only when the oscillation frequency is 270 kHz (one clock pulse of 3.7 μs).

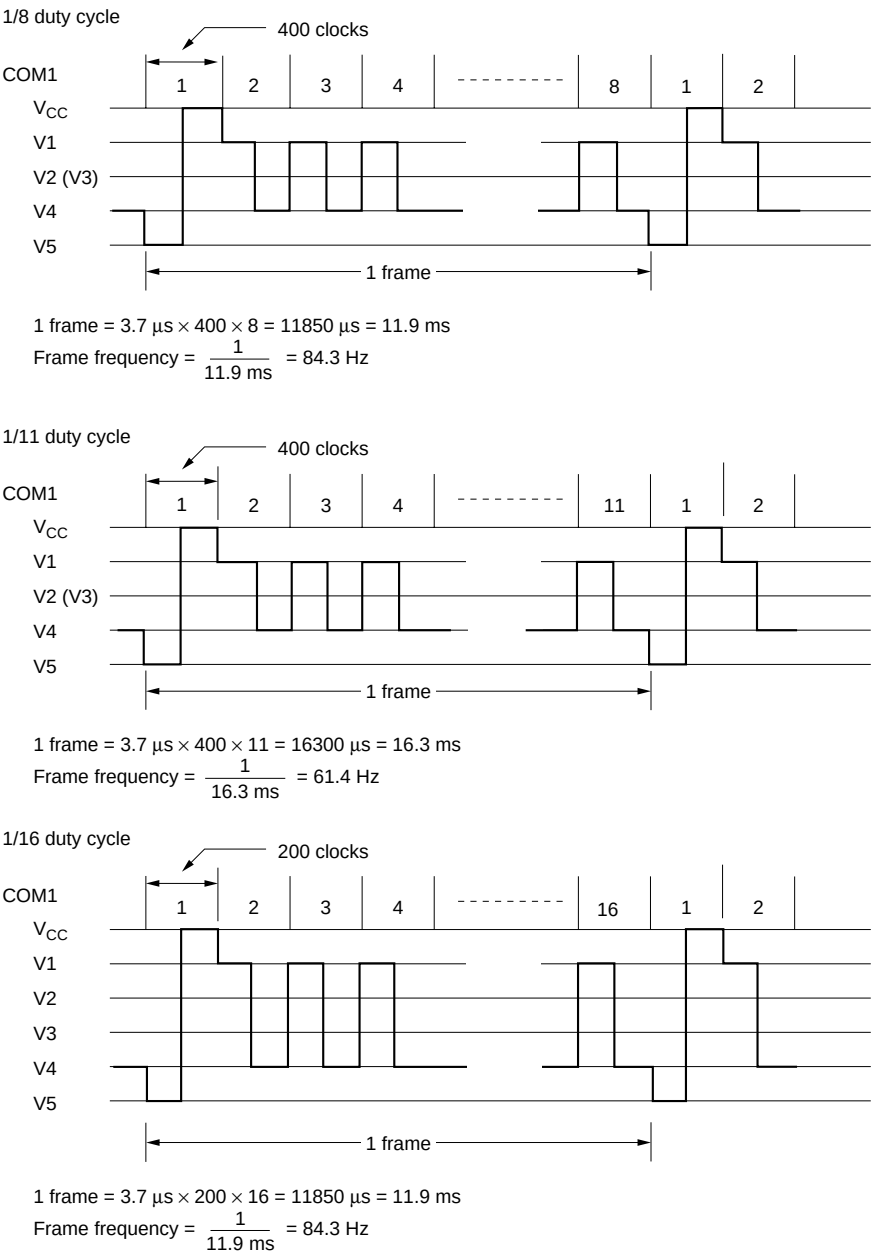


Figure 22 Frame Frequency

Instruction and Display Correspondence

- 8-bit operation, 8-digit $\times$ 1-line display with internal reset

Refer to Table 11 for an example of an 8-digit $\times$ 1-line display in 8-bit operation. The HD44780U functions must be set by the function set instruction prior to the display. Since the display data RAM can store data for 80 characters, as explained before, the RAM can be used for displays such as for advertising when combined with the display shift operation.

Since the display shift operation changes only the display position with DDRAM contents unchanged, the first display data entered into DDRAM can be output when the return home operation is performed.

- 4-bit operation, 8-digit $\times$ 1-line display with internal reset

The program must set all functions prior to the 4-bit operation (Table 12). When the power is turned on, 8-bit operation is automatically selected and the first write is performed as an 8-bit operation. Since DB0 to DB3 are not connected, a rewrite is then required. However, since one operation is completed in two accesses for 4-bit operation, a rewrite is needed to set the functions (see Table 12). Thus, DB4 to DB7 of the function set instruction is written twice.

- 8-bit operation, 8-digit $\times$ 2-line display

For a 2-line display, the cursor automatically moves from the first to the second line after the 40th digit of the first line has been written. Thus, if there are only 8 characters in the first line, the DDRAM address must be again set after the 8th character is completed. (See Table 13.) Note that the display shift operation is performed for the first and second lines. In the example of Table 13, the display shift is performed when the cursor is on the second line. However, if the shift operation is performed when the cursor is on the first line, both the first and second lines move together. If the shift is repeated, the display of the second line will not move to the first line. The same display will only shift within its own line for the number of times the shift is repeated.

Note: When using the internal reset, the electrical characteristics in the Power Supply Conditions Using Internal Reset Circuit table must be satisfied. If not, the HD44780U must be initialized by instructions. See the section, Initializing by Instruction.

Table 11 8-Bit Operation, 8-Digit × 1-Line Display Example with Internal Reset

Step No.	Instruction										Display	Operation
	RS	R/W	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0		
1	Power supply on (the HD44780U is initialized by the internal reset circuit)											Initialized. No display.
2	Function set 0 0 0 0 1 1 0 0 * *											Sets to 8-bit operation and selects 1-line display and 5 × 8 dot character font. (Number of display lines and character fonts cannot be changed after step #2.)
3	Display on/off control 0 0 0 0 0 0 1 1 1 0										_	Turns on display and cursor. Entire display is in space mode because of initialization.
4	Entry mode set 0 0 0 0 0 0 0 1 1 0										_	Sets mode to increment the address by one and to shift the cursor to the right at the time of write to the DD/CGRAM. Display is not shifted.
5	Write data to CGRAM/DDRAM 1 0 0 1 0 0 1 0 0 0										H_	Writes H. DDRAM has already been selected by initialization when the power was turned on. The cursor is incremented by one and shifted to the right.
6	Write data to CGRAM/DDRAM 1 0 0 1 0 0 1 0 0 1										HI_	Writes I.
7												
8	Write data to CGRAM/DDRAM 1 0 0 1 0 0 1 0 0 1										HITACHI_	Writes I.
9	Entry mode set 0 0 0 0 0 0 0 1 1 1										HITACHI_	Sets mode to shift display at the time of write.
10	Write data to CGRAM/DDRAM 1 0 0 0 1 0 0 0 0 0										ITACHI _	Writes a space.

Table 11 8-Bit Operation, 8-Digit × 1-Line Display Example with Internal Reset (cont)

Step No.	Instruction										Display	Operation
	RS	R/W	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0		
11	Write data to CGRAM/DDRAM										TACHI M_	Writes M.
	1	0	0	1	0	0	1	1	0	1		
12												
						.					.	
						.					.	
						.					.	
						.					.	
						.					.	
13	Write data to CGRAM/DDRAM										MICROKO_	Writes O.
	1	0	0	1	0	0	1	1	1	1		
14	Cursor or display shift										MICROKO	Shifts only the cursor position to the left.
	0	0	0	0	0	1	0	0	*	*		
15	Cursor or display shift										MICROKO	Shifts only the cursor position to the left.
	0	0	0	0	0	1	0	0	*	*		
16	Write data to CGRAM/DDRAM										ICROCO	Writes C over K. The display moves to the left.
	1	0	0	1	0	0	0	0	1	1		
17	Cursor or display shift										MICROCO	Shifts the display and cursor position to the right.
	0	0	0	0	0	1	1	1	*	*		
18	Cursor or display shift										MICROCO_	Shifts the display and cursor position to the right.
	0	0	0	0	0	1	0	1	*	*		
19	Write data to CGRAM/DDRAM										ICROCOM_	Writes M.
	1	0	0	1	0	0	1	1	0	1		
20												
						.					.	
						.					.	
						.					.	
						.					.	
						.					.	
21	Return home										HITACHI	Returns both display and cursor to the original position (address 0).
	0	0	0	0	0	0	0	0	1	0		

Table 12 4-Bit Operation, 8-Digit × 1-Line Display Example with Internal Reset

Step		Instruction						Display	Operation
No.	RS	R/ $\overline{W}$	DB7	DB6	DB5	DB4			
1	Power supply on (the HD44780U is initialized by the internal reset circuit)								Initialized. No display.
2	Function set 0 0 0 0 1 0								Sets to 4-bit operation. In this case, operation is handled as 8 bits by initialization, and only this instruction completes with one write.
3	Function set 0 0 0 0 1 0 0 0 0 0 * *								Sets 4-bit operation and selects 1-line display and 5 × 8 dot character font. 4-bit operation starts from this step and resetting is necessary. (Number of display lines and character fonts cannot be changed after step #3.)
4	Display on/off control 0 0 0 0 0 0 0 0 1 1 1 0							—	Turns on display and cursor. Entire display is in space mode because of initialization.
5	Entry mode set 0 0 0 0 0 0 0 0 0 1 1 0							—	Sets mode to increment the address by one and to shift the cursor to the right at the time of write to the DD/CGRAM. Display is not shifted.
6	Write data to CGRAM/DDRAM 1 0 0 1 0 0 1 0 1 0 0 0							H—	Writes H. The cursor is incremented by one and shifts to the right.

Note: The control is the same as for 8-bit operation beyond step #6.

Table 13 8-Bit Operation, 8-Digit × 2-Line Display Example with Internal Reset

Step No.	Instruction										Display	Operation
	RS	R/W	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0		
1	Power supply on (the HD44780U is initialized by the internal reset circuit)											Initialized. No display.
2	Function set 0 0 0 0 1 1 1 0 * *											Sets to 8-bit operation and selects 2-line display and 5 × 8 dot character font.
3	Display on/off control 0 0 0 0 0 0 1 1 1 0										—	Turns on display and cursor. All display is in space mode because of initialization.
4	Entry mode set 0 0 0 0 0 0 0 1 1 0										—	Sets mode to increment the address by one and to shift the cursor to the right at the time of write to the DD/CGRAM. Display is not shifted.
5	Write data to CGRAM/DDRAM 1 0 0 1 0 0 1 0 0 0										H—	Writes H. DDRAM has already been selected by initialization when the power was turned on. The cursor is incremented by one and shifted to the right.
6												
7	Write data to CGRAM/DDRAM 1 0 0 1 0 0 1 0 0 1										HITACHI—	Writes I.
8	Set DDRAM address 0 0 1 1 0 0 0 0 0 0										HITACHI —	Sets DDRAM address so that the cursor is positioned at the head of the second line.

Table 13 8-Bit Operation, 8-Digit × 2-Line Display Example with Internal Reset (cont)

Step		Instruction										Display	Operation
No.		RS	R/W	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0		
9	Write data to CGRAM/DDRAM	1	0	0	1	0	0	1	1	0	1	HITACHI M_	Writes M.
10						.							
11	Write data to CGRAM/DDRAM	1	0	0	1	0	0	1	1	1	1	HITACHI MICROCO_	Writes O.
12	Entry mode set	0	0	0	0	0	0	0	1	1	1	HITACHI MICROCO_	Sets mode to shift display at the time of write.
13	Write data to CGRAM/DDRAM	1	0	0	1	0	0	1	1	0	1	ITACHI ICROCOM_	Writes M. Display is shifted to the left. The first and second lines both shift at the same time.
14						.							
15	Return home	0	0	0	0	0	0	0	0	1	0	HITACHI MICROCOM	Returns both display and cursor to the original position (address 0).

Initializing by Instruction

If the power supply conditions for correctly operating the internal reset circuit are not met, initialization by instructions becomes necessary.

Refer to Figures 23 and 24 for the procedures on 8-bit and 4-bit initializations, respectively.

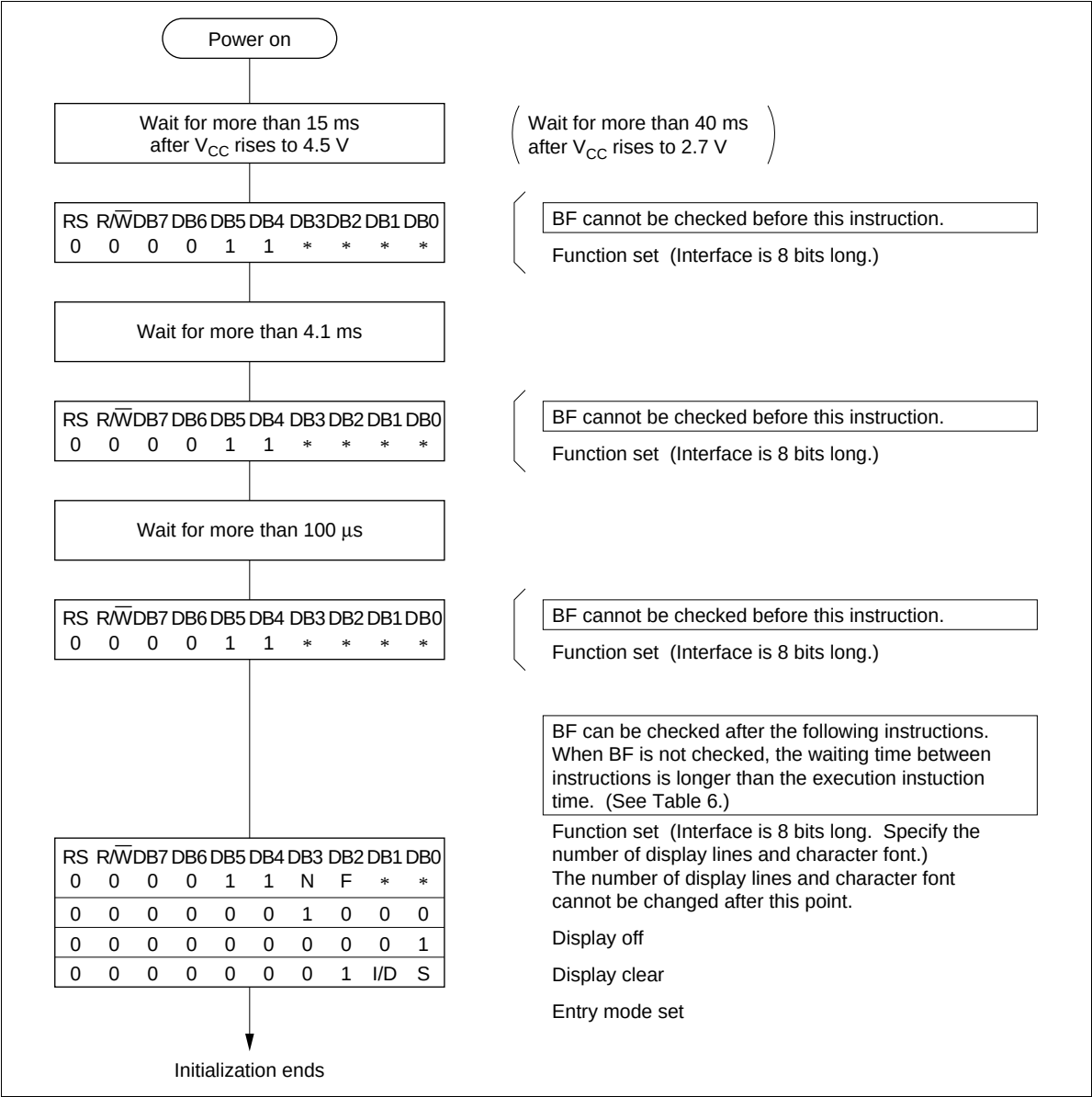


Figure 23 8-Bit Interface

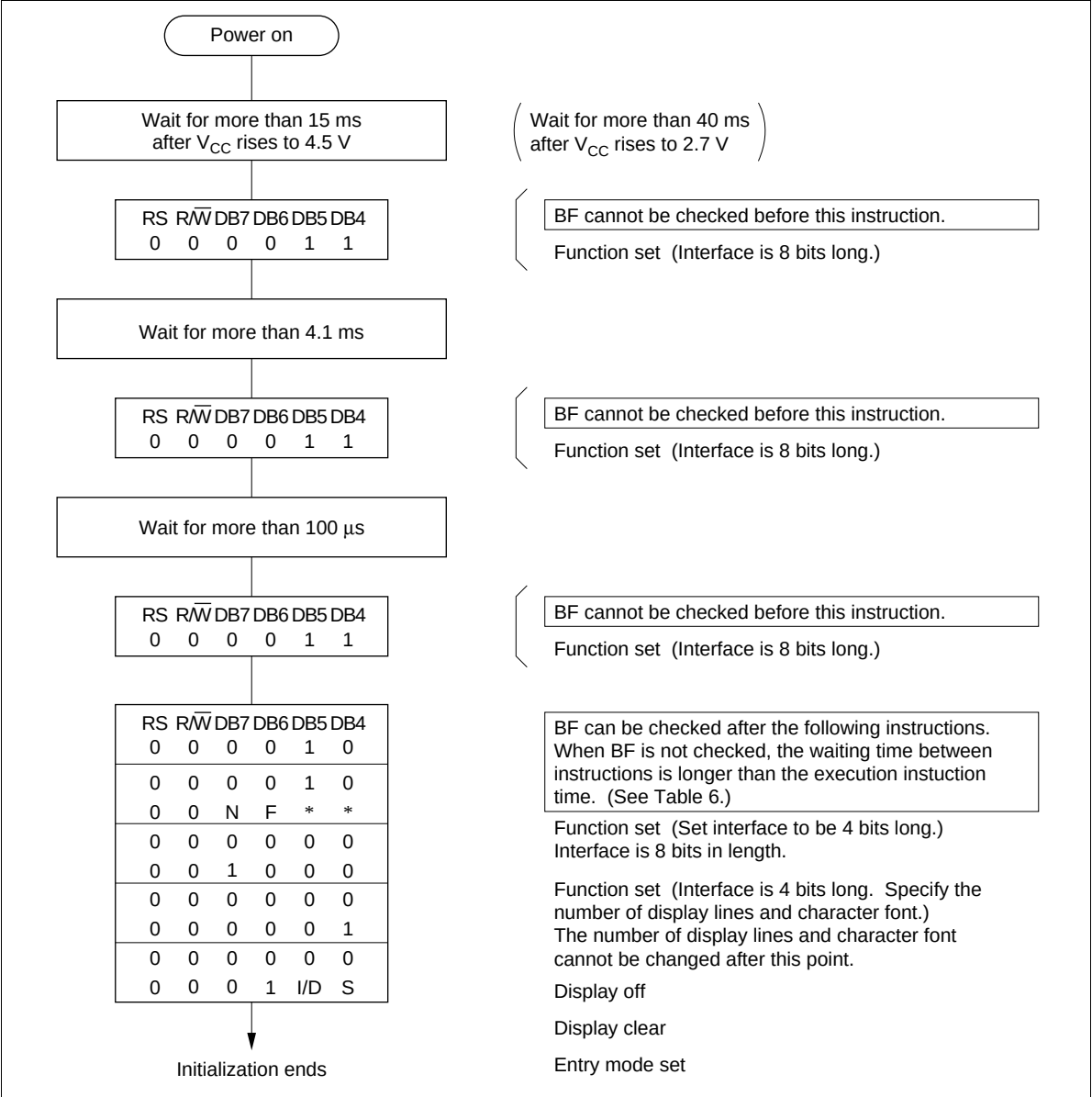


Figure 24 4-Bit Interface

Absolute Maximum Ratings*

Item	Symbol	Value	Unit	Notes
Power supply voltage (1)	V _{CC} –GND	–0.3 to +7.0	V	1
Power supply voltage (2)	V _{CC} –V5	–0.3 to +13.0	V	1, 2
Input voltage	Vt	–0.3 to V _{CC} +0.3	V	1
Operating temperature	T _{opr}	–30 to +75	°C	
Storage temperature	T _{stg}	–55 to +125	°C	4

Note: * If the LSI is used above these absolute maximum ratings, it may become permanently damaged. Using the LSI within the following electrical characteristic limits is strongly recommended for normal operation. If these electrical characteristic conditions are also exceeded, the LSI will malfunction and cause poor reliability.

DC Characteristics (V_{CC} = 2.7 to 4.5 V, T_a = -30 to +75°C*³)

Item	Symbol	Min	Typ	Max	Unit	Test Condition	Notes*
Input high voltage (1) (except OSC1)	VIH1	0.7V _{CC}	—	V _{CC}	V		6
Input low voltage (1) (except OSC1)	VIL1	-0.3	—	0.55	V		6
Input high voltage (2) (OSC1)	VIH2	0.7V _{CC}	—	V _{CC}	V		15
Input low voltage (2) (OSC1)	VIL2	—	—	0.2V _{CC}	V		15
Output high voltage (1) (DB0-DB7)	VOH1	0.75V _{CC}	—	—	V	-I _{OH} = 0.1 mA	7
Output low voltage (1) (DB0-DB7)	VOL1	—	—	0.2V _{CC}	V	I _{OL} = 0.1 mA	7
Output high voltage (2) (except DB0-DB7)	VOH2	0.8V _{CC}	—	—	V	-I _{OH} = 0.04 mA	8
Output low voltage (2) (except DB0-DB7)	VOL2	—	—	0.2V _{CC}	V	I _{OL} = 0.04 mA	8
Driver on resistance (COM)	R _{COM}	—	2	20	kΩ	±I _d = 0.05 mA, VLCD = 4 V	13
Driver on resistance (SEG)	R _{SEG}	—	2	30	kΩ	±I _d = 0.05 mA, VLCD = 4 V	13
Input leakage current	I _{LI}	-1	—	1	μA	VIN = 0 to V _{CC}	9
Pull-up MOS current (DB0-DB7, RS, R/W)	-I _p	10	50	120	μA	V _{CC} = 3 V	
Power supply current	I _{CC}	—	150	300	μA	R _f oscillation, external clock V _{CC} = 3 V, f _{OSC} = 270 kHz	10, 14
LCD voltage	VLCD1	3.0	—	11.0	V	V _{CC} -V5, 1/5 bias	16
	VLCD2	3.0	—	11.0	V	V _{CC} -V5, 1/4 bias	16

Note: * Refer to the Electrical Characteristics Notes section following these tables.

AC Characteristics ($V_{CC} = 2.7$ to 4.5 V, $T_a = -30$ to $+75^\circ\text{C}^{*3}$)

Clock Characteristics

Item		Symbol	Min	Typ	Max	Unit	Test Condition	Note*
External clock operation	External clock frequency	f_{cp}	125	250	350	kHz		11
	External clock duty	Duty	45	50	55	%		
	External clock rise time	t_{rcp}	—	—	0.2	μs		
	External clock fall time	t_{fcp}	—	—	0.2	μs		
R_f oscillation	Clock oscillation frequency	f_{OSC}	190	270	350	kHz	$R_f = 75\text{ k}\Omega$, $V_{CC} = 3\text{ V}$	12

Note: * Refer to the Electrical Characteristics Notes section following these tables.

Bus Timing Characteristics

Write Operation

Item	Symbol	Min	Typ	Max	Unit	Test Condition
Enable cycle time	t_{cycE}	1000	—	—	ns	Figure 25
Enable pulse width (high level)	PW_{EH}	450	—	—		
Enable rise/fall time	t_{Er} , t_{Ef}	—	—	25		
Address set-up time (RS, R/W to E)	t_{AS}	60	—	—		
Address hold time	t_{AH}	20	—	—		
Data set-up time	t_{DSW}	195	—	—		
Data hold time	t_H	10	—	—		

Read Operation

Item	Symbol	Min	Typ	Max	Unit	Test Condition
Enable cycle time	t_{cycE}	1000	—	—	ns	Figure 26
Enable pulse width (high level)	PW_{EH}	450	—	—		
Enable rise/fall time	t_{Er} , t_{Ef}	—	—	25		
Address set-up time (RS, R/W to E)	t_{AS}	60	—	—		
Address hold time	t_{AH}	20	—	—		
Data delay time	t_{DDR}	—	—	360		
Data hold time	t_{DHR}	5	—	—		

Interface Timing Characteristics with External Driver

Item		Symbol	Min	Typ	Max	Unit	Test Condition
Clock pulse width	High level	t _{CWH}	800	—	—	ns	Figure 27
	Low level	t _{CWL}	800	—	—		
Clock set-up time		t _{CSU}	500	—	—		
Data set-up time		t _{SU}	300	—	—		
Data hold time		t _{DH}	300	—	—		
M delay time		t _{DM}	−1000	—	1000		
Clock rise/fall time		t _{ct}	—	—	200		

Power Supply Conditions Using Internal Reset Circuit

Item		Symbol	Min	Typ	Max	Unit	Test Condition
Power supply rise time		t _{rCC}	0.1	—	10	ms	Figure 28
Power supply off time		t _{OFF}	1	—	—		

DC Characteristics ($V_{CC} = 4.5$ to 5.5 V, $T_a = -30$ to $+75^\circ\text{C}^{*3}$)

Item	Symbol	Min	Typ	Max	Unit	Test Condition	Notes*
Input high voltage (1) (except OSC1)	VIH1	2.2	—	V_{CC}	V		6
Input low voltage (1) (except OSC1)	VIL1	-0.3	—	0.6	V		6
Input high voltage (2) (OSC1)	VIH2	$V_{CC}-1.0$	—	V_{CC}	V		15
Input low voltage (2) (OSC1)	VIL2	—	—	1.0	V		15
Output high voltage (1) (DB0–DB7)	VOH1	2.4	—	—	V	$-I_{OH} = 0.205$ mA	7
Output low voltage (1) (DB0–DB7)	VOL1	—	—	0.4	V	$I_{OL} = 1.2$ mA	7
Output high voltage (2) (except DB0–DB7)	VOH2	$0.9 V_{CC}$	—	—	V	$-I_{OH} = 0.04$ mA	8
Output low voltage (2) (except DB0–DB7)	VOL2	—	—	$0.1 V_{CC}$	V	$I_{OL} = 0.04$ mA	8
Driver on resistance (COM)	RCOM	—	2	20	k Ω	$\pm I_d = 0.05$ mA, VLCD = 4 V	13
Driver on resistance (SEG)	RSEG	—	2	30	k Ω	$\pm I_d = 0.05$ mA, VLCD = 4 V	13
Input leakage current	I_{LI}	-1	—	1	μA	$V_{IN} = 0$ to V_{CC}	9
Pull-up MOS current (DB0–DB7, RS, R/W)	$-I_p$	50	125	250	μA	$V_{CC} = 5$ V	
Power supply current	I_{CC}	—	350	600	μA	R_f oscillation, external clock $V_{CC} = 5$ V, $f_{OSC} = 270$ kHz	10, 14
LCD voltage	VLCD1	3.0	—	11.0	V	$V_{CC}-V_5$, 1/5 bias	16
	VLCD2	3.0	—	11.0	V	$V_{CC}-V_5$, 1/4 bias	16

Note: * Refer to the Electrical Characteristics Notes section following these tables.

AC Characteristics (V_{CC} = 4.5 to 5.5 V, T_a = -30 to +75°C*³)

Clock Characteristics

Item		Symbol	Min	Typ	Max	Unit	Test Condition	Notes*
External clock operation	External clock frequency	f _{cp}	125	250	350	kHz		11
	External clock duty	Duty	45	50	55	%		11
	External clock rise time	t _{rcp}	—	—	0.2	μs		11
	External clock fall time	t _{fcp}	—	—	0.2	μs		11
R _f oscillation	Clock oscillation frequency	f _{OSC}	190	270	350	kHz	R _f = 91 kΩ V _{CC} = 5.0 V	12

Note: * Refer to the Electrical Characteristics Notes section following these tables.

Bus Timing Characteristics

Write Operation

Item	Symbol	Min	Typ	Max	Unit	Test Condition
Enable cycle time	t _{cycE}	500	—	—	ns	Figure 25
Enable pulse width (high level)	PW _{EH}	230	—	—		
Enable rise/fall time	t _{Er} , t _{Ef}	—	—	20		
Address set-up time (RS, R/W to E)	t _{AS}	40	—	—		
Address hold time	t _{AH}	10	—	—		
Data set-up time	t _{DSW}	80	—	—		
Data hold time	t _H	10	—	—		

Read Operation

Item	Symbol	Min	Typ	Max	Unit	Test Condition
Enable cycle time	t _{cycE}	500	—	—	ns	Figure 26
Enable pulse width (high level)	PW _{EH}	230	—	—		
Enable rise/fall time	t _{Er} , t _{Ef}	—	—	20		
Address set-up time (RS, R/W to E)	t _{AS}	40	—	—		
Address hold time	t _{AH}	10	—	—		
Data delay time	t _{DDR}	—	—	160		
Data hold time	t _{DHR}	5	—	—		

Interface Timing Characteristics with External Driver

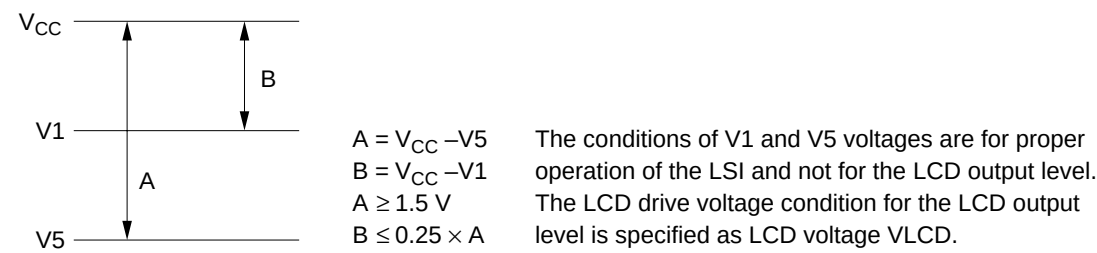
Item		Symbol	Min	Typ	Max	Unit	Test Condition
Clock pulse width	High level	t_{CWH}	800	—	—	ns	Figure 27
	Low level	t_{CWL}	800	—	—		
Clock set-up time		t_{CSU}	500	—	—		
Data set-up time		t_{SU}	300	—	—		
Data hold time		t_{DH}	300	—	—		
M delay time		t_{DM}	−1000	—	1000		
Clock rise/fall time		t_{ct}	—	—	100		

Power Supply Conditions Using Internal Reset Circuit

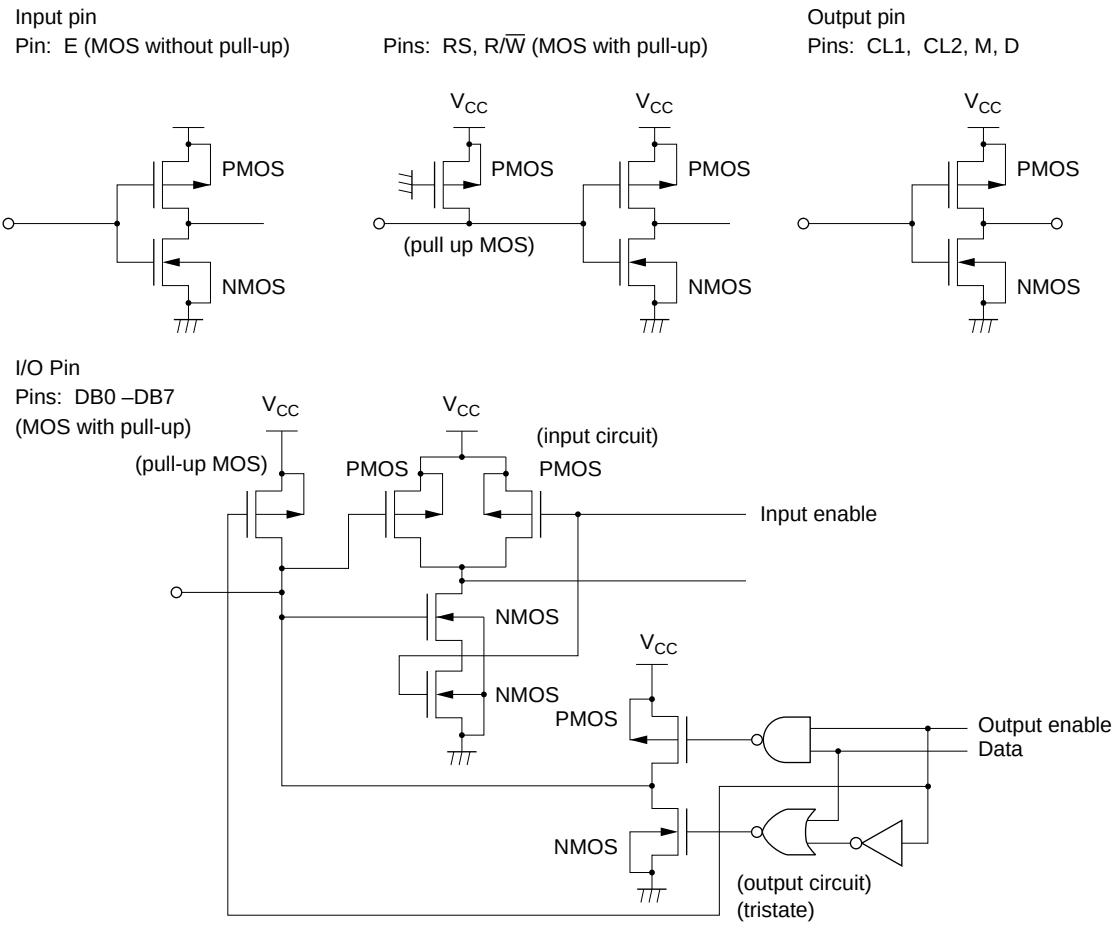
Item		Symbol	Min	Typ	Max	Unit	Test Condition
Power supply rise time		t_{rcc}	0.1	—	10	ms	Figure 28
Power supply off time		t_{OFF}	1	—	—		

Electrical Characteristics Notes

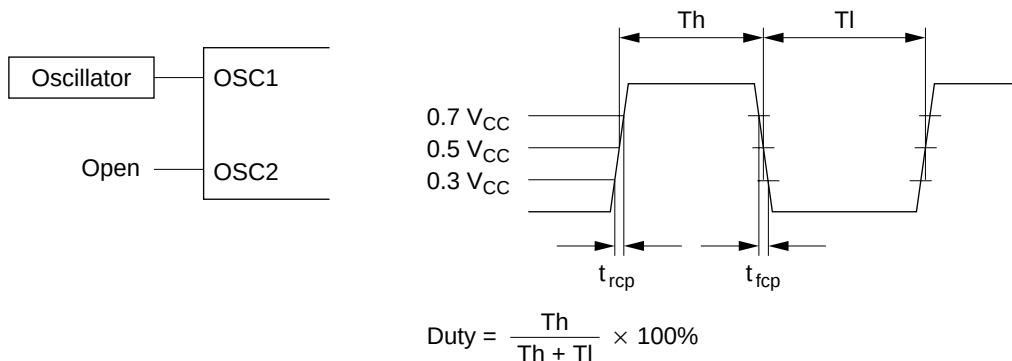
1. All voltage values are referred to GND = 0 V.



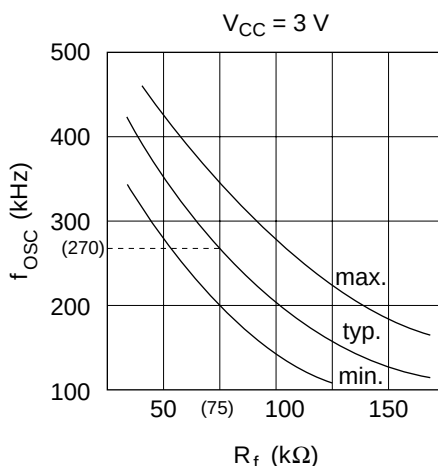
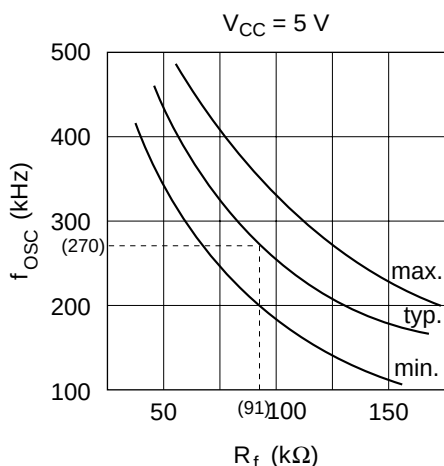
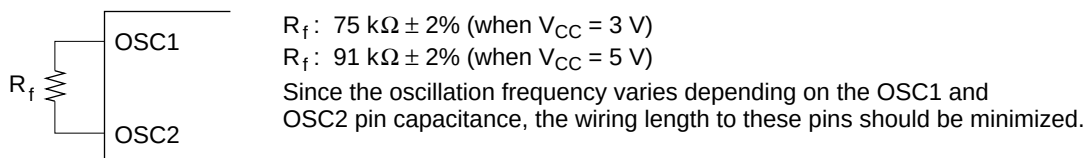
- 2. $V_{CC} \geq V1 \geq V2 \geq V3 \geq V4 \geq V5$ must be maintained.
- 3. For die products, specified at 75°C.
- 4. For die products, specified by the die shipment specification.
- 5. The following four circuits are I/O pin configurations except for liquid crystal display output.



6. Applies to input pins and I/O pins, excluding the OSC1 pin.
7. Applies to I/O pins.
8. Applies to output pins.
9. Current flowing through pull-up MOSs, excluding output drive MOSs.
10. Input/output current is excluded. When input is at an intermediate level with CMOS, the excessive current flows through the input circuit to the power supply. To avoid this from happening, the input level must be fixed high or low.
11. Applies only to external clock operation.



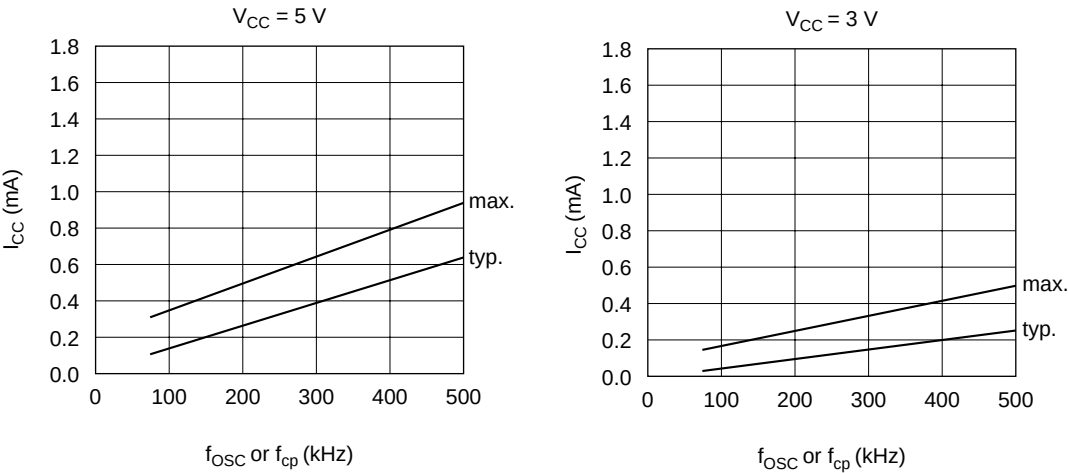
12. Applies only to the internal oscillator operation using oscillation resistor R_f.



13. RCOM is the resistance between the power supply pins (V_{CC} , V1, V4, V5) and each common signal pin (COM1 to COM16).

RSEG is the resistance between the power supply pins (V_{CC} , V2, V3, V5) and each segment signal pin (SEG1 to SEG40).

14. The following graphs show the relationship between operation frequency and current consumption.

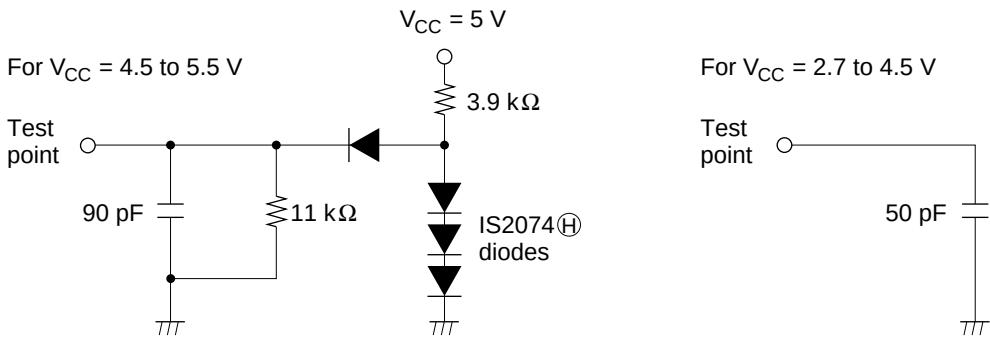


15. Applies to the OSC1 pin.

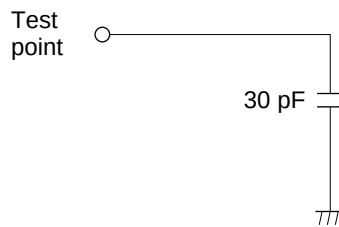
16. Each COM and SEG output voltage is within $\pm 0.15\text{ V}$ of the LCD voltage (V_{CC} , V1, V2, V3, V4, V5) when there is no load.

Load Circuits

Data Bus DB0 to DB7



External Driver Control Signals: CL1, CL2, D, M



Timing Characteristics

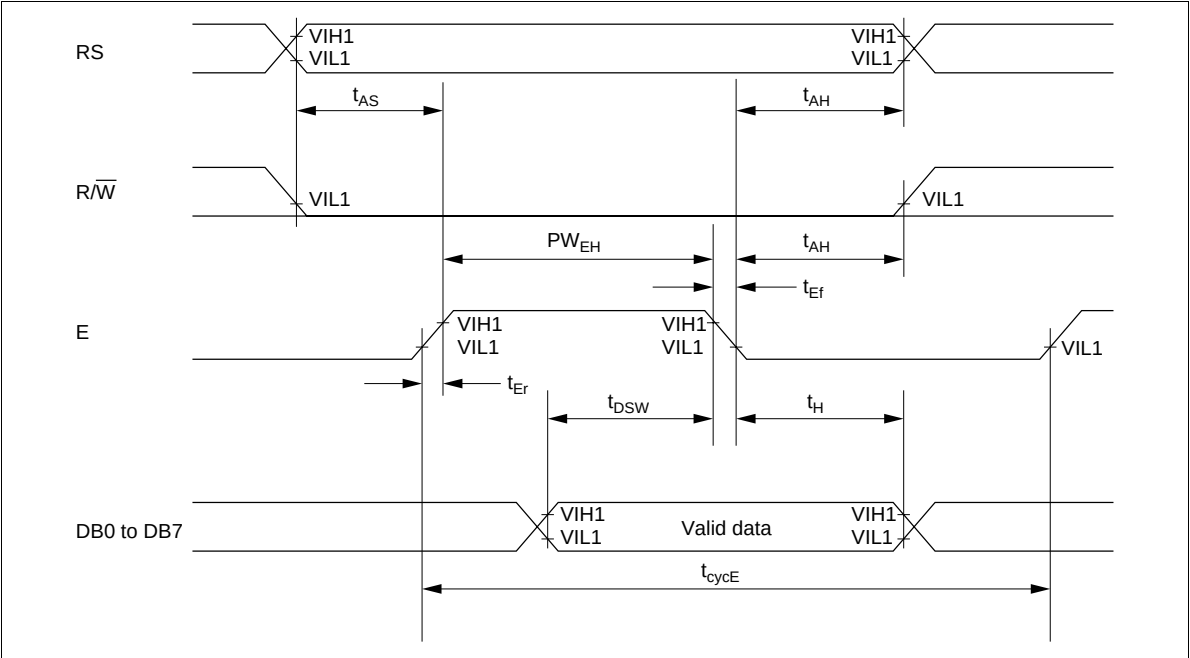


Figure 25 Write Operation

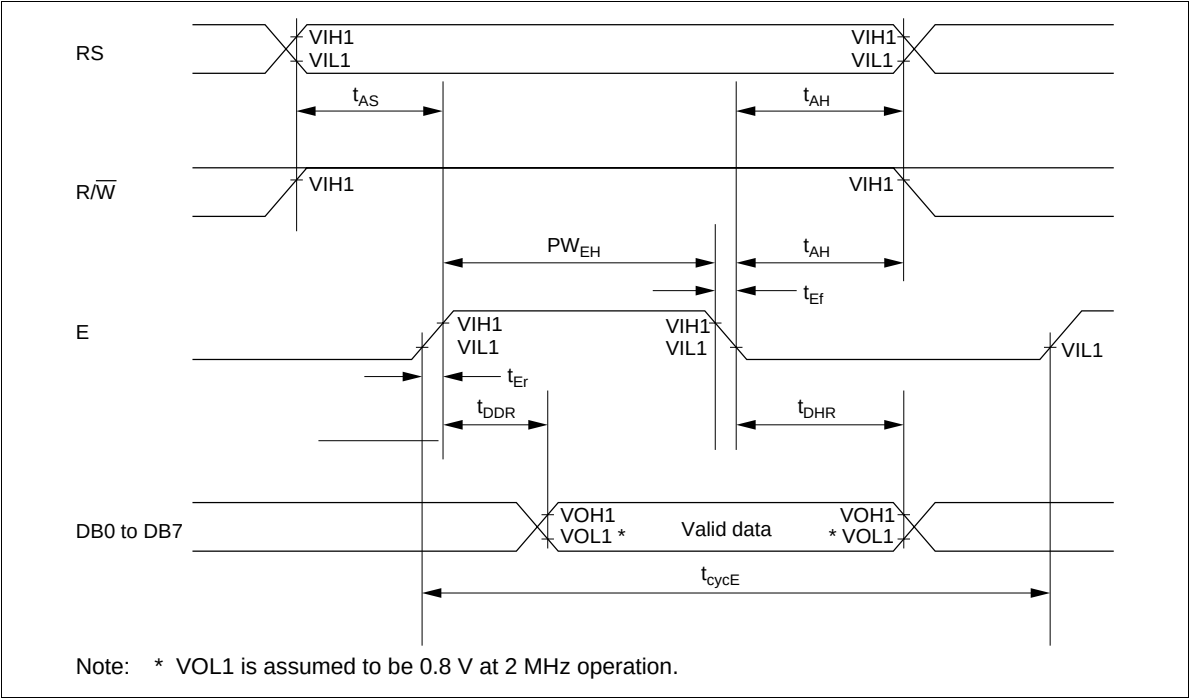


Figure 26 Read Operation

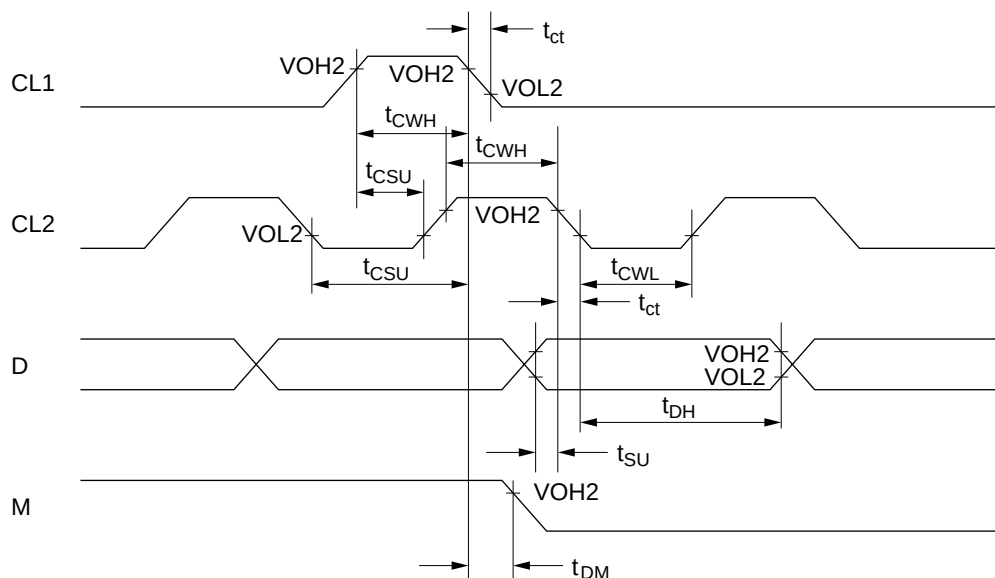
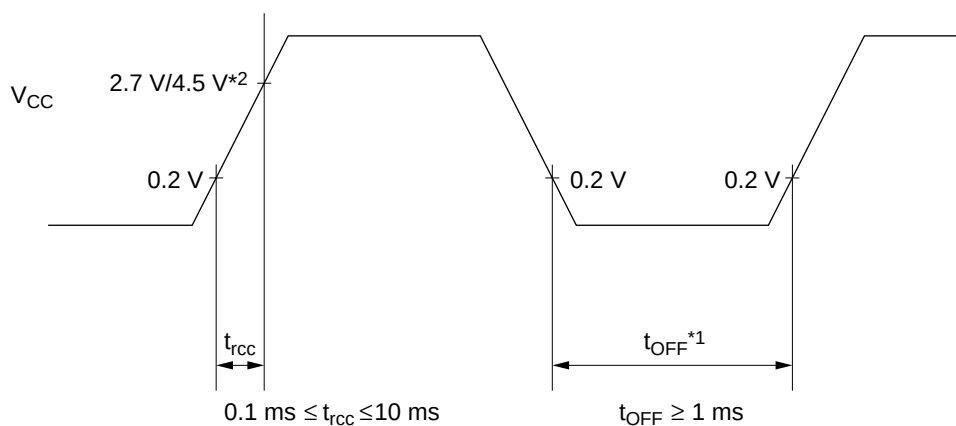


Figure 27 Interface Timing with External Driver



- Notes:
1. t_{OFF} compensates for the power oscillation period caused by momentary power supply oscillations.
 2. Specified at 4.5 V for 5-V operation, and at 2.7 V for 3-V operation.
 3. For if 4.5 V is not reached during 5-V operation, the internal reset circuit will not operate normally.
In this case, the LSI must be initialized by software. (Refer to the Initializing by Instruction section.)

Figure 28 Internal Power Supply Reset

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